Strategic Responses to College Admission Policies: Evidence from Chile

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This paper studies whether progressive college admission policies based on students' school can be undercut by the strategic pre-college behavior of sophisticated, more advantaged students. Using Chile's disclosure of a school-based affirmative action formula, I show that students who stood to gain from switching high schools became nearly three times as likely to switch in twelfth grade as before the disclosure. Switchers were more likely to graduate from highly selective and high-employability college programs and displaced disproportionately low-SES students toward college programs with weaker employment prospects.

Key Words: School choice, college admission, college outcomes, strategic responses

JEL Codes: D10, I24, I25, I28, J15, J24

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1. INTRODUCTION

College attendance is a key channel for upward mobility across developed and developing countries (Chetty et al. 2026; Barrios Fernández et al. 2024). Yet access to higher education—particularly at selective institutions—remains highly unequal, with a disproportionate share of college seats going to students from the highest income quintiles (Chetty et al. 2020; Michelman et al. 2022; Zimmerman 2019). To narrow this gap, governments have adopted a wide range of policies, one of the most widespread and contested of which is preferential admission for certain groups, or affirmative action (Arcidiacono et al. 2015; Neilson 2019).[1]

Because preferential admission based on race faces legal restrictions in several countries, including the United States, many policies instead base eligibility on characteristics common among underrepresented groups, such as the high school a student attends, as in the Texas Top Ten Percent plan (Antonovics and Backes 2014; Kapor 2025). Unlike race, however, these characteristics can be changed. Students can move to a different school or neighborhood to qualify for preferential admission without any change in effort or academic preparation. Such strategic responses could weaken the intended distributional effects by changing which students ultimately benefit, particularly when those who respond are more advantaged than the students the policy intends to benefit.

This paper shows that the intended effects of progressive college admission policies based on students' school can be weakened when sophisticated, more advantaged high school seniors respond strategically by switching schools. I exploit the disclosure of how a school-based ranking in Chile's centralized college admission system is computed, which made explicit that it depends on the school from which a student graduates, rather than every school attended, so students could raise their ranking, and thus their perceived admission chances, by transferring *at the last moment*, with no change in effort or preparation. I follow students from the switching decision through college graduation and identify which students lost seats to switchers and what losing a seat cost them. Switchers become much more likely to enroll in and graduate from highly selective and high-employability college programs. Because seats are fixed, these gains come at the expense of

[1] See Deming and Dynarski (2010) for a review. About one-quarter of countries use some form of affirmative action in higher education (Jenkins and Moses 2017).

other applicants, disproportionately students of low socioeconomic status (SES).

In Chile's centralized college admission system, each program admits students based on a weighted average of their admission exam scores, their standardized high school GPA, and a relative ranking (RR) score. The RR was introduced in 2012, but the official guidelines that students use to prepare their applications first presented its formula in 2014. The RR score transforms a student's GPA into an admission bonus based on how it compares with the average and the highest GPA of the previous three graduating cohorts at their school. While intended to expand access for high-achieving students in weaker schools, the formula implies that, for a given GPA, graduating from a school with lower historical grades yields a higher admission score.

The Chilean setting offers a clean case to study how a school-based admission rule changes students' behavior, and the distributional consequences of their response, for three main reasons. First, switching high schools is common in Chile, and administrative records show when each student moves and to which school.[2] Second, admission is centralized under common rules, and students rank specific programs (e.g., Economics at the University of Chile), which allows me to simulate counterfactual allocations and compare students just above and below each program's cutoff.[3] Third, administrative records follow students from high school through college graduation, which addresses a central data limitation in this literature (Bodoh-Creed and Hickman 2018).

By linking multiple sets of administrative records, I construct a dataset that tracks cohorts of students graduating from high school between 2010 and 2018. These data allow me to identify whether students transfer schools in twelfth grade, to distinguish transfers at the beginning of the academic year from those during the year, to compute the RR score each student would receive if they graduated from any school within their school market, and, for the cohorts graduating between 2011 and 2016, to observe where they apply, enroll, and graduate in college.

Using this information, I create two treatment measures for each student: a binary measure equal to one if the median potential gain across schools in the student's market is positive, and a continuous measure equal to that median gain, standardized within the graduating cohort.[4] I also construct a measure of SES based on mother's education, a measure of parents' aspirations

[2] Students may enroll in public, voucher, or private schools in any county.

[3] Centralized admission systems have become increasingly common worldwide (Neilson 2019), which enhances the external relevance of this setting.

[4] Given the RR formula, I can calculate potential gains for all cohorts in my sample.

from information reported in tenth-grade surveys, and a measure of school performance based on schools' average scores on the tenth-grade standardized test before the disclosure. This dataset is used throughout the analysis to estimate the causal effect of the disclosure of the RR formula on twelfth-grade school switching and subsequent college outcomes, and to identify who bore the cost of the response.

I divide the analysis into three parts. First, I develop a simple theoretical framework in which students decide whether to transfer high schools. The model clarifies when an admission rule alters the composition of admitted students and when its effects are attenuated by strategic relocation. I use the model's main predictions to structure the empirical analysis. For students to switch schools, the programs they apply to must be selective enough that a higher RR score changes where they think they can be admitted, and they must live in school markets with alternatives that would raise their ranking. Because of the formula's nonlinearity, incentives to switch are concentrated among students whose GPA is in neither the lower nor the upper tail of their school's distribution, particularly at high-threshold schools, where these thresholds generate larger potential gains from graduating elsewhere.

The second part of the paper estimates the causal effect of the disclosure of the RR formula on strategic school switching during twelfth grade and on subsequent college outcomes. Exploiting variation in students' potential gains from switching schools and a difference-in-differences event-study design, I show that the disclosure led to a meaningful increase in school transfers during twelfth grade. I focus on students living in Gran Santiago, the only metropolitan area with both competitive school and university markets. Students there who stood to gain from switching schools were 2.5 percentage points (pp) more likely to transfer in 2014 than similar students without a gain, relative to 2013, corresponding to approximately 600 additional strategic transfers in a single cohort, and an increase of 177% over the pre-disclosure mean. Monthly data further show that these transfers occurred in the second half of the academic year—when strategic switching is most valuable and least costly.[5] The response remains positive and significant under alternative definitions of the potential gain and of school markets.

[5]The magnitude of this response exceeds that documented by Cullen et al. (2013) in the context of Texas' Top Ten Percent plan, where 211 students per cohort strategically chose less competitive high schools within their estimation sample, a number the authors describe as a lower bound—despite that setting requiring families to commit to a school years in advance. The lower cost of switching in the Chilean context, where students can transfer mid-year during twelfth grade, is consistent with a larger behavioral response.

Under both treatment measures, responses follow the model's predictions. They are concentrated at high-performing schools, which have higher thresholds, where the disclosure raised the probability of switching by 5.0 pp (3.6 pp per standard deviation of the potential gain), and in the academic track, where most students who plan to attend college are enrolled. Students respond at every level of mother's education, though least when the mother did not complete high school. The largest difference is by parents' aspirations in tenth grade: students whose parents hoped they would attend college were 3.3 pp more likely to switch (2.1 pp per standard deviation) than similar students without a gain, while the rest barely respond. Responses thus depend less on family resources, which correlate with aspirations, than on whether college is the plan. Still, most students with a gain did not switch, consistent with the cost of switching and of leaving their school and peers exceeding, for most, the value of a selective program, and possibly with information frictions. Students in private and voucher schools barely respond, and those without internet at home respond less.

I then examine how these strategic responses translate into postsecondary outcomes for students with a potential gain. A positive potential gain does not increase overall admission to or enrollment in universities participating in the centralized admission system. Instead, students with positive gains are 1.1 pp more likely to enroll in and graduate from highly selective programs they entered in their first admission process, and 0.7 pp more likely to enroll in and 1.1 pp more likely to graduate from high-employability programs, with no detectable increase in dropout or delay in graduation. For the students who switched in response to the disclosure, this implies a 45 pp higher probability of graduating from a highly selective program and a 42 pp higher probability of graduating from a high-employability one, worth about 90,000 and 32,000 dollars, respectively, over a 35-year career.

The third part of the paper studies the distributional consequences of strategic switching. I exploit the structure of the admission score formula and the centralized student-program matching algorithm to identify which students are displaced. Using students' ranked application lists and each program's admission rules, I simulate the assignments under a counterfactual scenario in which no student switches schools during twelfth grade and compare these allocations to the observed matches that incorporate strategic switching. Switchers displace about 1% of admitted non-switchers, and low-SES students lose seats in highly selective and high-employability programs,

while students from high-performing and public schools gain them. Using a regression discontinuity (RD) design at the admission cutoffs of the affected programs, I find that most displaced students still enroll in college, and their main loss is a lower probability of graduating from a high-employability program. Losing a highly selective seat, by contrast, does not reduce the probability of graduating from such a program, since most students admitted at the margin of these programs drop out.

This study makes three contributions. The first contribution is to the literature on how college admission policies affect pre-college behavior, such as effort (Grau 2018; Bodoh-Creed and Hickman 2018; Tincani et al. 2025; González and Johnson 2018), attendance and dropout (Akhtari et al. 2024; Cáceres-Delpiano et al. 2018), exam performance (Antonovics and Backes 2014; Bleemer 2021; Laajaj et al. 2022), and school choice (Cullen et al. 2013; Mello 2023; Estevan et al. 2019; González and Johnson 2018). The closest papers are Cullen et al. (2013) and Mello (2023), who show that school-based admission rules in Texas and Brazil change which high school students attend, and González and Johnson (2018), who document that Chilean students below the bonus range at their own school changed schools more often after the RR was introduced. I extend this work in three ways. First, I follow students from the switching decision through college completion, which Cullen et al. (2013) do not observe and Mello (2023) observes only as far as enrollment in federal universities. Second, I identify the students displaced by switchers and estimate what losing a seat costs them, whereas Cullen et al. (2013) count displaced minority students but cannot follow them into college. Third, students in Chile move in their final year, most of them in its last months, rather than years before applying.[6] This limits effects on learning and lets me compute each student's gain directly from their GPA and the thresholds of the schools in their market, rather than from ability proxies or area-level exposure.[7]

The second contribution is to the literature evaluating the effects of college admission policies more broadly. This literature has shown that both the context and design of these policies shape outcomes for disadvantaged students (Andrews and Stange 2019; Angrist et al. 2022; Harris and Mills 2021; Kapor 2025; Long et al. 2010) and has examined their distributional effects (Otero et al.

[6] In Texas, schools typically require transferring students to attend for some time before counting them in the class rank, so transfers after the spring of tenth grade may not be rewarded (Cullen et al. 2013, p. 34). In Brazil, the quota applies only to students who attended all three years of upper secondary education in a public school (Mello 2023, p. 2).

[7] Cullen et al. (2013) predict each student's probability of reaching the top 10% of each school from eighth-grade test scores, and describe their estimates as lower bounds due to "relatively crude proxies for individual incentives" (p. 44). Mello (2023) assigns each student the quota exposure of the federal universities in their microregion of residence.

2021; Bleemer 2021; Black et al. 2023; Mello 2023; Reyes 2022; Bucarey 2017; Bertrand et al. 2010). I contribute by providing evidence that the design of the RR policy—specifically, its dependence on relative performance within the school a student graduates from—induced strategic pre-college responses that reallocated selective and high-employability seats away from low-SES students.

The third contribution is to the literature on relative grading and rank-based incentives. Research documents that students' ordinal rank affects their peer interactions and educational choices (Calsamiglia and Loviglio 2019), college attendance (Elsner and Isphording 2017; Diamond and Persson 2016; Rangvid 2015), and later earnings (Diamond and Persson 2016). I show that when college admission scores depend on relative performance within the school a student graduates from, students in higher-performing schools may receive lower admission scores than those in lower-performing schools despite having identical GPAs. This feature creates an incentive to transfer to lower-performing schools to improve admission prospects.

The rest of the paper is organized as follows. Section 2 describes the institutional setting. Section 3 introduces a stylized model rationalizing the incentives for students to switch schools and their equilibrium effects. Section 4 presents the data, descriptive statistics, and construction of the main variables. Section 5 outlines the identification strategies. Section 6 presents the empirical results on strategic switching and college outcomes. Section 7 studies the distributional consequences of strategic switching. Section 8 concludes.

2. CONTEXT: FROM SECONDARY TO HIGHER EDUCATION IN CHILE

Chile is a middle-income country with persistently high income and educational inequality.[8] Narayan et al. (2018) rank it among the least socially mobile countries in the world, based on the share of individuals born into the bottom half of the income distribution in the 1980s who reached the top quartile as adults.

[8] According to World Bank estimates, Chile's Gini coefficient was 44.4 in 2017—comparable to that of the United States, which stood at 41.4 in 2018. Educational attainment statistics also reveal striking disparities: in 2019, 85.2% of adults aged 25 to 34 had completed at least high school, but only 33.7% had attained a higher education degree (Education at a Glance, 2020).

2.1. Secondary Education

Chile has the highest secondary education graduation rate in Latin America, with relatively little variation across socioeconomic groups. In 2022, 89% of young people completed upper secondary education—85% among those in the lowest income quintile and 95% among those in the highest (CASEN 2022). Yet, students from low socioeconomic backgrounds attend schools with significantly lower academic performance. Figure 1 presents the relationship between schools' ranking and the school vulnerability index.[9] The figure shows a strong positive correlation ($\rho = 0.79$), indicating that lower-ranked schools tend to serve a larger proportion of socioeconomically disadvantaged students.

In contrast to the United States, Chile's education system allows parents to enroll their children in any K–12 school, irrespective of residential location, resulting in one of the most robust school-choice systems globally. This high degree of choice applies not only to private schools—which served roughly 7% of high school students in 2010—but also to public and voucher-subsidized schools, which together accounted for about 93% of all high school students nationwide.[10,11] In the Santiago metropolitan area, average scores on the national college admission test differ systematically between private schools and public and voucher schools (see Table A1).

Differences in school performance explain part of the large gap in tertiary education attendance. In 2022, about 70% of individuals from the highest income quintile attended some form of tertiary education, compared with only 25% from the lowest quintile.[12] The link between students' SES, school quality, and access to higher education has been a central concern of education policy reform in Chile. In 2011, the country witnessed one of the longest student-led protests in its history, known as the "Chilean Winter," which called for structural changes to the education system.[13] Advocates for reform argued that students from low-quality schools face significantly lower chances of

[9] School ranking was calculated as the average score on a standardized test taken by tenth-grade students in 2010, with 1 assigned to the school with the highest average.

[10] The share of high school students enrolled in private schools increased by 1.4 pp (about 20%) between 2010 and 2022.

[11] This paper focuses on the Santiago metropolitan area, where about 40% of the country's population lives. Table A1 shows that about half of the students in the Santiago metropolitan area attended a school located outside their county of residence, with similar patterns across sectors.

[12] The net attendance rate indicator measures the percentage of people effectively attending tertiary education over the population that should be attending according to their age (Arias Ortiz et al. 2024).

[13] Gabriel Boric, one of the leaders of this movement, was president of Chile from 2022 to 2026. See: `https://www.theguardian.com/world/2021/nov/18/a-fairer-chile-ex-student-leader-bids-to-reshape-country-in-divisive-election`.

attending college—regardless of their individual ability or academic performance—due to systemic disadvantages tied to school quality. The RR policy examined in this paper was introduced in response to these concerns. It was designed to expand college access for high-performing students by accounting for the quality of the school they attended (see Section 2.3 for details).

2.2. College Application System

In the 2013 admission process, all 25 traditional universities and 8 private universities in Chile admitted students through a centralized admissions system. Since that process, each applicant's admission scores (AS) is a weighted sum of three components—the national standardized test (PSU), the student's standardized high school GPA (SGPA; NEM in Spanish), and the relative ranking score (RR)—with weights that can vary by program. DEMRE, the institution in charge of the centralized college admission system, publishes detailed information about the admission process, including weights and seats by program, in a preliminary version around June of each year, six months before students' applications are submitted, and a final version around October. Although the weights are set internally by each university for each of its programs, they do not vary substantially within universities.[14] For example, the Universidad de Chile (UCH), one of the two most selective universities in the country, offered 50 programs in the 2014 admission process, using only two different weights for the SGPA and two for the RR across all the programs. Table A2 presents the average weights of the GPA and the RR across the years; the national college entrance exams (language, mathematics, and history or science) account for the remainder.

From the student's perspective, applying through the centralized system involves three main steps. First, students must complete high school with a GPA of at least 4.0 on a 1–7 scale.[15] Second, they must take the PSU,[16] which was administered annually in mid- to late December during the period I study.[17] Third, applicants submit an ordered list of programs. To support students and families in this process, DEMRE maintains an official website with comprehensive information on

[14] University fixed effects account for 73% of the variance in the GPA weight and 83% in the RR weight in the 2014 admission process.

[15] The centralized system standardizes students' high school GPA into a score ranging from 208 (for a GPA of 4.0) to 826 (for a GPA of 7.0), based on the average across all four years of high school.

[16] The PSU includes mandatory language and mathematic sections, as well as an elective test in either history or science. Each test score is normally distributed with a mean of 500 and a standard deviation of 110, truncated at 150 and 850 points.

[17] The academic year in Chile runs from March to early December.

processes, timelines, and participating institutions.[18]

Applicants to the centralized system are matched to programs using a deferred acceptance (DA) algorithm.[19] In the period I study, students may rank at most ten programs, and the two most selective universities (Pontificia Universidad Católica de Chile and Universidad de Chile) further restrict where their programs may be ranked, considering only those listed among a student's first four preferences.[20] Table A3 summarizes application behavior and match outcomes in the 2013 admission process. Column (1) shows that more than 92% of applicants rank fewer than ten programs, and Column (3) shows that 93.9% of admitted students are placed within their first four preferences.

2.3. The Relative Ranking Policy

In June 2012, to support students from low-ranked high schools with strong grades but lower PSU scores, the Consejo de Rectores de las Universidades Chilenas (CRUCH) introduced a new component to the college admission criteria, the *relative ranking* (RR).[21] The RR compares a student's GPA with the average GPA of the three previous graduating cohorts from the same school. Because students are evaluated relative to earlier cohorts at their own school, they do not compete with peers from their own class for a higher RR score.

The standardized RR score is derived from a nonlinear transformation of a student's high school GPA. When a student's GPA is below the average of the three previous cohorts at their school, their RR score equals their standardized GPA score. When the GPA lies between that average and the highest GPA observed among those cohorts, the student receives a proportional bonus. Finally, students whose GPA exceeds the highest GPA of the prior three cohorts receive the maximum RR score of 850 points. Figure 2 illustrates this mapping, with the blue line depicting the linear transformation from GPA to SGPA, and the gold line showing the nonlinear relationship between GPA and the standardized RR score. In the figure, $\underline{r}_s$ denotes the average GPA across the three

[18] See `https://demre.cl/index` for more information.

[19] In each round, students apply to their highest-ranked available program, which tentatively accepts applicants up to capacity based on their AS. Rejected students move to their next choice. The process repeats until no applicant can improve their assignment.

[20] Source: DEMRE's annual publication *Oferta Definitiva de Carreras, Vacantes y Ponderaciones*.

[21] Among other studies, the policy drew on "Determinantes del desempeño universitario: ¿Importa la habilidad relativa?" (Contreras et al. 2009).

previous cohorts, and $\bar{r}_s$ the maximum GPA within those cohorts. I use the same notation throughout the paper.

For students to respond strategically, the RR formula must be salient. I review the documents available on the university application system's official website, together with press coverage from the period, to establish when the relevant information was public and in what form. Although the policy was implemented in 2012,[22] the official guidelines did not clearly explain how the RR was computed until 2014. The policy itself, unlike its formula, was discussed openly from the start, with some experts supporting it and others raising its potential unintended consequences.[23]

Figure A1 reproduces the corresponding sections of the official guidelines for the 2014 and 2015 admission processes (i.e., students applying in 2013 and 2014, respectively). In the earlier document, the RR score is described in prose, within the section on test-score normalization. In the later one, it has its own section, which states what happens to students below their school's historical average, and which includes a figure plotting the RR score against a student's GPA. The later document also makes explicit that the comparison is with previous cohorts of the school from which the student graduates. In August 2014, the rule was changed, effective from the 2016 admission process, to weight scores across all schools attended, limiting the gains from last-minute switches (Figure A2 shows the RR policy timeline).

The introduction of a clearer description of the RR policy in the students' guidelines does not constitute direct evidence on families' information sets. What it establishes is that the incentive was stated clearly only from 2014 onward, in the document students most likely consult when applying to college. This is consistent with the identifying assumption that the disclosure of the RR formula marked an inflection point in students' ability to act strategically on the policy by switching schools. I do not rule out the possibility that the information was also spread through other types of sources, such as peers and schools, but I argue that the clarification in the guidelines made such diffusion possible.

[22] That is, the admission process for the 2013 academic year.

[23] For supporters, see, for example, "Los datos que respaldan la utilidad del ranking de notas," published by CIPER on July 17, 2012: `https://www.ciperchile.cl/2012/07/17/los-datos-que-respaldan-la-utilidad-del-ranking-de-notas/`. For critics, see, for example, "Ranking de notas: ¿equidad o confusión?" a compilation of press coverage published by José Joaquín Brunner on July 15, 2012: `https://brunner.cl/2012/07/ranking-de-notas-equidad-o-confusion/`.

2.4. Tertiary Education

Table 1 summarizes key characteristics of postsecondary institutions for the 2014 academic year. Within the centralized system, public universities place greater weight on the RR component and less on the quantitative PSU section compared with private universities. Programs in the centralized system offer about 46 seats on average, 38 at public and 52 at private universities.

Among higher education institutions, universities in the centralized system are generally considered the most selective and prestigious.[24] I classify selectivity at the university and program levels. Universities are classified using the tiers of Kapor et al. (2024), which group them by the mean admission scores of the students admitted in 2012. All tier 1–2 universities offer their programs in the Santiago metropolitan area, and only one of them, the Universidad Adolfo Ibáñez, also offers programs in another metropolitan area, Gran Valparaíso (see Table A4 for selectivity tiers). Similarly, programs are ranked by the mean admission scores of the students they accepted in the 2014 admission process, the last before the disclosure of the RR formula. A program is classified as highly selective if it is in the top quartile of that ranking among the 1,406 programs with a selectivity measure, out of the 1,419 offered that year. Of the highly selective programs, 46% are at tier 1–2 universities, and 35% are high-employability programs (Table A5).

3. CONCEPTUAL FRAMEWORK

I develop a simple theoretical model to identify which students have incentives to game the policy by switching schools and to assess how switching may affect the policy's overall effectiveness. In the first part of this section, I derive the main drivers of students' potential gains under the policy. I then present a model of school-switching decisions and its main testable predictions.[25]

3.1. admission scoress

Suppose there are two high schools, L and H, and two colleges: a selective college and a less selective college. A fraction μ_H of students attend school H, and the remaining $\mu_L = 1 - \mu_H$ attend school L.

[24] According to the QS World University Rankings: Latin America & The Caribbean 2027, the Pontificia Universidad Católica de Chile and the Universidad de Chile ranked first and seventh, respectively, and all other Chilean universities in the top 50 also belong to the centralized admission system.

[25] In Appendix B, I analyze how the admission cutoff and the composition of the admitted student body change in equilibrium before and after the introduction of the RR policy.

Each high school $j \in \{H, L\}$ is characterized by two predetermined, publicly known parameters: a mean threshold $\underline{r}_j$ and a maximum threshold $\bar{r}_j$.

While the initial school choice decision is important and has been widely studied,[26] I take students' initial assignment as given and focus exclusively on the switching decision. Each student i is characterized by a GPA $g_i \in (\underline{g}, \bar{g})$ and an initial high school $s \in \{L, H\}$.

The selective college has capacity $K < 1$ and admits students based on their admission scoress; the remaining students enroll in the less selective college, which has no capacity constraint. This structure mirrors Chile's centralized admission system, with institutions outside it as the less selective option.[27]

Before the policy, student i's admission scores is a linear function of their GPA:[28]

$$AS(g_i) = \gamma g_i + \theta, \tag{1}$$

where γ is a slope parameter and θ is a constant. Because PSU scores and GPA are unaffected by switching, I write the admission scores as a function of GPA alone.

After the policy, the mapping from GPA to admission scoress depends on the school $e \in \{H, L\}$ from which the student graduates:

$$AS_e(g_i) = \begin{cases} \gamma g_i + \theta, & \text{if } g_i < \underline{r}_e, \\ \gamma_e g_i + \theta_e, & \text{if } \underline{r}_e \le g_i < \bar{r}_e, \\ 850, & \text{if } g_i \ge \bar{r}_e, \end{cases} \tag{2}$$

where γ_e and θ_e denote school-specific slope and intercept parameters, and 850 is the maximum possible score, with $\gamma_e > \gamma$. A student with GPA below $\underline{r}_e$ receives the same score as under the pre-policy rule, and a student with GPA between $\underline{r}_e$ and $\bar{r}_e$ receives a school-specific bonus. A student with GPA above $\bar{r}_e$ receives the maximum score.

[26] See, for example, Alves et al. (2015); Pop-Eleches and Urquiola (2013); Hastings et al. (2012); Neilson (2013); Allende (2019).

[27] Chile has a large number of universities, Institutos Profesionales and Centros de Formación Técnica outside the centralized system that serve as additional options for students who are not placed centrally. Therefore, the relevant margin for the vast majority of applicants is which institution they attend, not whether they attend college at all.

[28] This formula follows the official conversion table published by DEMRE. The admission scores does not vary across schools in equation (1).

Figure 3 illustrates the case in which school H has higher thresholds than school L.[29] Let $s \in \{H, L\}$ denote the school in which a student starts twelfth grade and $e \in \{H, L\}$ the school from which they graduate, so that $e = s$ if the student does not switch. A student has a potential gain from switching if the school they could switch to gives them a higher admission scores than their own:

$$AS_e(g_i) > AS_s(g_i), \text{ with } e \neq s.$$

3.2. Switching Decision

To model the switching decision, I adapt the frameworks of Cullen et al. (2013) and Estevan et al. (2019). A continuum of students of mass 1 applies from both schools and is placed into one of the two colleges. The selective college admits students whose admission scores is at or above an equilibrium cutoff AS^*, while the remaining students enroll in the less selective college.

A student admitted to the selective college derives utility $U_i^S > 0$ relative to the less selective college (normalized to zero).[30] Students pay a direct cost $c_{ise} \geq 0$ when switching from school s to school e and derive school-specific utility b_{is} from school s with $b_{is} > b_{ie}$.[31] The student's utility from staying in school s is $V_{is} = b_{is} + U_i^S \cdot \mathbf{1}\{AS_s(g_i) \geq AS^*\}$, while the utility from switching to school e is $V_{ie} = b_{ie} - c_{ise} + U_i^S \cdot \mathbf{1}\{AS_e(g_i) \geq AS^*\}$. Because $b_{is} > b_{ie}$ and $c_{ise} \geq 0$, switching can raise a student's utility only when they are below the selective-college cutoff at school s but above it at school e: $AS_e(g_i) \geq AS^* > AS_s(g_i)$. In that case, switching is optimal if and only if

$$U_i^S \geq b_{is} - b_{ie} + c_{ise}, \tag{3}$$

where $b_{is} - b_{ie} + c_{ise} \equiv \tilde{c}_i$ bundles the school-value differential and the direct switching cost.

Testable predictions. The model generates three main predictions. First, potential gains from switching are concentrated among students whose GPA falls in the bonus region of a school they could switch to, i.e., between the mean and maximum thresholds of that school, rather than in the

[29] Although other combinations are possible, I do not discuss them in detail for two reasons. First, in Chile, schools with higher $\underline{r}$ tend to also have higher $\bar{r}$ (see Table A1). Second, I do not impose any restrictions in the empirical analysis when calculating students' potential gains.

[30] U_i^S captures the premium from attending the selective college over the less selective college—for example, the difference in expected returns to a degree from an elite versus a non-elite institution.

[31] This inequality reflects the fact that families chose school s at the beginning of secondary education.

bonus region of their own school. Second, students from high-threshold schools are more likely to have a positive gain (see Figure 3). Third, a positive score gain is necessary but not sufficient for switching: the gain must move the student from the less selective college into the selective college, and the selective-college premium must exceed the cost of switching. In practice, students do not know the cutoff when they decide to switch, switching should be more attractive in college markets with highly selective programs, where a given score gain is more likely to change admission.

4. DATA

4.1. Administrative Data Sources

This paper combines four administrative data sources from the Chilean education system, covering students enrolled in twelfth grade nationwide between 2010 and 2018. The Ministry of Education (MINEDUC) provides student-level information such as gender, school enrollment history, and high school GPA, as well as school-level characteristics including geolocation, school type (public, voucher, or private), and other institutional attributes. MINEDUC also compiles DEMRE's college program information, including the weights assigned to each college admission component, the number of available seats, and program-level characteristics. The second dataset comes from the Higher Education Information Service (SIES), which records the yearly enrollment and graduation of every student in higher education.

The third dataset comes from DEMRE (Departamento de Evaluación, Medición y Registro Educacional). These data include national test scores by subject, students' ranked lists of up to ten program preferences, admission results, and household characteristics at the student level.

Finally, the fourth set of administrative records is provided by the Education Quality Agency (Agencia de Calidad de la Educación), including students' scores on the tenth-grade standardized test (SIMCE), self-reported SES, parental education, and parental and student aspirations. One limitation of the SIMCE data is that the test and questionnaire were administered in 2006, 2008, 2010, and every year from 2012 onward, so students who finished twelfth grade in 2011 and 2013 have no tenth-grade standardized scores or questionnaire responses, and those two cohorts are excluded from any analysis that relies on them.[32] Together, the MINEDUC records identify school

[32]The student questionnaire was first administered in 2010, so the cohort that finished twelfth grade in 2010 also lacks

switching and potential gains, the DEMRE records cover applications and admission, the SIES records follow enrollment, dropout and graduation, and the SIMCE records provide tenth-grade achievement, family aspirations and background.

4.2. Data Limitations

Although Chilean administrative data offer rich student-level information, the data used in this paper have two main limitations. First, I only observe self-reported income brackets in the SIMCE questionnaires and, for college applicants, in DEMRE records. The latter are used to determine financial aid eligibility and are therefore likely under-reported. To characterize students by SES, I rely on maternal education as a proxy. This measure has two advantages over self-reported income: it is less likely to be biased and is available for most cohorts regardless of whether they apply to college. Because self-reported income is itself noisy, I validate maternal education at the school level. Figure A3 shows a strong association between maternal education and the school vulnerability index.

Second, I do not have access to students' home addresses, which limits my ability to define students' local school markets. I approximate residential locations using the geolocation of each student's primary school. Using this proxy, I define each student's relevant school market as the set of schools within 2 km of their primary school.[33] The 2-km school market is broadly consistent with how far students travel in practice: in the Santiago metropolitan area (Gran Santiago), half of the students start twelfth grade in a school within 2 km of their primary school and two thirds within 4 km. Similarly, 78% of twelfth graders attend primary and secondary school in the same county.[34] Both shares are stable across cohorts and differ little by treatment status (see Figures A4 and A5). Moreover, in Appendix D, I re-estimate the switching event study using buffers of 4, 6, and 8 km, and the results are similar.

student aspirations.

[33]This approach is motivated by findings in the school choice literature, which document a strong relationship between distance and school selection in primary education (Neilson 2013; Allende 2019).

[34]The median county in the Santiago metropolitan area covers about 38 km^2.

4.3. Descriptive Statistics and Variables Construction

Using the census of schools and students for the cohorts entering twelfth grade between 2010 and 2018, I construct two measures of potential gain, one discrete and one continuous, and the first outcome of interest: whether a student switches schools during twelfth grade. For college outcomes, such as enrollment, dropout and graduation, I use the census of applications and enrollment and graduation for the cohorts graduating from high school between 2011 and 2016. All descriptive statistics refer to the main sample, i.e., twelfth graders in Gran Santiago, the college market with the highest share of programs in the national top 100 (Appendix H, Table A19).

I classify schools into performance quartiles by funding type and region, based on average scores from the national standardized test taken by tenth-grade students in 2010.[35] Figure A6 shows that schools with higher tenth-grade scores also tend to have higher GPA, and therefore higher mean thresholds ($\underline{r}$) with a pre-policy correlation of 0.7 between the two.[36]

Table 2 presents descriptive statistics for twelfth graders in the switching sample. About 4% switch schools in their final year, about 40% of them during the school year. About 85% attend either public (28%) or voucher (57%) high schools. In terms of socioeconomic background, 74% have a mother with at least a high school diploma, and a similar share (79%) report having access to the internet at home.

Table 3 presents descriptive statistics for the students who apply to college. Of these, 75% list a highly selective program among their three most preferred. Around 86% are admitted to a program on their list and 28% to a high-employability program. As expected given the selectivity of the centralized admission system, students applying to college perform academically better than twelfth graders as a whole and come disproportionately from private schools and from schools in the top two performance quartiles, which account for about 35% and 84% of applicants, respectively.

I. Potential Gain

A student's *potential gain* is the difference between the RR score they would obtain if they had graduated from an alternative school e, one they were not attending at the beginning of twelfth grade, and the score they would receive from graduating from their initial school s. I compute

[35] I also observe tenth-grade scores for 2012–2018. I use scores in 2010 to have a pre-policy measure.

[36] Table A1 reports additional descriptive statistics.

this gain for every school in the student's school market, defined as the set of high schools located within a 2 km radius of their primary school (see Section 4.2 for data limitations, Figure A7, and Appendix D for school market validation).

I summarize each student's *potential gains* across their school market using their *median*. The median captures whether a student faces a generally favorable environment for strategic switching, rather than depending on the existence of a single attractive option. It is also robust to outliers that may arise from measurement error in individual schools' thresholds or from schools that are geographically proximate but practically inaccessible. Most importantly, it does not require families to have detailed knowledge of every school in their market—a strong assumption given evidence that parents often hold imperfect information about available schools and their attributes (Hastings and Weinstein 2008; Kapor et al. 2020).[37]

Because the RR formula relies only on predetermined variables (see Section 2.3), I can compute the *median potential gain* for every cohort, including those that finished high school before the policy implementation. For cohorts graduating from 2012 to 2015, the RR score I compute matches the one in the administrative records closely. Figure A8 presents the distribution of the differences between the two, with a mean of 0.20 points on a scale that runs from 208 to 850, a standard deviation of 125, and a median of zero.

For the *median potential gain* to identify switching driven by the disclosure of the RR formula, two conditions must hold. First, families should not sort residentially into different school markets in response to the disclosure. Second, any error in constructing the *median potential gain* or the school markets must be unrelated to unobserved determinants of switching and college admission outcomes.

One feature of the Chilean context mitigates these concerns. First, students are not assigned to schools based on residence, which limits incentives for selective migration following policy changes. Additionally, following Chetty et al. (2014), I define each student's school market around their primary school rather than the school where they start twelfth grade, so the treatment variable does not depend on students' subsequent switching or migration decisions.

Using the *median potential gain*, I classify students into two groups: those with a positive gain

[37] I re-estimate the switching analysis using different measures of treatment and the results are robust to alternative definitions (see Appendix C).

from switching schools and those without. I also use a continuous measure, the *median potential gain* standardized within each graduating cohort. On average, 38% of twelfth-grade students in Gran Santiago have a positive median gain within their school market. Before the standardization, the median potential gain has a mean of 4.5 points. Lastly, the share of students with a positive gain and the distribution of the median gain are stable across cohorts (Figures 4 and A9). Tables 2 and 3 report descriptive statistics for twelfth graders and college applicants separately for students with no gain (Columns 3 and 4) and with a gain (Columns 5 and 6).

II. **Switching Schools during Twelfth Grade**

I focus on switching during the academic year because, after the disclosure and until its modification, the RR formula rewarded students who changed schools late (see Section 2.3). I define switching during the academic year as one if the student is enrolled in one school in March, at the start of the academic year, and in a different one in December, when grades are published. Because enrollment is recorded at both points, these moves can be distinguished from switches before the year begins. In addition, students are permitted to switch schools during the year up to two months before it ends, provided that parents request the switch and the original school completes a form reporting the student's GPA up to that point. For additional details on the construction of the switching variable, including its monthly definition, see Appendix G.

Figure 5 shows the number of students switching schools in twelfth grade between 2010 and 2018 in the Santiago metropolitan area. On average, around 2,600 students switch schools in their final year, about 40% of them during the academic year. Both panels in Figure 5 present two spikes: one in mid-2011, due to the student protests, which extended to the start of 2012, and another in 2014, when the RR formula was clarified in students' guidelines. The 2014 spike comes from mid-year switches, particularly in high-performing schools, consistent with a strategic response to the policy and with the model's prediction that students in schools with higher thresholds should react more (see Appendix F for a detailed discussion of the differences between the two spikes).

III. **College Outcomes**

For college outcomes, I restrict the analysis to cohorts graduating from high school between 2011 and 2016 for two reasons. First, in the 2012 admission process, the first for the 2011 cohort, eight additional universities joined the centralized admission system, increasing the number of seats

by approximately 40% (Kapor et al. 2024). Second, graduation is not yet fully observed for later cohorts. By 2025, the last year of the data, 66% of the 2016 cohort who enrolled in a centralized university had graduated, compared with 51% of the 2018 cohort.

The college outcomes studied in the paper are whether the student applies to, is admitted to, and enrolls in a program in the centralized system, whether they drop out of or switch out of their first program within two years, and whether they graduate from their first program and do so on time. I measure each outcome for all programs and for programs that are highly selective, high-employability, or high-earnings. For students who do not apply through the centralized system, I do not distinguish between enrolling in an institution outside it and entering the labor market. For additional details on the construction of college outcomes, including program characteristics, see Appendix G.

5. EMPIRICAL SPECIFICATION

5.1. Switching Schools during Twelfth Grade

To examine the impact of the disclosure of the RR formula on students' strategic behavior during twelfth grade, I estimate the following specification:

$$switch_during12_{istm} = \sum_{\substack{\tau=2010 \\ \tau\neq 2013}}^{2018} \beta_\tau \mathbb{1}\{t=\tau\} \cdot positive_gain_{ism} + \beta_{pg}\, positive_gain_{ism} + \delta_t + \delta_s + \delta_{g\,pa_decile} + \varepsilon_{istm}, \tag{4}$$

where $switch_during12_{istm}$ equals 1 if student i, attending school s at the beginning of twelfth grade in year t and belonging to school market m, switches schools during that year, and 0 otherwise. $positive_gain_{ism}$ is a dummy equal to 1 if the student has a positive median potential gain. I also estimate the same equation using the continuous standardized gain, $standardized_gain_{ism}$.

The treatment variable, binary or continuous, is interacted with event-year dummies, $\mathbb{1}\{t=\tau\}$, to test whether the disclosure of the RR formula affected students' behavior only in the year it was made public (i.e., 2014). The omitted category corresponds to the cohort in twelfth grade in

2013, the year immediately before the disclosure. With the binary measure, each coefficient β_τ measures the difference in the probability of switching schools between students with a positive gain and those without in cohort τ, relative to the same difference in the 2013 cohort. With the continuous treatment, β_τ measures the change in the probability of switching associated with a one-standard-deviation increase in the gain, relative to 2013.

The main specification controls for several sets of fixed effects. δ_t denotes twelfth-grade cohort dummies, which capture year-specific shocks to the likelihood of switching schools for all students. δ_s represents beginning-of-twelfth-grade school fixed effects, which absorb time-invariant differences across schools, such as disciplinary structure or administrative rigidity, that may influence switching. Finally, δ_{gpa_decile} includes dummies for each decile of the within-school GPA distribution at the start of twelfth grade, which control for the differences in the propensity to switch schools across the GPA distribution that are constant over time. With these fixed effects, β_τ is identified from the change, relative to 2013, in the switching gap between students with and without a positive gain, net of time-invariant differences across schools and GPA deciles. Students in the same school and GPA decile differ in potential gain for two reasons: they face different local school markets, and their own GPA places them at different distances from the ranking thresholds of their own school and of the schools in their school market. Standard errors are clustered by primary school, which defines the school market.

To interpret the estimated coefficients as causal effects, two identifying assumptions must hold. First, absent the disclosure, switching would have evolved in parallel for students with and without a positive gain. Second, treatment assignment must be stable over time, that is, the likelihood of being classified as having a positive gain should not change for reasons unrelated to the disclosure of the RR formula. When these assumptions are satisfied, Equation (4) identifies the causal effect of the disclosure on students' strategic decision to switch schools.

Two concerns could undermine a causal reading of the estimates. First, because the RR policy was implemented in 2012 and the weight of the RR component in the AS increased for the 2014 admission process, the estimates could reflect the growing bite of the policy rather than the disclosure of the formula. The timing of the response separates the two. The average weight of the RR in admission scoress more than doubled for the 2013 cohort, from 10.0 to 22.2 points, and rose by less than one point for the 2014 cohort (Table A2). If students responded to the weight of the RR

rather than to the disclosure of its formula, switching would have increased in 2013, rather than in 2014. Appendix I presents further evidence that the response is unlikely to reflect the policy's bite. Second, because treatment is built from GPA and school thresholds, schools that inflate grades differentially could change which students are classified as having a positive gain. Appendix E re-estimates Equation (4) with measures rebuilt from deflated grades following Fajnzylber et al. (2019), and the estimates are essentially unchanged because only 2.4% of students change treatment status.

5.2. College Outcomes

For college outcomes, I estimate the reduced form of the same specification as in Section 5.1:

$$(5)\quad college_outcome_{istm} = \sum_{\substack{\tau=2011 \\ \tau\neq 2013}}^{2016} \beta_\tau \mathbb{1}\{t=\tau\} \cdot positive_gain_{ism} + \beta_{pg}\, positive_gain_{ism} + \delta_t + \delta_s + \delta_{g\,pa_decile} + \nu_{istm},$$

where $college_outcome_{istm}$ is a college outcome for student i, starting twelfth grade in school s, and belonging to school market m in year t. The college outcomes of interest are whether the student applies to, is admitted to, and enrolls in a program in the centralized system, whether they drop out of or switch out of their first program within two years, and whether they graduate from it and do so on time. I measure each outcome for all programs and separately for highly selective, high-employability, and high-earnings programs.

As before, $positive_gain_{ism}$ is interacted with the event-year dummies, with 2013 as the omitted cohort, and the specification includes the same set of fixed effects as in Equation (4). Unlike for switching, I estimate the college outcomes with the positive gain only. The effect on the students who switched, recovered 6.2, is the ratio of the reduced form to the first stage, and with a binary instrument this ratio is the effect on those who switched in response to the disclosure. With the standardized gain it would be a weighted average of such effects across values of the gain, with weights that are harder to interpret (Imbens and Angrist 1994).

The same assumptions as in Section 5.1 must hold to recover the reduced-form effects on college outcomes. In addition, the reduced form requires no interference between students: a

student's college outcome should not depend on whether other students switch. Because seats in the centralized system are fixed, a switcher admitted to a program can displace a non-switcher, including students without a positive gain who form the comparison group, either from that program or, through a chain of displacements, from another one. Section 7 shows that in the 2015 admission process switchers displaced about 1% of non-switchers from a program they would otherwise have been admitted to, at similar rates among students with and without a positive gain.

Recovering a local average treatment effect requires the parallel trend assumptions stated in Sections 5.1 and 5.2 to hold jointly, together with two further assumptions (Hudson et al. 2017). The first is that the disclosure led no student to stay who would otherwise have moved (*monotonicity*). The second is that a positive potential gain affects college outcomes only through the decision to switch schools (*exclusion restriction*). The exclusion restriction fails only if a positive gain changes college outcomes directly or through a channel that is correlated with unobservables. Although unlikely, two violations of this assumption could occur. First, students might exert more effort because they learn they have a positive potential gain in their school market. The switching timeline limits this concern, since most switchers move a few months before graduating, leaving little time for a change in effort to affect their GPA or the national exam. Second, the potential gain could be correlated with neighborhood characteristics that affect potential college outcomes. Adding county fixed effects to Equation (5) leaves the estimates essentially unchanged (Table A12).

6. RESULTS

6.1. Switching Schools during Twelfth Grade

I begin by estimating the effect of the disclosure of the RR formula on switching schools during twelfth grade following Equation (4), using both the positive gain, an indicator for a positive median potential gain, and the standardized gain, the median potential gain standardized within each cohort.

Figure 6 presents the results nationwide for students who lived in the same region in primary school and at the start of twelfth grade, and for the narrower sample of those who lived in the same county. Students with a positive gain in their school market are 1.2 pp more likely to switch schools during twelfth grade in 2014 than in 2013, a 72% increase over the 2010–2013 mean of 1.7%. Under

the continuous measure, a one-standard-deviation increase in the gain raises the probability of switching by 0.7 pp, or 40% of that mean. Before the disclosure, the coefficients are close to zero except in 2011, when student protests kept many schools closed for several months and led some students to change schools for reasons unrelated to the disclosure of the RR formula (see Section 2.1). Appendix F separates the two episodes: the 2011 switches concentrate in July, while the 2014 response concentrates in August and in Gran Santiago. After 2014, the difference disappears, as expected once the formula began to compare students across all the schools they attended rather than only the one from which they graduated (see Section 2.3 for details).

The national estimate, however, averages over very different settings. The theoretical framework in Section 3 implies that switching pays only when two conditions hold together (Equation 3): the student has a school within reach where graduating would raise their ranking score, and applies to programs selective enough that a higher score changes their perceived chances of admission. Across Chile's three main metropolitan areas, Gran Santiago, Gran Concepción, and Gran Valparaíso, only Gran Santiago meets both (Appendix H).

On the first condition, Gran Santiago differs from each of the other two areas on different margins. Within 2 km, a student in Gran Santiago in 2013 has 18 alternatives on average and a median gain of 4.9 RR points. Gran Concepción offers a similar gain, 5.4 points, but from only 10 alternatives, while Gran Valparaíso offers as many alternatives, 17 on average, but a median gain of only 0.9 points. On the second condition, Gran Santiago differs from both. Only 7% of its seats are left unfilled, against 18% in Gran Concepción and 12% in Gran Valparaíso. In addition, 14% of its programs are among the hundred most selective in the country, against 6% and 5%, and 71% of its applicants list a tier 1 or 2 university among their first five preferences, against 7% in Gran Concepción and 22% in Gran Valparaíso.

Figure 7 estimates Equation (4) separately for each area, and, as the model predicts, only students in Gran Santiago respond in 2014. There, a positive gain raises the probability of switching schools during twelfth grade in 2014 by 2.5 pp, a 177% increase over the 2010–2013 mean of 1.4%, and a one-standard-deviation increase in the gain raises it by 1.5 pp. The estimates for Gran Concepción, −0.11 pp, and Gran Valparaíso, −0.20 pp, are not distinguishable from zero, and a formal joint test rejects equality with the Gran Santiago coefficient. In Gran Santiago, the coefficients before the disclosure are close to zero, including 2011, and the effect disappears after the formula changed.

The rest of the paper therefore focuses on Gran Santiago.

The timing of the switches supports a strategic interpretation. Using monthly attendance records from 2011 to 2015, Figure 8 estimates Equation (4) month by month.[38] Students with a positive gain become more likely to switch schools during the 2014 school year, with a peak in August. Switches at the beginning of the year, in contrast, respond much less, by about a fifth of the mid-year response under the discrete measure and a third under the continuous one (Figure A10). If the 2014 increase reflected factors unrelated to the formula, switching at the start of the year should have risen as much.

I also conduct several robustness checks. First, I replace the median potential gain with alternative summaries of the gains in the student's school market to test whether the results are specific to the median (Appendix C). Second, because the 2 km market is a proxy for the schools a student can reach, I re-estimate the event study with markets of 4, 6, and 8 km (Appendix D). Third, I rebuild the treatment from deflated grades to rule out that grade inflation changes which students are treated (Appendix E). Fourth, I drop one county at a time to verify that no single county drives the estimate (Figure A11). In all four cases, the estimates remain similar to the main specification. Following Rambachan and Roth (2023), I show that the 2014 effect remains significant under both standard relaxations of parallel trends. Under relative magnitudes, it holds even if violations after the disclosure are up to 3.9 times as large as the largest violation before it, and under smoothness restrictions, it holds if the slope of the trend changes by up to 1.9 pp from one year to the next (Figure A12).

I next add progressively more demanding fixed effects. Allowing schools to trend differently by their pre-disclosure threshold, and also by their baseline SIMCE score, leaves the 2014 estimate at 2.2 and 2.1 pp, both significant at the 1% level (Tables A6 and A7). With school-by-year fixed effects, it falls to 0.6 pp, still significant at the 1% level. A Shapley decomposition of the treatment's variance shows that 70% of its explained variation involves the origin school's threshold, which is constant within school and year (Table A8). School-by-year fixed effects therefore absorb most of the variation that identifies the effect, which explains the drop. Adding school-by-GPA-decile-by-year fixed effects removes the remaining variation, and the estimate is indistinguishable from zero.

[38] See Appendix G for the construction of monthly switches.

I. Heterogeneity in the Response to the Disclosure of the RR Formula

I next examine which students drive the response. Switching pays off only when the two conditions in Equation (3) hold together: a nearby school would raise the student's ranking score, and the student applies to college in a competitive college market. Figures 9 and 10 estimate Equation (4) separately by school and student characteristics that can shift these conditions.

The first condition depends on the student's own school. Schools with higher standardized tenth-grade scores also have higher GPAs (see Section 4.3), so their students are the ones most likely to find a nearby school where the same GPA gives them a higher RR score. In the top performance quartile, a positive gain increases the probability of switching schools during twelfth grade by 5.0 pp in 2014, and a one-standard-deviation increase in the gain raises it by 3.6 pp. In the other three quartiles, both estimates are below 0.5 pp and indistinguishable from zero (Figure 9, Panels A and B). The response is also concentrated in public schools, where a positive gain raises switching by 8.5 pp, against less than 1 pp in voucher and private schools (Figure A13). Further, the response falls at both ends of the GPA distribution (Figure A14), as the model predicts.

The second condition depends on whether the student plans to apply to a university in the centralized admission system. Only students in the academic track, which prepares students for the PSU, respond to the disclosure, as Figure 9, Panels C and D, show. For them, a positive gain raises the probability of switching by 3.5 pp in 2014, and a one-standard-deviation increase in the gain raises it by 2.5 pp, while the estimates for the vocational track are close to zero. For both tracks, there are no differences before the disclosure of the RR formula.

Even among top-quartile public schools, the response is largest where students already applied to selective universities. Within Gran Santiago, these schools include a small group of long-established, high-performing schools, known in Chile as *emblematic* schools, whose students have traditionally applied to the most selective universities. Before the disclosure, 52% of their students listed a tier 1 or 2 program among their first five preferences, against 17% at the other top-quartile public schools. After the disclosure of the RR formula, students with a positive gain at emblematic schools respond about twice as much, 13.9 pp against 7.4 pp, but the response at the other top-quartile public schools is still large and precisely estimated and consistent with their students' pre-disclosure applications (Appendix J).

Family background also shapes students' responses. Among low-SES students, a positive gain

raises the probability of switching by 1.6 pp in 2014, against 3.0 pp for medium-SES and 2.5 pp for high-SES students, and both differences with the low-SES group are statistically significant (Figure 10, Panels A and B). The continuous measure shows the same gradient: 0.7, 1.7, and 2.0 pp per standard deviation. The split by parents' aspirations is even sharper. Only students whose parents hope they will attend university respond, by 3.3 pp against 0.6 pp for the rest, with a similar pattern for the continuous measure (Figure 10, Panels C and D).[39]

The students who switched are therefore those for whom the model predicts that switching pays: students at high-performing schools, in the academic track, from higher-SES families, and whose parents hope they will attend a university. The response does not differ by gender and is larger among students with internet access at home (Figure A13), suggesting that better-informed students, who tend to share these characteristics, were the ones who responded.

II. **Additional Evidence of Strategic Behavior**

If students switched to raise their ranking score, they should have moved to schools where the same GPA gives them a higher RR score, that is, schools with lower RR thresholds, which also tend to be weaker on other dimensions. I estimate Equation (4) on characteristics of the destination school. Each outcome is an indicator equal to one if the student switches schools during in twelfth grade and the destination school, compared with the school of origin, has a lower RR mean threshold ($\underline{r}_e < \underline{r}_s$), a lower average PSU score, fewer students admitted to college or to a tier 1 university, a lower performance quartile, or a lower socioeconomic group, or is closer to the student's primary school, and zero otherwise.[40]

Figure 11 reports the 2014 coefficients. Students with a positive gain moved to schools with lower thresholds, as the model predicts. In 2014, a positive gain raises the probability of switching to a school with a lower RR threshold, a lower average PSU score, or fewer students admitted to college by 2.5 pp each. Moves to schools with lower performance, serving a more disadvantaged population, and closer to the student's primary school also increase in 2014, although by less. The standardized gain shows the same pattern. Before the disclosure, the coefficients are close to zero, except for small positive ones in 2011, the year of the student protests (Appendix F). Tables A9 and

[39] Parents' aspirations come from the tenth-grade SIMCE questionnaire, which is not available for the 2011 and 2013 cohorts, so this split uses 2012 as the base year.

[40] The RR threshold is the one used to compute the potential gain (Section 4.3). The average PSU score and the number of students admitted to college are school averages over the years before the disclosure.

A10 report the full event studies.

6.2. Admission, Enrollment, and Completion in Postsecondary Education

I next estimate the effect of the disclosure of the RR formula on college outcomes using Equation (5), for the cohorts graduating from high school between 2011 and 2016 (Section 4.3).[41] A positive gain does not change whether students apply to, are admitted to, or enroll in a university in the centralized system (Table A11). However, it changes which programs they enter. Students with a positive gain move toward high-employability and highly selective programs and graduate from them, with no increase in dropout or delay in graduation.[42]

I report results for high-employability and highly selective programs because they capture two distinct dimensions of program quality. Selectivity is where a higher ranking score changes admission, while employability measures the labor-market prospects of graduates, and the two are only weakly correlated across programs (0.27). Average earnings four years after graduation are correlated with both, at 0.57 with selectivity and 0.44 with employability.[43]

In 2014, students with a positive gain are no more likely to list a high-employability program among their first three preferences, but they are 0.7 pp more likely to be admitted to one and 0.7 pp more likely to enroll in it, both significant at the 10% level (Figure 12, Panels A, B, and C). For highly selective programs, the response starts in the application process. Students with a positive gain are 0.9 pp more likely to list a highly selective program among their first three preferences, consistent with evidence that students list programs strategically in this setting (Larroucau and Ríos 2018), 0.7 pp more likely to be admitted to one, and 1.1 pp more likely to enroll in it (Figure 13, Panels A, B, and C).

If a higher score placed students in programs they were not prepared for, they would leave those programs more often or would take longer to graduate. I do not find evidence that they do. Dropout from high-employability and highly selective programs does not rise. Students with a positive gain are instead 1.1 pp more likely to graduate from a high-employability program, where on-time

[41] On this sample, a positive gain raises switching during twelfth grade by 2.5 pp, the same estimate as in Figure 7 for the switching sample (see Table A11).

[42] As explained in Section 5.2, the results in this section use the positive gain only.

[43] I prefer employability to earnings for two reasons. First, it adds information that selectivity does not, while earnings largely overlap with it. Second, it relies less on predicted values (see Appendix G.3).

graduation also rises, by 0.5 pp (Figure 12, Panels E and F), and 1.1 pp more likely to graduate from a highly selective program (Figure 13, Panel E).[44] The estimates for highly selective programs are similar when adding county fixed effects to account for neighborhood unobservables (Table A12). Graduation from high-earnings programs follows the same pattern (Figure A15).

Using the reduced form results, together with the first stage on the same sample, I recover the Local Average Treatment Effect (LATE) for the group of students who switched schools in response to the disclosure of the RR formula (the compliers). I do so by computing the ratio of the reduced form, obtained by regressing college outcomes on having a positive potential gain, to the first stage, which regresses an indicator equal to one if the student switched schools during twelfth grade on the same instrument. Students who switched schools during twelfth grade in response to the disclosure are 42 pp more likely to graduate from a high-employability program and 45 pp more likely to graduate from a highly selective program.

A back-of-the-envelope calculation expresses the return to switching in monetary terms using the program-specific average monthly income four years after graduation (see Appendix G for variable constructions).[45] A highly selective program pays 465,000 Chilean pesos a month more than a program in the adjacent selectivity quartile.[46] Switching schools was therefore worth about 209,000 pesos a month to the students who switched, or around 2,600 US dollars a year and some 90,000 dollars over a 35-year career. Measured by employability, the return is smaller but still substantial. A high-employability program pays 178,000 pesos a month more than one in the adjacent employability quartile, so the 42 pp effect on graduating from such a program is worth about 75,000 pesos a month, around 900 dollars a year and some 32,000 dollars over a career. Using the average program as the comparison instead, the return would be a little more than 100,000 dollars by selectivity and about 67,000 by employability, about three and two times, respectively, what an average worker accumulates over a working life.[47]

[44] For graduation from any program, the 2014 estimate is 0.3 pp and not significant. The gain is therefore in what students graduate from rather than in whether they graduate at all.

[45] I do not consider a discount rate or promotions over the work cycle in this calculation.

[46] In pesos of September 2025, among programs with reported income.

[47] Cecilia Arroyo A., "Ahorro acumulado de los nuevos pensionados alcanza a $34 millones promedio en las AFP", *Diario Financiero*, 16 March 2023, reporting figures from Ciedess for new pensioners in January 2023.

7. DISTRIBUTIONAL CONSEQUENCES OF STRATEGIC SWITCHING

Switching paid off for the students who switched, who became 45 pp more likely to graduate from a highly selective program and 42 pp more likely to graduate from a high-employability one. In a centralized system with a fixed number of seats, these gains come at the expense of other applicants. Even if few students lose a seat, how much switching undermined the RR policy depends on who those students are and on what losing the seat costs them.

7.1. Simulating Admission Matches with and without Switching

I compare the admissions of the 2015 cycle with those the same applicants would have received had no student switched schools. For the counterfactual, I recompute each switcher's ranking score as if they had graduated from the school they attended at the start of twelfth grade, holding their GPA, PSU scores, and application lists fixed, and re-run the admission algorithm on these scores. I run the counterfactual for three groups of switchers: all students who switched that year, only those with a positive gain nationwide, and only those with a positive gain in Gran Santiago, the group whose response I identify in Section 6.1.

Because only the switchers' ranking scores change, any difference between the two allocations is attributable to switching, provided that switching did not change students' PSU scores or GPA. This is plausible because most switchers spend only a few months at the destination school. Holding application portfolios fixed, however, understates the selective seats that would have gone to other students, because students with a positive gain are more likely to apply to highly selective programs, though not to high-employability or high-earnings ones (Section 6.2 and Figure A15). The estimates in this section are therefore conservative.

7.2. Assignments and Distributional Effects under the Simulation Analysis

Relative to the counterfactual in which no student switches in 2014, switchers displace 920 non-switchers from the program they would otherwise have been admitted to, four in five of them to a lower preference on their list. The number of displaced students is smaller when the counterfactual undoes only the switches of students with a positive gain, and only those of students with a gain in Gran Santiago (709 and 567 displaced students, respectively). Displacement rates are very similar

among non-switchers with and without a positive gain (1.05% and 0.95%).

Figure 14 reports the net change in seats per 1,000 applicants by student and school characteristics. Low-SES students lose seats at tier 1–2 universities and in highly selective and high-employability programs, while students from high-performing schools gain seats in highly selective programs and, to a lesser extent, in high-employability ones, and students from public schools gain the most. These patterns mirror the heterogeneous responses in Section 6.1.

7.3. What Does a Lost Seat Cost?

The simulations show that some students are displaced from high-employability and highly selective programs, although most of them are admitted to another program. The question is then whether the switches harmed displaced students, and by how much. Answering it requires comparing displaced students with otherwise similar students who kept their seat. I take advantage of the fact that only students close to the cutoff can be displaced. The same holds along a chain of displacements, since a student who loses a seat in a different program only because someone was displaced from their preferred program by someone who switched must also have a score close to the second program's cutoff.

I use a regression discontinuity (RD) design at the admission cutoffs of the programs where switching displaces students, directly or through such a chain. I pool 2014 and 2015 admission processes for precision, since they use nearly the same weights, and keep only students who did not switch schools. I weight each program by its number of displaced students, so that programs with more displacement count more. The density of scores is smooth at the cutoff, and applicants' characteristics are balanced except that those just above are somewhat more likely to come from public schools and, at highly selective programs, to have a positive gain (Table 4).[48]

Across all affected programs, an offer raises enrollment in the program by 49 pp but enrollment in any program by only 4.6 pp, so about nine in ten students who would take the seat enroll elsewhere without it (Table A14). An offer to a high-employability program raises the probability of enrolling in one by 16.4 pp and of graduating from one by 6.2 pp, while dropout from these programs also rises, by 4.8 pp. An offer to a highly selective program raises enrollment in one by 10 pp, but most

[48]The validity checks are similar when the programs with displacement come from counterfactuals that undo only the switches of students with a positive gain, nationwide or in Gran Santiago (Table A13).

of these additional enrollees leave, since dropout rises by 7.2 pp and graduation from a highly selective program does not change (Table 5).

The cost of displacement therefore falls on high-employability seats. Valued with the earnings gap between high-employability programs and those in the next employability quartile, a lost offer is worth about 4,700 dollars over a 35-year career. Because an offer raises enrollment in such a program by only 16 pp, a seat the student would have taken is worth about 29,000 dollars, close to the 32,000 dollars switchers gain from theirs. Students displaced from highly selective seats, in contrast, lose no selective degree, since most students admitted at the margin of these programs drop out.

8. CONCLUSION

This paper shows that changes in centralized college admission rules can trigger strategic responses during high school that weaken the policies behind them, moving valuable seats from the students who would have held them to those who respond. After the disclosure of the formula of Chile's relative ranking, which compares a student's GPA with that of previous cohorts at the school from which they graduate, students with a positive potential gain switched schools during twelfth grade.

As the model predicts, among Chile's three main metropolitan areas the response appears only in Gran Santiago, the only metropolitan area with both competitive school and university markets, where a positive gain raised the probability of switching by 2.5 pp in 2014. The response was concentrated among students at high-performing schools, in the academic track, and from families who hoped they would attend university, and switchers moved to schools with lower ranking thresholds.

Students who switched in response to the disclosure were 42 pp more likely to graduate from a high-employability program and 45 pp more likely to graduate from a highly selective one, a return of about 32,000 to 90,000 dollars over a career, with no increase in dropout or delay in graduation. Part of these gains came at the expense of low-SES students. Switchers displaced about 1% of admitted non-switchers, most of whom were admitted to another program, but those displaced became less likely to graduate from a high-employability program.

These results have two implications for school-based affirmative action. First, evaluating such

policies by the school from which students graduate overstates their effect, because switchers are counted as students of the weaker schools they moved to. Second, the scope for gaming depends on how easy it is to change schools. Rules that require a minimum stay before a student counts in a school's ranking, or that compute the ranking across all schools attended, as Chile did from 2016, make late switching less attractive, although they may shift the response to the choice of high school at the start of ninth grade.

FIGURES AND TABLES

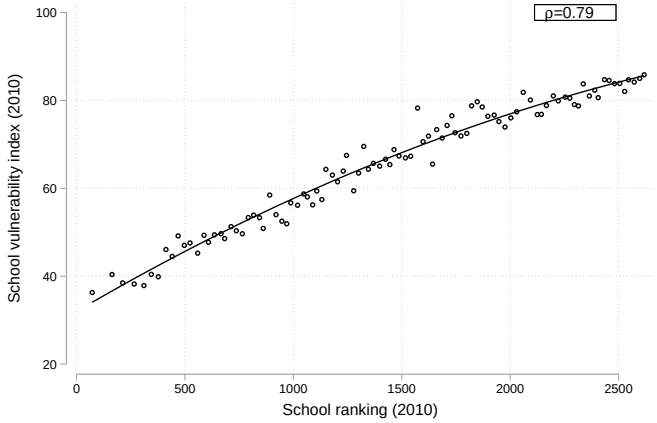


FIGURE 1. School Vulnerability Index & School Ranking

Note: This figure presents the association between school ranking and the school vulnerability index (IVE) in 2010. The school ranking is calculated using the school's average score on the standardized tenth-grade mathematics test (SIMCE) in 2010, with 1 assigned to the school with the highest average. The school vulnerability index measures the percentage of students facing socioeconomic vulnerability and educational barriers within each school in 2010. Each dot is the mean vulnerability index within one of 97 bins of the school ranking, and the line is a quadratic fit. ρ is the correlation between the two variables across schools.

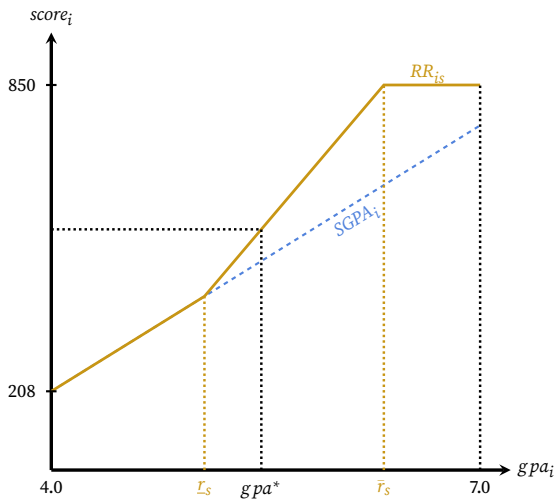


FIGURE 2. Construction of the Relative Ranking Score

Note: This figure shows how the RR score is constructed from a student's GPA. The horizontal axis is the student's GPA on Chile's 1.0–7.0 scale, shown from 4.0, the minimum passing grade, to 7.0. The vertical axis is the 208–850 scale of university admission scores. The blue dashed line is the SGPA, a linear function of GPA. The gold solid line is the RR score. $\underline{r}_s$ is the mean GPA of the three previous cohorts that graduated from school s, and $\bar{r}_s$ is the highest GPA among them. Below $\underline{r}_s$, the RR score equals the SGPA. Between $\underline{r}_s$ and $\bar{r}_s$, it rises more steeply, and the gap between the two lines is the RR bonus. At and above $\bar{r}_s$, it is capped at 850. Dotted lines trace the RR score of an illustrative student with GPA gpa^*.

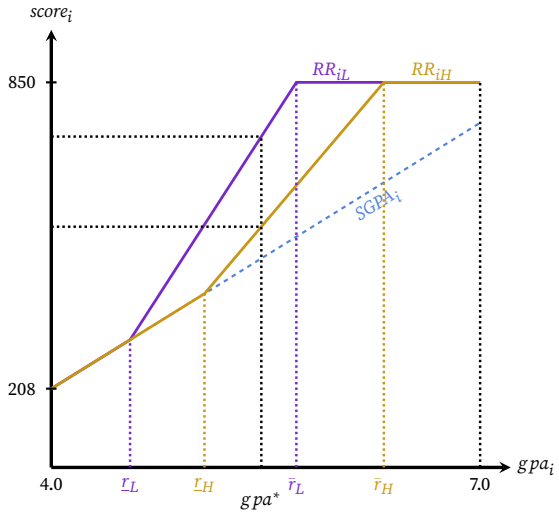


FIGURE 3. Relative Ranking Score at Two Schools: Thresholds in the Same Order

Note: This figure shows the RR score at two schools. The horizontal axis is the student's GPA on Chile's 1.0–7.0 scale, shown from 4.0, the minimum passing grade, to 7.0. The vertical axis is the 208–850 scale of university admission scores. The blue dashed line is the SGPA, a linear function of GPA. The gold solid line is the RR score at school H, and the purple solid line is the RR score at school L. $\underline{r}_s$ is the mean GPA of the three previous cohorts that graduated from school s, and $\bar{r}_s$ is the highest GPA among them. Here $\underline{r}_L < \underline{r}_H$ and $\bar{r}_L < \bar{r}_H$, so students at L start earning the bonus at a lower GPA and reach the 850 cap sooner. Dotted lines trace the RR scores of two students with GPA gpa^*: the one at school L receives the higher score.

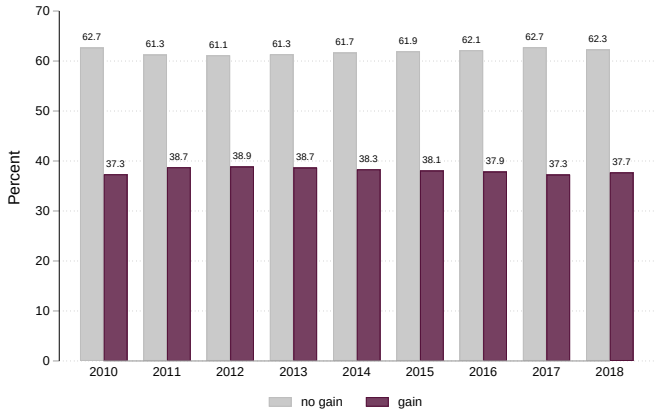
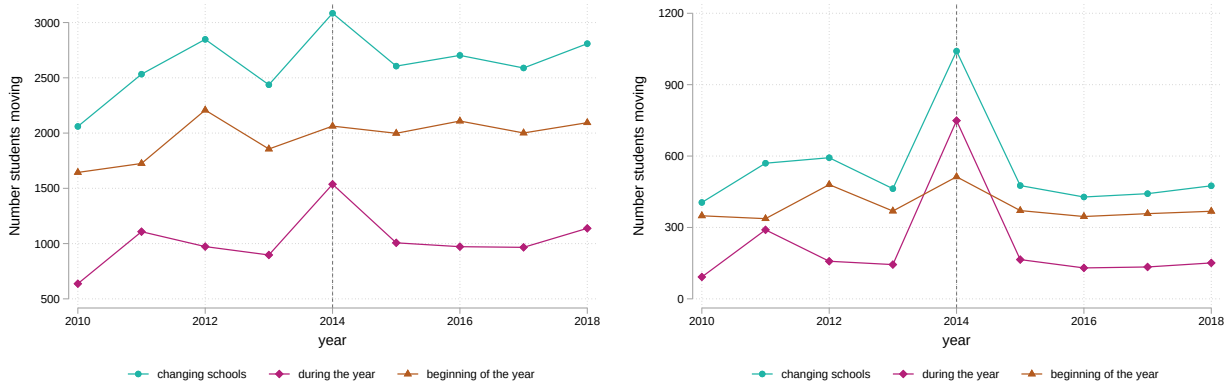


FIGURE 4. Share of Students with a Positive Median Potential Gain

Note: This figure shows, for each year from 2010 to 2018, the share of twelfth graders in Gran Santiago with a positive median potential gain, in purple, and without one, in gray. A student's choice set is the schools within 2 km of their primary school, and the student has a positive median potential gain if the median of the potential gains across those schools is positive.

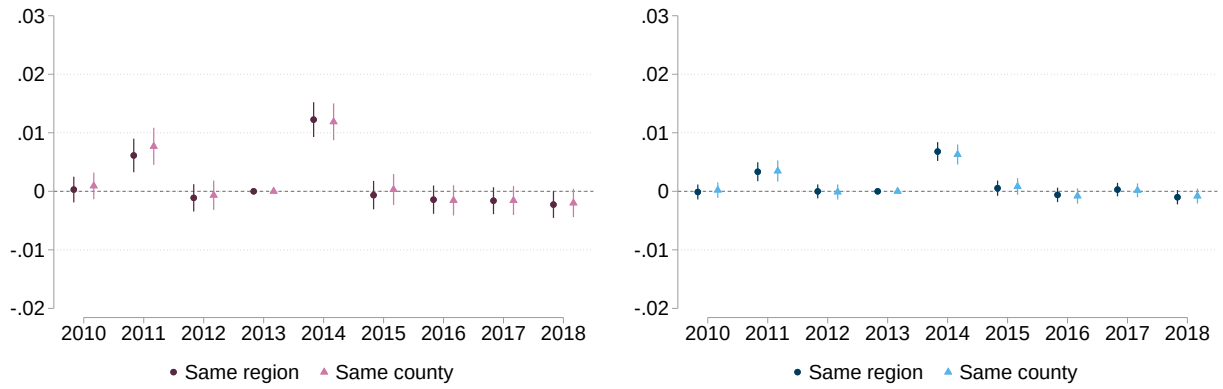


A. All schools

B. High-performing schools

FIGURE 5. Number of Students Switching Schools in Twelfth Grade, 2010–2018

Note: This figure shows the number of twelfth graders switching schools each year, for all schools in Panel A and for high-performing schools in Panel B. Circles show all switches, triangles switches at the beginning of twelfth grade, and diamonds switches during twelfth grade, from April to October. A student who switched at both times in the same year counts in both series but only once in the total. The sample is students living in Gran Santiago in primary school and at the start of twelfth grade and attending a school there at the start of twelfth grade. High-performing schools are those in the top quartile of the 2010 tenth-grade SIMCE by school type and region. The vertical line marks 2014, the first cohort after the disclosure of the RR formula.

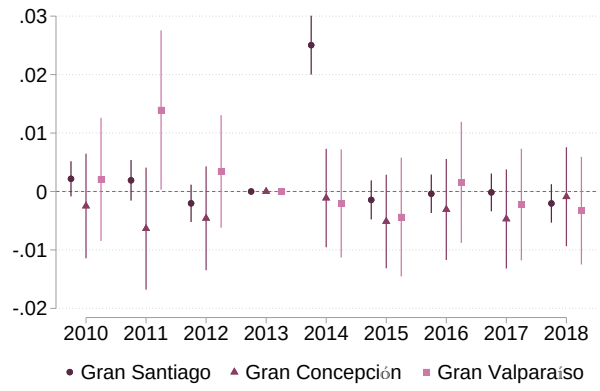


A. Positive median gain

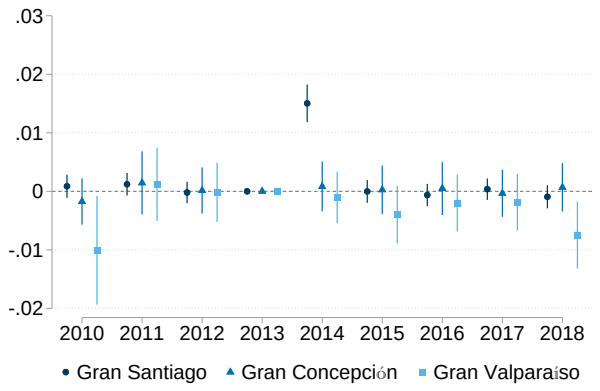
B. Standardized median gain

FIGURE 6. Impact of the RR Formula Disclosure on Students Switching Schools during Twelfth Grade: Full Country

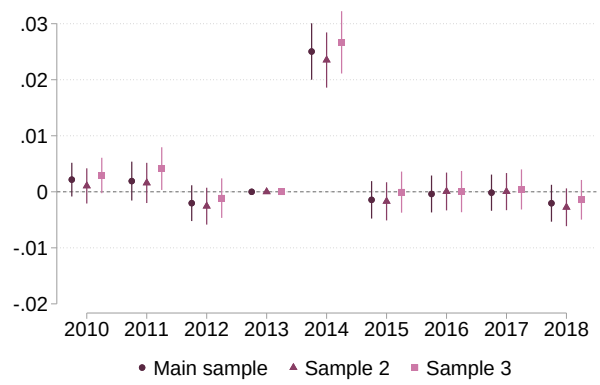
Note: This figure shows the coefficients from Equation (4) for the whole country, where the outcome is an indicator for switching schools during twelfth grade. In Panel A the key regressors are the interactions of an indicator for a positive median potential gain with year dummies; in Panel B, the interactions of the standardized median potential gain with year dummies. Both panels include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. Circles show students who lived in the same region in primary school and at the start of twelfth grade, and triangles students who lived in the same county at both times. Coefficients are relative to 2013, the omitted year; in Panel B they are per standard deviation of the median potential gain within each cohort. Standard errors are clustered at the school-market level, and the bars are 99% confidence intervals.



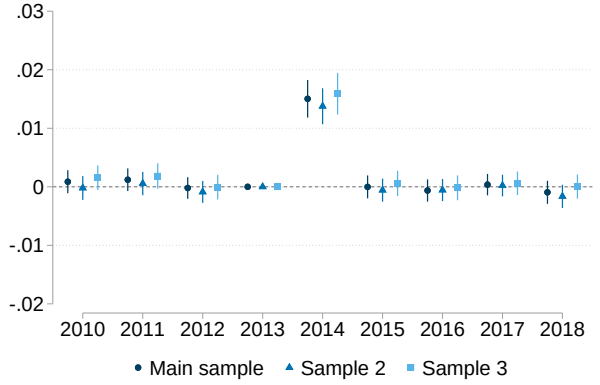
A. Main Metropolitan Areas: Positive median gain



B. Main Metropolitan Areas: Standardized median gain



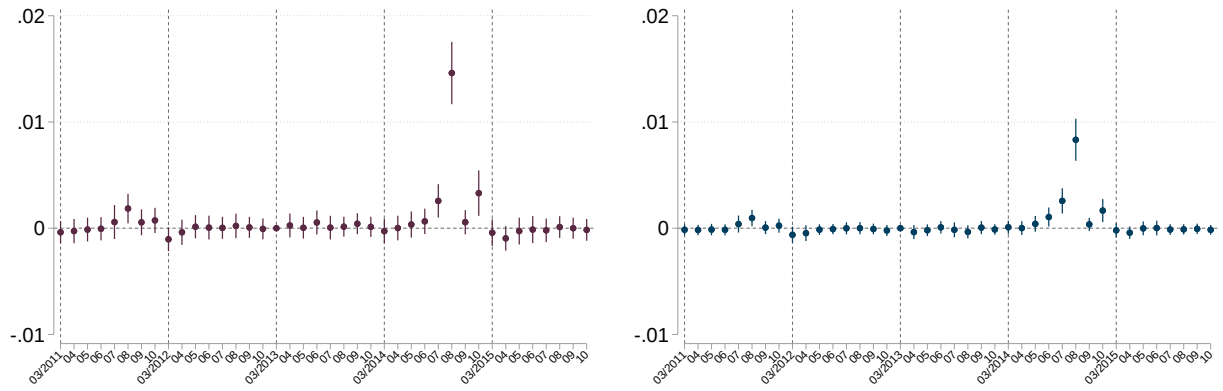
C. Gran Santiago: Positive median gain



D. Gran Santiago: Standardized median gain

FIGURE 7. Impact of the RR Formula Disclosure on Students Switching Schools during Twelfth Grade by Metropolitan Area

Note: This figure shows the coefficients from Equation (4) by metropolitan area, where the outcome is an indicator for switching schools during twelfth grade. In Panels A and C the key regressors are the interactions of an indicator for a positive median potential gain with year dummies; in Panels B and D, the interactions of the standardized median potential gain with year dummies. All panels include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. Panels A and B estimate the equation separately for Gran Santiago, Gran Concepción and Gran Valparaíso, the three largest metropolitan areas as defined by Law 21.074, using students who lived in the same area in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade. Panels C and D use Gran Santiago only, with three samples: the main sample; students who lived in Gran Santiago in primary school; and students in the main sample who lived in the same county at both times. In Panels A and B, circles show Gran Santiago, triangles Gran Concepción and squares Gran Valparaíso; in Panels C and D, circles, triangles and squares show the three samples in that order. Coefficients are relative to 2013, the omitted year; in Panels B and D they are per standard deviation of the median potential gain within each cohort. Standard errors are clustered at the school-market level, and the bars are 99% confidence intervals.



A. Positive median gain

B. Standardized median gain

FIGURE 8. Impact of the RR Formula Disclosure on Students Switching Schools Month by Month during Twelfth Grade

Note: This figure shows the coefficients from a monthly version of Equation (4) for Gran Santiago, from March to October of 2011 to 2015. The outcome equals one in the last month a student attends a school before switching to another. In Panel A the key regressors are the interactions of an indicator for a positive median potential gain with month dummies; in Panel B, the interactions of the standardized median potential gain with month dummies. Both panels include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade. Vertical dashed lines mark March, the start of each school year. Coefficients are relative to March 2013, the omitted month; in Panel B they are per standard deviation of the median potential gain within each cohort. Standard errors are clustered at the school-market level, and the bars are 99% confidence intervals.

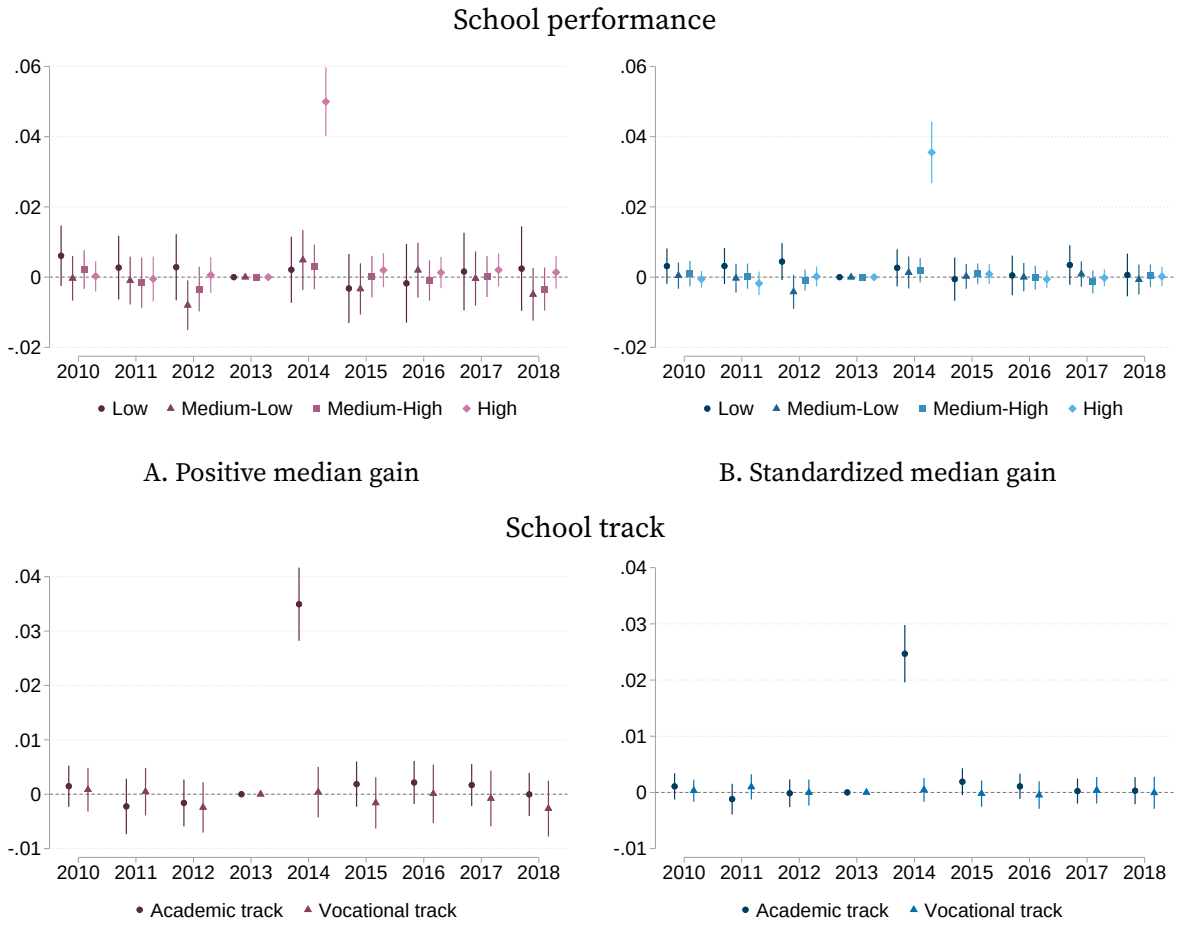


FIGURE 9. Impact of the RR Formula Disclosure on School Switching, by School Performance and Track

Note: This figure shows the coefficients from Equation (4) estimated separately by group, where the outcome is an indicator for switching schools during twelfth grade. In Panels A and C the key regressors are the interactions of an indicator for a positive median potential gain with year dummies; in Panels B and D, the interactions of the standardized median potential gain with year dummies. Panels A and B split students by the quartile of their school's 2010 tenth-grade SIMCE mathematics score by school type and region, with circles, triangles, squares and diamonds from the bottom to the top quartile. Panels C and D split them by the track of their school, with circles for academic and triangles for vocational. All panels include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade. Coefficients are relative to 2013, the omitted year; in Panels B and D they are per standard deviation of the median potential gain within each cohort. Standard errors are clustered at the school-market level, and the bars are 99% confidence intervals.

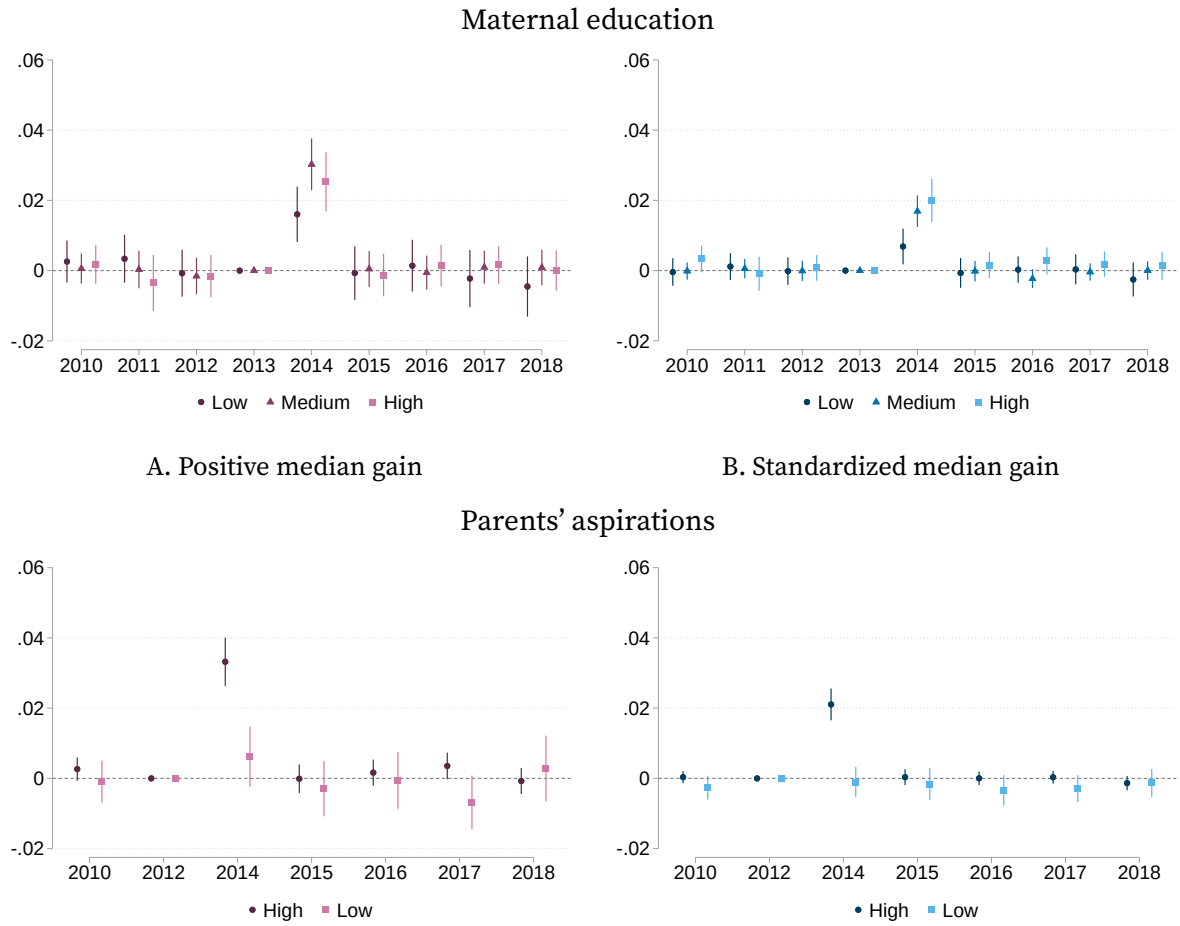
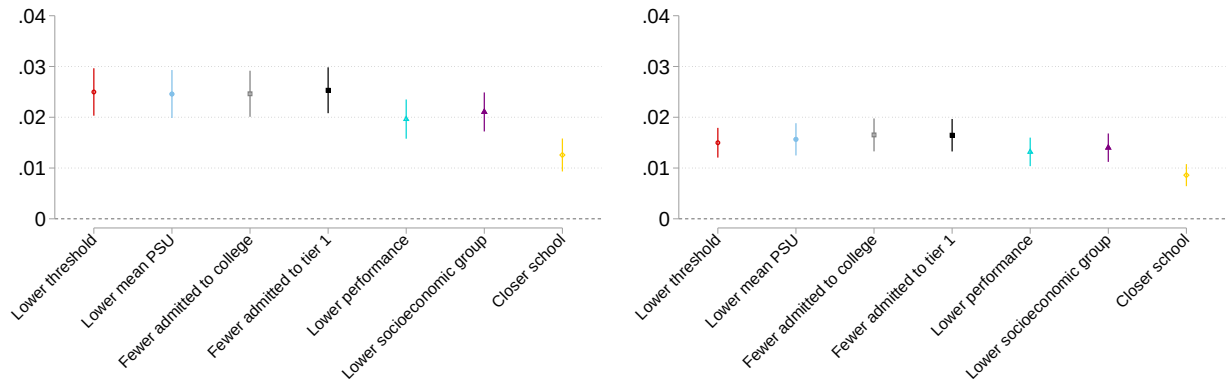


FIGURE 10. Heterogeneous Impact of the RR Formula Disclosure on School Switching, by Family Characteristics

Note: This figure plots the coefficients from estimating Equation (4) separately by group, where the outcome is an indicator for switching schools during twelfth grade. In Panels A and C, the key regressors are the interactions between an indicator for having a positive median potential gain and event-year dummies; in Panels B and D, the interactions between the median potential gain, standardized within the graduating cohort, and event-year dummies. Panels A and B present the results by mother's education. Panels C and D present the results by whether the parents expected the student to attend university, which comes from SIMCE data, and therefore it is not available for cohorts 2011 and 2013 (see Section **??**). The event of interest is the disclosure of the RR formula in 2014. The sample includes students who lived in Gran Santiago while in primary school and at the start of twelfth grade, and who attended a school in Gran Santiago at the start of twelfth grade. All regressions include fixed effects for within-school GPA deciles and for the school at the start of twelfth grade. The y-axis reports the change in the probability of switching relative to the omitted year (2013 for Panels A and B and 2012 for Panels C and D); in Panels B and D, this is the change associated with a one standard deviation higher potential gain. Standard errors are clustered at the school-market level, and 99% confidence intervals are shown.



A. Positive median gain

B. Standardized median gain

FIGURE 11. Impact of the RR Formula Disclosure on School Switching, by Destination School Characteristics

Note: This figure shows the 2014 coefficients from Equation (4), estimated separately for each outcome. Each outcome is an indicator for switching schools during twelfth grade to a school that, relative to the origin school, has a lower mean GPA threshold, a lower pre-2014 mean PSU score, fewer graduates admitted to college or to tier 1 before 2014, a lower 2010 SIMCE quartile, a lower socioeconomic group, or is closer to the student's primary school. In Panel A the coefficient is on the interaction of an indicator for a positive median potential gain with the 2014 dummy; in Panel B, on the interaction of the standardized median potential gain with the 2014 dummy. All regressions include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade. Coefficients are relative to 2013, the omitted year; in Panel B they are per standard deviation of the median potential gain within each cohort. Standard errors are clustered at the school-market level, and the bars are 99% confidence intervals.

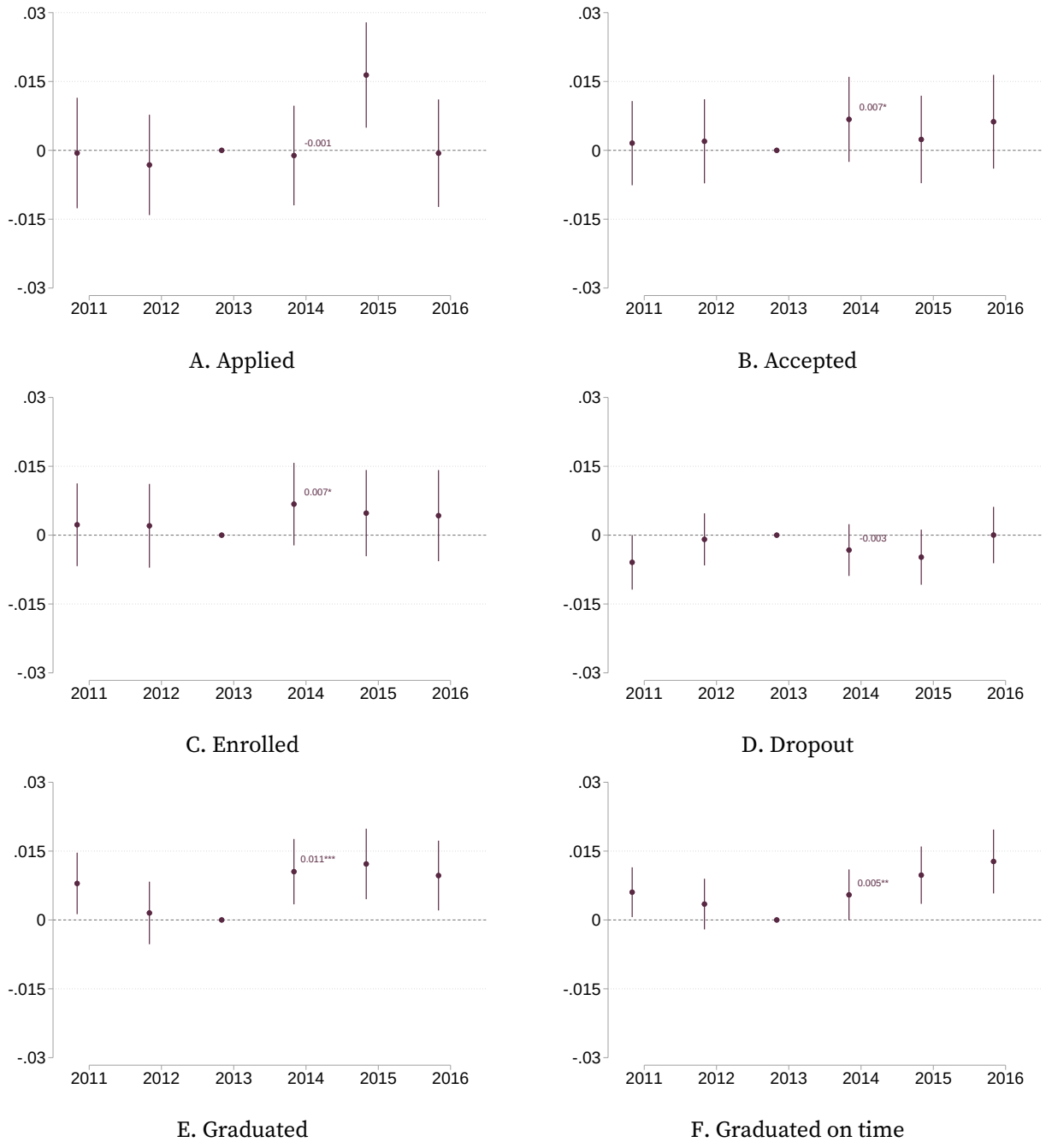


FIGURE 12. Impact of the RR Formula Disclosure on College Outcomes: High-Employability Programs

Note: This figure shows the coefficients from Equation (5) for high-employability programs. The outcomes are indicators for listing such a program among the first three preferences in Panel A, being admitted to one in Panel B, enrolling in one in Panel C, dropping out of one in Panel D, graduating from one in Panel E, and graduating from one on time in Panel F. The key regressors are the interactions of an indicator for a positive median potential gain with year dummies. High-employability programs are those in the top quartile of the share of graduates employed one year after graduation, among those who graduated between 2013 and 2015. All panels include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade, in the cohorts finishing twelfth grade between 2011 and 2016. Coefficients are relative to 2013, the omitted year. Standard errors are clustered at the school-market level, and the bars are 99% confidence intervals. Stars indicate significance at the 1% (***), 5% (**), and 10% (*) levels.

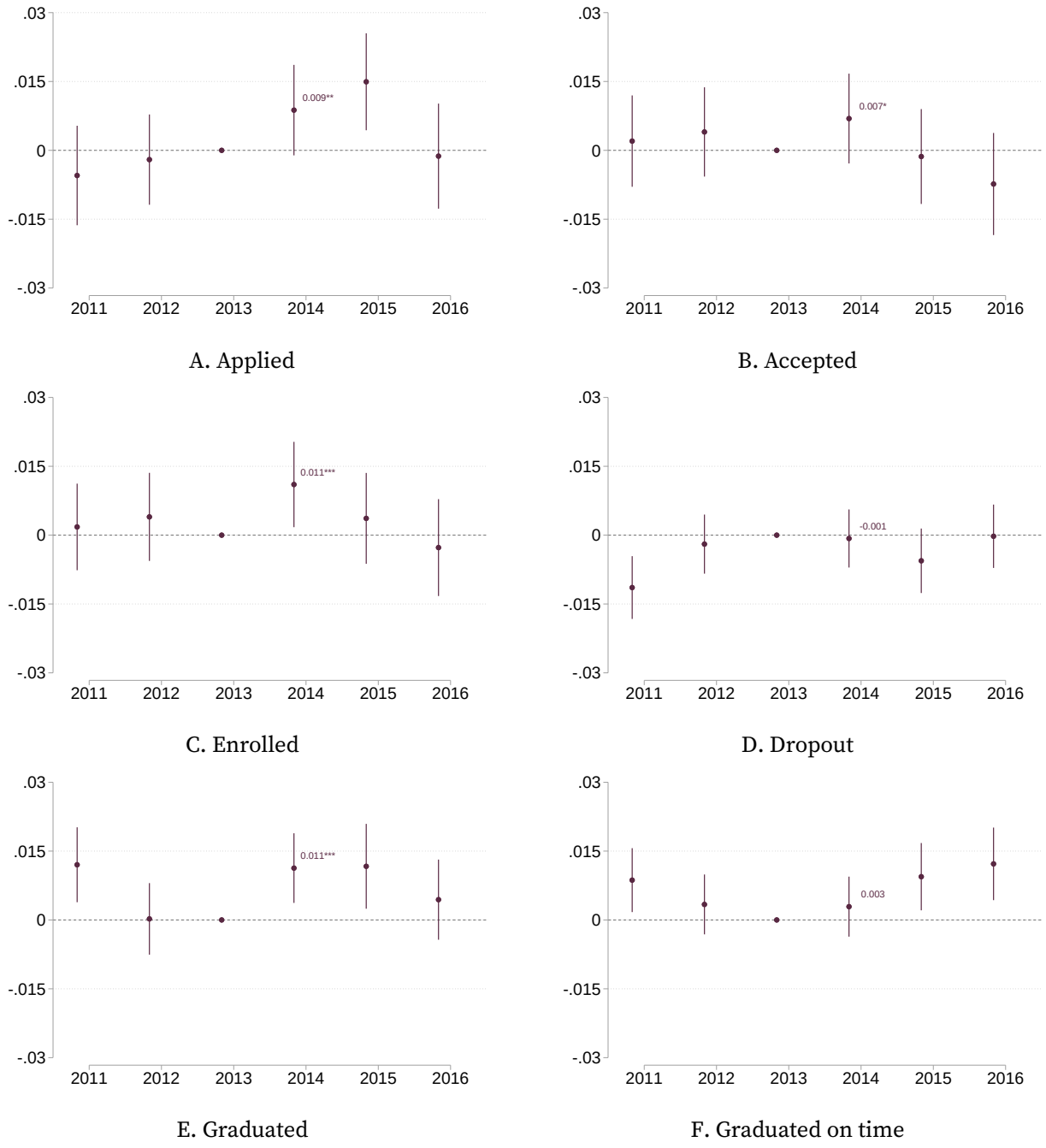


FIGURE 13. Impact of the RR Formula Disclosure on College Outcomes: Highly Selective Programs

Note: This figure shows the coefficients from Equation (5) for highly selective programs, using each student's first admission process. The outcomes are indicators for listing such a program among the first three preferences in Panel A, being admitted to one in Panel B, and enrolling in one in Panel C. Panels D to F refer to the first program the student enrolls in, if it is highly selective: dropping out of college, graduating from it, and graduating within one year of its nominal length. Students who never applied or enrolled are zero. The key regressors are the interactions of an indicator for a positive median potential gain with year dummies. Highly selective programs are those in the top quartile of the mean admission score of students admitted in the 2014 admission process. All panels include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade, in the cohorts finishing twelfth grade between 2011 and 2016. Coefficients are relative to 2013, the omitted year. Standard errors are clustered at the school-market level, and the bars are 99% confidence intervals. The label reports the 2014 coefficient, with stars at the 1% (***), 5% (**) and 10% (*) levels.

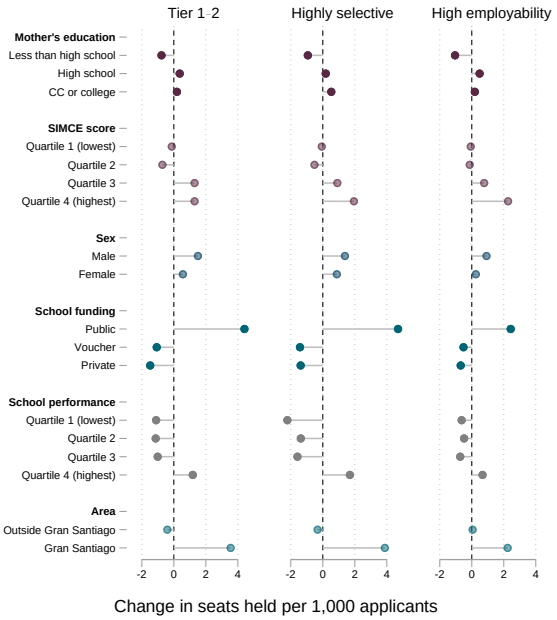


FIGURE 14. Net Change in Seats Held through School Switching, by Applicant Characteristics

Note: This figure shows, for each category of applicants, the net change in seats held when switching is allowed relative to the allocation without it, per 1,000 applicants in the category, in the 2015 admission process. Both allocations are simulated with the 2015 assignment mechanism for all 112,824 applicants with a valid application, switchers and non-switchers alike. School characteristics refer to the school at the start of twelfth grade. SIMCE score, sex and area are available only for the class of 2014, so gains and losses across their categories need not offset. Tier 1–2 universities follow Kapor et al. (2024). Highly selective programs are those in the top quartile of the mean admission score of students admitted in the 2014 admission process; high-employability programs are those in the top quartile of the share of graduates employed one year after graduation, among those who graduated between 2013 and 2015.

TABLE 1. Characteristics of Higher Education Institutions by Admission System and Type of Institution, Academic Year 2014

	Centralized Admission		Decentralized Admission		
	Public Univ.	Private U. in the system	Private U. outside the system	IP	CFT
	(1)	(2)	(3)	(4)	(5)
Panel 1: *College characteristics*					
Slots	37.91	52.30	39.36	43.83	47.98
Annual tuition (in CLP)	2,197,323	2,846,087	2,072,740	1,237,337	1,185,249
Share of STEM programs	0.31	0.30	.	.	.
Share of accredited programs	0.37	0.36	0.33	0.24	0.21
Number of programs	849	1,002	1,575	3,185	1,630
Panel 2: *Enrolled students*					
Total enrollment	39,383	58,612	47,380	126,572	65,447
Total female enrollment	19,208	28,456	27,432	62,429	32,979
Panel 3: *Application requirements' weights*					
Average weight high school GPA	15.80	16.36	58.61	.	.
Average weight relative ranking	26.05	18.83	38.44	.	.
Average weight PSU language	20.69	21.22	46.96	.	.
Average weight PSU mathematics	24.63	29.98	47.35	.	.

Note: This table reports the characteristics of higher education institutions in academic year 2014, by type of institution. Columns 1 and 2 are the public and private universities in the centralized system; columns 3 to 5 are institutions with decentralized admission: private universities outside the system, professional institutes and technical training centers. Enrollment and female enrollment count first-year students. All other rows except the number of programs are averages across programs.

TABLE 2. Summary Statistics for Students in Twelfth Grade

	All		No positive gain		Positive gain	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Panel A: *Outcomes*						
Switched schools in twelfth grade, beginning or during the year	0.04	0.20	0.05	0.22	0.03	0.16
Switched schools during twelfth grade	0.02	0.13	0.02	0.14	0.01	0.10
Panel B: *GPA and test score*						
GPA (non-standardized)	5.51	0.52	5.30	0.44	5.86	0.44
Tenth-grade SIMCE score, standardized	0.24	1.01	-0.06	0.94	0.67	0.95
Panel C: *Other characteristics*						
Public school	0.28	0.45	0.29	0.46	0.26	0.44
Voucher school	0.57	0.50	0.60	0.49	0.51	0.50
Private school	0.15	0.36	0.11	0.31	0.22	0.42
Academic track	0.60	0.49	0.50	0.50	0.72	0.45
Vocational track	0.40	0.49	0.50	0.50	0.28	0.45
School performance, quartile 1 (lowest)	0.16	0.37	0.21	0.41	0.10	0.30
School performance, quartile 2	0.23	0.42	0.27	0.45	0.18	0.38
School performance, quartile 3	0.27	0.44	0.27	0.45	0.26	0.44
School performance, quartile 4 (highest)	0.33	0.47	0.25	0.43	0.46	0.50
Mother's education: less than high school	0.26	0.44	0.29	0.46	0.20	0.40
Mother's education: high school	0.45	0.50	0.47	0.50	0.43	0.50
Mother's education: more than high school	0.29	0.45	0.24	0.43	0.37	0.48
Male	0.49	0.50	0.52	0.50	0.43	0.49
Female	0.51	0.50	0.48	0.50	0.57	0.49
Low aspirations	0.30	0.46	0.39	0.49	0.18	0.39
High aspirations	0.70	0.46	0.61	0.49	0.82	0.39
No internet access	0.21	0.41	0.24	0.43	0.17	0.37
Internet access	0.79	0.41	0.76	0.43	0.83	0.37
Observations	577,026		356,624		220,402	

Note: This table reports summary statistics for twelfth graders in the sample. Panel A reports the outcomes: switching schools in twelfth grade, at the beginning or during the year, and switching during the year. Panel B reports GPA, used as a control, and the tenth-grade SIMCE score. Panel C reports characteristics used to study heterogeneity but not as controls. School characteristics refer to the school at the start of twelfth grade; aspirations and internet access come from the SIMCE parents' survey, and mother's education from that survey and the PSU registration form. Shares are among students with the characteristic observed, and track shares exclude adult education. Columns split students by whether they have a positive median potential gain in their choice set, the schools within 2 km of their primary school. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade, in the cohorts finishing twelfth grade between 2010 and 2018.

TABLE 3. Summary Statistics for Students Applying to Universities in the Centralized System

	All		No positive gain		Positive gain	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Panel A: *Outcomes*						
Applied to a university (share of all students)	0.34	0.47	0.21	0.41	0.54	0.50
High-employability program among the first three preferences	0.58	0.49	0.50	0.50	0.63	0.48
High-earnings program among the first three preferences	0.64	0.48	0.59	0.49	0.67	0.47
Highly selective program among the first three preferences	0.75	0.43	0.62	0.49	0.83	0.37
Accepted in a university	0.86	0.35	0.82	0.39	0.88	0.32
Accepted in a high-employability program	0.28	0.45	0.23	0.42	0.30	0.46
Accepted in a high-earnings program	0.36	0.48	0.32	0.47	0.39	0.49
Accepted in a highly selective program	0.40	0.49	0.24	0.43	0.49	0.50
Enrolled in a university	0.72	0.45	0.67	0.47	0.76	0.43
Enrolled in a high-employability program	0.29	0.45	0.23	0.42	0.33	0.47
Enrolled in a high-earnings program	0.36	0.48	0.31	0.46	0.39	0.49
Enrolled in a highly selective program	0.43	0.49	0.28	0.45	0.52	0.50
Dropped out of a high-employability first program	0.08	0.27	0.07	0.26	0.09	0.28
Dropped out of a high-earnings first program	0.09	0.29	0.09	0.29	0.09	0.29
Dropped out of a highly selective first program	0.10	0.31	0.07	0.26	0.13	0.33
Graduated from the first program	0.37	0.48	0.31	0.46	0.40	0.49
Graduated from a high-employability first program	0.15	0.36	0.11	0.31	0.18	0.38
Graduated from a high-earnings first program	0.19	0.39	0.15	0.36	0.22	0.41
Graduated from a highly selective first program	0.22	0.42	0.14	0.35	0.27	0.44
Graduated on time from a high-employability first program	0.08	0.28	0.06	0.24	0.10	0.30
Graduated on time from a high-earnings first program	0.11	0.31	0.08	0.28	0.13	0.33
Graduated on time from a highly selective first program	0.13	0.34	0.09	0.29	0.16	0.37
Panel B: *GPA and test score*						
GPA (non-standardized)	5.92	0.46	5.69	0.50	6.06	0.37
Tenth-grade SIMCE score, standardized	1.03	0.71	0.80	0.74	1.16	0.66
Panel C: *Other characteristics*						
Public school	0.19	0.39	0.15	0.36	0.22	0.41
Voucher school	0.45	0.50	0.50	0.50	0.42	0.49
Private school	0.35	0.48	0.34	0.48	0.36	0.48
Academic track	0.91	0.29	0.87	0.34	0.93	0.26
Vocational track	0.09	0.29	0.13	0.34	0.07	0.26
School performance, quartile 1 (lowest)	0.04	0.20	0.07	0.26	0.02	0.15
School performance, quartile 2	0.12	0.32	0.16	0.37	0.09	0.29
School performance, quartile 3	0.27	0.44	0.29	0.45	0.25	0.44
School performance, quartile 4 (highest)	0.57	0.50	0.48	0.50	0.63	0.48
Mother's education: less than high school	0.11	0.31	0.12	0.33	0.10	0.30
Mother's education: high school	0.41	0.49	0.42	0.49	0.40	0.49
Mother's education: more than high school	0.48	0.50	0.45	0.50	0.50	0.50
Male	0.48	0.50	0.53	0.50	0.45	0.50
Female	0.52	0.50	0.47	0.50	0.55	0.50
Low aspirations	0.07	0.25	0.10	0.30	0.05	0.21
High aspirations	0.93	0.25	0.90	0.30	0.95	0.21
No internet access	0.09	0.29	0.10	0.30	0.09	0.28
Internet access	0.91	0.29	0.90	0.30	0.91	0.28
Switched schools during twelfth grade	0.01	0.11	0.01	0.10	0.01	0.11
Observations	129,081		49,301		79,780	

Note: This table reports summary statistics for applicants to universities in the centralized system. Panel A reports the outcomes, measured in each student's first admission process; the first program is the one the student enrolled in through that process. The first row reports the share of all students in the sample who applied; every other statistic is among applicants. Panel B reports GPA, used as a control through within-school decile fixed effects, and the tenth-grade SIMCE score, a weighted average of the language and mathematics scores standardized nationally each year. Panel C reports switching schools during twelfth grade and characteristics used to study heterogeneity but not as controls. School characteristics refer to the school at the start of twelfth grade; aspirations and internet access come from the SIMCE parents' survey, and mother's education from that survey and the PSU registration form. Shares are among students with the characteristic observed, and track shares exclude adult education. Highly selective programs are those in the top quartile of the mean admission score of students admitted in the 2014 admission process. High-employability and high-earnings programs are those in the top quartile of the share of graduates employed one year after graduation and of their average monthly income four years after graduation, among those who graduated between 2013 and 2015. Dropout is not defined for students still enrolled in their first program at the end of the data. Columns split students by whether they have a positive median potential gain in their choice set, the schools within 2 km of their primary school. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade, in the cohorts finishing twelfth grade between 2011 and 2016.

TABLE 4. Regression Discontinuity Validity at the Cutoffs of Programs with Displacement: All Switchers

	All programs	Tier 1–2	Highly selective	High employability
Positive potential gain	0.033	0.040	0.042**	0.032
	(0.023)	(0.034)	(0.025)	(0.037)
Female	-0.008	-0.018	-0.021	0.001
	(0.025)	(0.033)	(0.030)	(0.037)
Mother completed high school	-0.004	0.002	-0.015	-0.007
	(0.014)	(0.021)	(0.017)	(0.020)
SIMCE score	-0.018	-0.003	-0.020	-0.058
	(0.033)	(0.044)	(0.032)	(0.050)
Public school	0.023*	0.027	0.040**	0.046**
	(0.016)	(0.025)	(0.023)	(0.026)
School performance, top quartile	0.006	-0.014	0.005	0.023
	(0.021)	(0.032)	(0.024)	(0.031)
Gran Santiago	-0.025	-0.034	-0.031	-0.012
	(0.023)	(0.031)	(0.026)	(0.033)
Density test (p-value)	0.247	0.863	0.376	0.647

Note: This table reports regression discontinuity estimates of the change in applicant characteristics at the admission cutoffs of programs where at least one student lost a seat to switching, treating every student who switched schools in twelfth grade in 2014 as strategic. Columns use the cutoffs of all such programs and of those at tier 1–2 universities, highly selective programs and high-employability programs. The sample is applications in the 2015 admission process by students who did not switch schools in twelfth grade. Each row is a separate regression of the characteristic on the distance between the admission score and the cutoff, within 50 points of it, with a triangular kernel and the mean squared error optimal bandwidth of Calonico et al. (2014). All characteristics are defined only for the class of 2014. The last row reports the p-value of the density test of Cattaneo et al. (2020). Estimates are conventional; standard errors are robust bias-corrected and clustered by student. Stars indicate significance at the 1% (***), 5% (**), and 10% (*) levels based on robust bias-corrected p-values, so they need not match the ratio of the conventional estimate to its standard error.

TABLE 5. Effect of a Seat Offer at the Cutoffs of Programs with Displacement: High-Employability and Highly Selective Programs

	Programs with displaced students when the switchers are:		
	All	With a gain	With a gain, in Gran Santiago
	(1)	(2)	(3)
Panel A: *High-employability programs*			
Accepted, high-employability	0.478***	0.490***	0.492***
	(0.019)	(0.019)	(0.019)
Enrolled, high-employability	0.164***	0.166***	0.173***
	(0.023)	(0.022)	(0.023)
Dropped out, high-employability first program	0.048***	0.053***	0.049***
	(0.017)	(0.017)	(0.018)
Graduated, high-employability first program	0.062**	0.053	0.065**
	(0.026)	(0.026)	(0.027)
Graduated on time, high-employability first program	0.016	0.013	0.014
	(0.022)	(0.022)	(0.023)
Panel B: *Highly selective programs*			
Accepted, highly selective	0.289***	0.275***	0.261***
	(0.012)	(0.013)	(0.013)
Enrolled, highly selective	0.102***	0.095***	0.089***
	(0.016)	(0.017)	(0.017)
Dropped out, highly selective first program	0.072***	0.074***	0.076***
	(0.019)	(0.020)	(0.022)
Graduated, highly selective first program	0.012	0.002	0.006
	(0.021)	(0.021)	(0.022)
Graduated on time, highly selective first program	0.016	0.014	0.015
	(0.015)	(0.016)	(0.016)

Note: This table reports regression discontinuity estimates of the effect of a seat offer at the admission cutoffs of programs where at least one student lost a seat to switching. Columns use the programs with displaced students when the switchers are all students who switched schools in twelfth grade in 2014, those with a positive potential gain, and those with a positive potential gain in Gran Santiago. Panel A uses the cutoffs of high-employability programs and Panel B those of highly selective programs. The sample is applications in the 2014 and 2015 admission processes, at or above the preference where the student was accepted, by students who did not switch schools in twelfth grade. Outcomes are measured in the admission process of the application, and the first program is the one the student enrolled in through that process. Each row is a separate regression of the outcome on the distance between the admission score and the cutoff, within 50 points of it, with a triangular kernel, the mean squared error optimal bandwidth of Calonico et al. (2014), and weights equal to the number of displaced students in the program. Estimates are conventional; standard errors are robust bias-corrected and clustered by student. Stars indicate significance at the 1% (***), 5% (**), and 10% (*) levels based on robust bias-corrected p-values, so they need not match the ratio of the conventional estimate to its standard error.

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Appendix A. Additional Figures and Tables

TABLE A1. High School Characteristics by School Performance, Gran Santiago, 2010–2013

	Public Schools					Voucher Schools					Non-Voucher Private Schools				
	Overall	Q_1	Q_2	Q_3	Q_4	Overall	Q_1	Q_2	Q_3	Q_4	Overall	Q_1	Q_2	Q_3	Q_4
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
Panel A: *Infrastructure*															
Classroom size	36.39	32.95	34.06	36.24	39.96	37.53	36.13	38.06	37.30	38.63	27.78	22.53	26.44	28.85	29.31
	(6.62)	(5.46)	(6.92)	(5.72)	(5.98)	(6.28)	(5.61)	(5.97)	(5.08)	(7.79)	(6.01)	(6.13)	(5.92)	(4.89)	(5.65)
Students per teacher	20.59	16.46	21.07	20.67	21.98	15.32	19.72	16.82	14.28	10.59	6.05	6.29	4.79	6.11	6.70
	(7.35)	(6.36)	(6.86)	(8.10)	(6.72)	(8.84)	(8.59)	(8.48)	(9.04)	(6.53)	(14.40)	(6.92)	(1.66)	(5.78)	(23.94)
Panel B: *Population served*															
Share of students from the same county	0.45	0.63	0.44	0.51	0.32	0.57	0.46	0.56	0.64	0.63	0.57	0.51	0.54	0.60	0.55
	(0.33)	(0.22)	(0.35)	(0.33)	(0.32)	(0.32)	(0.35)	(0.31)	(0.30)	(0.27)	(0.24)	(0.19)	(0.22)	(0.25)	(0.25)
Father is household head	0.53	0.53	0.53	0.53	0.53	0.56	0.54	0.56	0.57	0.59	0.71	0.58	0.64	0.73	0.78
	(0.09)	(0.11)	(0.09)	(0.08)	(0.07)	(0.09)	(0.09)	(0.09)	(0.09)	(0.08)	(0.14)	(0.14)	(0.15)	(0.12)	(0.11)
Mother education < HS	0.51	0.66	0.61	0.52	0.31	0.36	0.56	0.43	0.29	0.16	0.01	0.02	0.01	0.01	0.00
	(0.19)	(0.10)	(0.13)	(0.14)	(0.17)	(0.21)	(0.15)	(0.15)	(0.15)	(0.13)	(0.02)	(0.03)	(0.02)	(0.01)	(0.01)
Has access to internet	0.42	0.28	0.35	0.41	0.59	0.56	0.38	0.48	0.63	0.76	0.97	0.95	0.96	0.97	0.99
	(0.20)	(0.14)	(0.16)	(0.15)	(0.19)	(0.23)	(0.18)	(0.18)	(0.18)	(0.17)	(0.04)	(0.06)	(0.05)	(0.04)	(0.02)
High aspirations	0.47	0.28	0.37	0.42	0.74	0.60	0.38	0.53	0.68	0.82	0.96	0.86	0.96	0.98	0.99
	(0.24)	(0.10)	(0.14)	(0.16)	(0.20)	(0.22)	(0.15)	(0.15)	(0.16)	(0.14)	(0.07)	(0.11)	(0.04)	(0.02)	(0.01)
Panel C: *Ex-post results*															
Mean $\underline{r}_s$	5.39	5.28	5.34	5.30	5.57	5.41	5.26	5.35	5.44	5.58	5.89	5.61	5.75	5.91	6.05
	(0.21)	(0.09)	(0.15)	(0.13)	(0.24)	(0.20)	(0.14)	(0.15)	(0.15)	(0.20)	(0.22)	(0.18)	(0.17)	(0.15)	(0.17)
Mean $\bar{r}_s$	6.51	6.40	6.50	6.48	6.60	6.49	6.39	6.48	6.51	6.60	6.74	6.51	6.66	6.78	6.81
	(0.22)	(0.20)	(0.21)	(0.20)	(0.21)	(0.22)	(0.23)	(0.20)	(0.22)	(0.20)	(0.17)	(0.26)	(0.15)	(0.08)	(0.10)
College admission test mean score	547.47	541.17	541.97	539.18	561.87	547.72	538.37	542.34	549.32	560.95	591.91	566.12	577.42	592.20	609.52
	(20.39)	(13.30)	(16.68)	(13.50)	(23.29)	(18.14)	(16.61)	(15.25)	(14.72)	(17.65)	(23.69)	(20.46)	(18.45)	(17.22)	(17.51)
Took the college admission test	0.73	0.57	0.66	0.69	0.89	0.81	0.61	0.79	0.89	0.96	0.98	0.94	0.98	0.98	1.00
	(0.20)	(0.18)	(0.15)	(0.17)	(0.13)	(0.19)	(0.19)	(0.16)	(0.11)	(0.07)	(0.05)	(0.06)	(0.02)	(0.07)	(0.01)
Applied to college, among test takers	0.23	0.06	0.08	0.12	0.47	0.26	0.08	0.16	0.27	0.50	0.75	0.46	0.66	0.77	0.87
	(0.25)	(0.06)	(0.06)	(0.09)	(0.28)	(0.21)	(0.07)	(0.10)	(0.14)	(0.19)	(0.22)	(0.20)	(0.19)	(0.19)	(0.15)
Accepted in a high-earnings program	0.30	0.28	0.28	0.28	0.32	0.31	0.33	0.29	0.30	0.32	0.39	0.33	0.34	0.38	0.44
	(0.22)	(0.31)	(0.28)	(0.21)	(0.12)	(0.19)	(0.28)	(0.18)	(0.14)	(0.11)	(0.13)	(0.22)	(0.12)	(0.10)	(0.10)
Accepted in a high-employability program	0.28	0.28	0.26	0.28	0.28	0.27	0.29	0.27	0.27	0.27	0.28	0.19	0.23	0.28	0.35
	(0.21)	(0.30)	(0.27)	(0.21)	(0.12)	(0.18)	(0.28)	(0.17)	(0.13)	(0.10)	(0.11)	(0.15)	(0.10)	(0.08)	(0.09)
Accepted in a highly selective program	0.21	0.12	0.16	0.16	0.33	0.22	0.15	0.17	0.20	0.32	0.58	0.39	0.46	0.58	0.72
	(0.21)	(0.21)	(0.21)	(0.16)	(0.20)	(0.17)	(0.21)	(0.15)	(0.13)	(0.13)	(0.19)	(0.23)	(0.17)	(0.15)	(0.11)
Number of high schools	188	45	48	51	44	551	131	133	135	152	187	46	42	50	49

Note: This table reports characteristics of high schools in Gran Santiago, cohorts graduating 2010–2013. Columns 1 to 5 are public, 6 to 10 voucher and 11 to 15 non-voucher private schools; within each group, the first column covers all schools and the next four split them by quartile of the 2010 tenth-grade SIMCE mathematics score within funding type and region. Statistics are weighted by twelfth-grade enrollment and standard deviations are displayed in parentheses. Mother's education, internet access and aspirations come from the SIMCE parents' survey, available only for the 2010 and 2012 cohorts. Acceptance shares are among the school's applicants. High-earnings and high-employability programs are those in the top quartile of graduates' average monthly income four years after graduation and of the share employed one year after graduation, among 2013–2015 graduates; highly selective programs are those in the top quartile of the mean application score of students admitted in the 2014 admission process.

TABLE A2. RR and GPA Weights in the Centralized Admission System, Cohorts 2010–2018

	All programs		Tier 1		Tier 2		Tier 3	
Year	RR	GPA	RR	GPA	RR	GPA	RR	GPA
2010	0.00	30.20	0.00	24.42	0.00	29.53	0.00	28.90
2011	0.00	29.33	0.00	24.56	0.00	27.30	0.00	27.66
2012	10.00	23.33	10.00	19.29	10.00	26.63	10.00	20.52
2013	22.21	16.05	19.90	17.65	29.52	11.15	18.43	17.02
2014	22.98	15.78	20.00	17.62	29.25	11.27	19.89	16.55
2015	23.87	15.73	20.00	17.62	29.17	11.42	20.83	16.70
2016	24.70	15.26	21.24	16.49	29.02	11.56	21.70	16.14
2017	24.05	14.32	21.13	16.60	28.41	11.42	22.29	15.79
2018	24.35	14.07	20.56	16.48	28.82	11.32	22.24	15.87
Difference 2014 – 2013	0.763		0.102		-0.274		1.460	
	(0.393)		(0.753)		(1.931)		(0.354)	
p-value	0.052		0.892		0.887		0.000	

Note: This table reports the mean weight of the relative ranking (RR) and of GPA in the admission score across programs in the centralized system, overall and by university tier, following Kapor et al. (2024). The year is the graduating cohort, which applies in the admission process of the following year. The RR did not exist before the 2013 process, so its weight is zero for the 2010 and 2011 cohorts. The last rows report the 2014 minus 2013 difference in the mean RR weight, its standard error and the *p*-value of a two-sample test with unequal variances. The number of programs ranges from 980 to 1,806 overall, 86 to 108 in Tier 1, 64 to 116 in Tier 2, and 445 to 630 in Tier 3.

TABLE A3. Submitted Preference Lists and Admission Outcomes, 2013 Admission Process

	Fraction reporting (1)	Admission rate (2)	Share of admitted (3)
Choice 1	1.000	0.429	0.532
Choice 2	0.933	0.196	0.227
Choice 3	0.831	0.123	0.126
Choice 4	0.627	0.069	0.054
Choice 5	0.454	0.051	0.029
Choice 6	0.316	0.038	0.015
Choice 7	0.221	0.030	0.008
Choice 8	0.156	0.022	0.004
Choice 9	0.106	0.019	0.003
Choice 10	0.078	0.016	0.002
Nb. applicants / admitted	118,208		95,300

Note: This table reports application and admission outcomes by preference rank for all applicants to the centralized system in the 2013 admission process, the first under the relative ranking. Column 1 is the share of applicants listing at least k preferences; column 2 is the share admitted at preference k among those listing it, so rows have different denominators; column 3 is the distribution of admitted students by the preference at which they were placed, and sums to one.

TABLE A4. Selectivity by Institution, 2012 Admission Process

University	Mean	Std. Dev.	P10	P25	P50	P75	P90	Admitted	Tier
UCH	691.56	46.09	630.90	658.25	688.90	723.55	751.90	5,424	1
PUC	694.75	52.53	630.60	652.80	690.40	734.20	769.90	4,754	1
USACH	633.90	36.64	591.80	610.70	631.75	654.90	676.60	4,725	2
UAI	656.25	37.10	610.30	633.70	655.55	678.30	702.30	1,868	2
UANDE	657.23	53.55	586.00	613.60	657.60	694.90	723.90	1,221	2
UDEC	602.88	62.86	521.30	554.30	601.85	645.05	687.45	6,377	3
UTFSM	615.26	78.90	506.90	547.70	629.20	680.50	708.20	4,346	3
UV	596.58	54.47	532.70	558.70	589.88	625.30	671.50	3,984	3
PUCV	610.95	43.06	557.40	581.40	608.30	637.70	667.30	3,448	3
UACH	582.65	58.14	514.55	541.60	577.15	617.02	660.40	3,076	3
UDD	605.96	55.06	537.85	566.20	603.00	636.10	673.90	2,903	3
UDP	617.22	40.02	571.20	590.00	613.33	638.85	668.60	2,726	3
UFRO	592.74	56.16	525.10	549.90	586.30	627.10	669.20	2,354	3
UCN	589.36	55.32	520.40	546.35	583.60	625.10	664.80	2,322	3
UTAL	616.17	52.12	551.40	582.92	615.02	646.90	685.90	1,904	3
UCM	581.56	48.03	527.80	546.10	573.25	606.75	644.30	1,639	3
UAH	595.67	38.86	547.40	567.50	593.90	619.60	646.80	1,539	3
UFT	580.95	59.61	509.70	540.85	575.70	613.70	660.05	1,145	3
UMCE	605.76	33.62	570.00	580.77	603.33	624.95	649.05	1,092	3
UNAB	550.38	53.70	484.50	511.90	546.80	583.70	618.60	12,615	4
UMAYO	571.86	56.56	501.40	529.00	568.00	604.40	645.80	4,895	4
UCSC	563.23	51.92	502.75	525.60	557.60	593.40	625.45	2,451	4
UBB	577.58	43.70	523.40	544.50	572.45	604.35	639.70	2,449	4
UTEM	548.76	36.01	503.20	524.55	548.35	574.25	595.05	2,392	4
UCT	543.95	43.09	492.05	509.60	540.05	571.35	598.80	2,206	4
ULS	575.44	45.06	521.40	542.50	569.40	605.70	635.40	1,791	4
UPA	556.07	37.49	512.05	530.03	553.02	578.55	604.00	1,748	4
UTA	557.07	51.94	493.70	517.17	549.02	594.40	627.80	1,720	4
UANT	566.13	65.25	487.20	515.70	558.30	606.70	662.15	1,433	4
UNAP	550.89	47.10	493.80	514.08	541.30	583.88	616.45	928	4
UDA	550.61	50.05	486.85	511.90	549.50	585.70	616.60	815	4
ULAG	541.04	44.78	484.40	508.20	535.80	571.00	603.75	785	4
UMAG	554.09	50.70	492.10	516.15	547.45	586.00	626.05	499	4

Note: This table reproduces Table A-2 of Kapor et al. (2024), which reports the distribution of admission scoress among students admitted through the centralized system in the 2012 admission process, for each participating university. Universities are grouped into tiers. Tier 1 includes universities with an average admitted score in $[660, 700)$, tier 2 in $[620, 660)$, tier 3 in $[580, 620)$ and tier 4 in $[540, 580)$. Within each tier, universities are ordered by the number of students they admitted. The 2012 process includes eight private universities that joined the platform that year; the number of universities in the centralized system then remained stable until the 2017 process.

TABLE A5. Selectivity by Program, 2014 Admission Process

	Selectivity quartile				
	Q1 (least selective)	Q2	Q3	Q4 (highly selective)	All
	(1)	(2)	(3)	(4)	(5)
Panel A: *Admission scores*					
Number of programs	352	350	352	352	1,406
Mean admitted score	527.7	560.3	600.4	672.8	590.4
Range of mean admitted scores	487.0–545.5	545.6–576.2	576.3–626.7	626.8–803.1	487.0–803.1
Students admitted	17,182	21,649	24,590	31,333	94,754
Students admitted per program	49	62	70	89	67
Panel B: *Program characteristics*					
Share at tier 1–2 universities	0.00	0.02	0.09	0.46	0.14
Share high-employability	0.20	0.23	0.24	0.35	0.26
Share high-earnings	0.11	0.19	0.27	0.50	0.27

Note: This table groups the programs in the 2014 admission process by quartile of the mean application score of the students they admitted; column 4 is the highly selective programs used throughout the paper. The mean admitted score averages each program's mean across programs, the range gives the lowest and highest program mean, and students admitted is the total in each quartile. Shares are over programs. Tier 1–2 universities follow Kapor et al. (2024). High-employability and high-earnings programs are those in the top quartile of the share of graduates employed one year after graduation and of graduates' average monthly income four years after graduation, among 2013–2015 graduates.

TABLE A6. Alternative Specifications: Discrete Measure of Potential Gain

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Positive gain × 2010	0.0022*	0.0022*	0.0014	0.0026**	0.0010	0.0013	0.0002	0.0009	0.0013	0.0003
	(0.0012)	(0.0012)	(0.0014)	(0.0013)	(0.0014)	(0.0014)	(0.0014)	(0.0014)	(0.0016)	(0.0022)
Positive gain × 2011	0.0019	0.0019	0.0025	0.0011	-0.0008	-0.0001	-0.0011	-0.0007	-0.0026	-0.0030
	(0.0013)	(0.0013)	(0.0016)	(0.0015)	(0.0016)	(0.0016)	(0.0016)	(0.0016)	(0.0018)	(0.0025)
Positive gain × 2012	-0.0020	-0.0020	-0.0020	-0.0018	-0.0022	-0.0019	-0.0022	-0.0034**	-0.0034*	-0.0066***
	(0.0012)	(0.0012)	(0.0014)	(0.0014)	(0.0016)	(0.0015)	(0.0015)	(0.0015)	(0.0018)	(0.0023)
Positive gain × 2013	[base]	[base]	[base]	[base]	[base]	[base]	[base]	[base]	[base]	[base]
Positive gain × 2014	0.0250***	0.0250***	0.0287***	0.0274***	0.0222***	0.0209***	0.0159***	0.0056***	0.0043**	-0.0006
	(0.0020)	(0.0020)	(0.0022)	(0.0022)	(0.0021)	(0.0021)	(0.0020)	(0.0017)	(0.0019)	(0.0025)
Positive gain × 2015	-0.0014	-0.0014	-0.0018	-0.0014	-0.0010	-0.0001	-0.0009	-0.0012	-0.0012	-0.0020
	(0.0013)	(0.0013)	(0.0015)	(0.0015)	(0.0016)	(0.0015)	(0.0015)	(0.0015)	(0.0018)	(0.0024)
Positive gain × 2016	-0.0004	-0.0004	-0.0010	-0.0008	-0.0002	0.0008	0.0005	0.0014	0.0008	0.0020
	(0.0013)	(0.0013)	(0.0015)	(0.0015)	(0.0016)	(0.0015)	(0.0015)	(0.0015)	(0.0018)	(0.0025)
Positive gain × 2017	-0.0002	-0.0001	-0.0005	0.0009	0.0010	0.0024	0.0013	-0.0008	0.0007	0.0013
	(0.0013)	(0.0013)	(0.0015)	(0.0014)	(0.0016)	(0.0015)	(0.0015)	(0.0015)	(0.0018)	(0.0024)
Positive gain × 2018	-0.0020	-0.0020	-0.0022	-0.0010	0.0004	0.0020	0.0005	-0.0023	-0.0007	-0.0010
	(0.0013)	(0.0013)	(0.0015)	(0.0015)	(0.0016)	(0.0015)	(0.0015)	(0.0015)	(0.0019)	(0.0024)
School FE	✓	✓	✓	✓	✓	✓	✓			
GPA decile FE	✓	✓	✓					✓		
Market size		✓								
Market × year FE			✓							
Decile × year FE				✓	✓	✓	✓		✓	
Baseline threshold × year					✓	✓				
Baseline SIMCE × year						✓				
Contemporaneous threshold × year							✓			
School × year FE								✓	✓	
School × decile × year FE										✓
Observations	577,022	577,022	576,239	577,022	539,794	495,754	577,022	576,999	576,999	568,830

Note: This table reports the coefficients from Equation (4) under alternative fixed effects and controls, marked in the lower panel. The outcome is an indicator for switching schools during twelfth grade and the key regressors are the interactions of an indicator for a positive median potential gain with year dummies. Column 1 is the main specification, with fixed effects for the school at the start of twelfth grade and within-school GPA deciles. Column 2 adds the number of schools in the student's market and column 3 market × year fixed effects. Column 4 replaces the decile fixed effects with decile × year; columns 5 and 6 add the school's baseline threshold and then its baseline SIMCE score, each interacted with year, and column 7 instead its contemporaneous threshold × year. Columns 8 and 9 absorb school × year with decile and with decile × year fixed effects, and column 10 school × decile × year. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade, in the cohorts finishing twelfth grade between 2010 and 2018. Coefficients are relative to 2013, the omitted year. Standard errors are clustered at the school-market level, are in parentheses. Stars indicate significance at the 1% (***), 5% (**), and 10% (*) levels.

TABLE A7. Alternative Specifications: Continuous Measure of Potential Gain

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Potential gain × 2010	0.0009	0.0009	0.0006	0.0009	0.0003	0.0004	-0.0003	-0.0006	-0.0006	-0.0010
	(0.0008)	(0.0008)	(0.0009)	(0.0008)	(0.0009)	(0.0008)	(0.0009)	(0.0009)	(0.0009)	(0.0014)
Potential gain × 2011	0.0012	0.0012	0.0019**	0.0011	0.0004	0.0008	0.0001	-0.0010	-0.0013	-0.0005
	(0.0007)	(0.0008)	(0.0009)	(0.0008)	(0.0009)	(0.0008)	(0.0009)	(0.0010)	(0.0010)	(0.0016)
Potential gain × 2012	-0.0002	-0.0002	0.0003	-0.0002	-0.0000	-0.0000	-0.0002	-0.0007	-0.0007	-0.0022
	(0.0007)	(0.0007)	(0.0008)	(0.0007)	(0.0009)	(0.0008)	(0.0008)	(0.0010)	(0.0010)	(0.0016)
Potential gain × 2013	[base]	[base]	[base]	[base]	[base]	[base]	[base]	[base]	[base]	[base]
Potential gain × 2014	0.0150***	0.0150***	0.0178***	0.0149***	0.0146***	0.0150***	0.0100***	0.0020*	0.0015	-0.0003
	(0.0012)	(0.0012)	(0.0013)	(0.0013)	(0.0014)	(0.0014)	(0.0012)	(0.0011)	(0.0011)	(0.0018)
Potential gain × 2015	-0.0000	-0.0000	0.0000	-0.0000	0.0004	0.0005	0.0004	-0.0003	-0.0003	-0.0012
	(0.0008)	(0.0008)	(0.0009)	(0.0008)	(0.0009)	(0.0008)	(0.0008)	(0.0010)	(0.0010)	(0.0016)
Potential gain × 2016	-0.0006	-0.0006	-0.0008	-0.0007	-0.0002	-0.0001	-0.0003	-0.0005	-0.0007	-0.0001
	(0.0007)	(0.0007)	(0.0009)	(0.0008)	(0.0008)	(0.0008)	(0.0008)	(0.0009)	(0.0010)	(0.0015)
Potential gain × 2017	0.0004	0.0004	0.0005	0.0004	0.0004	0.0005	0.0004	-0.0007	-0.0007	-0.0015
	(0.0007)	(0.0007)	(0.0008)	(0.0007)	(0.0008)	(0.0008)	(0.0008)	(0.0009)	(0.0009)	(0.0014)
Potential gain × 2018	-0.0009	-0.0009	-0.0008	-0.0009	0.0003	0.0009	-0.0004	-0.0014	-0.0013	0.0001
	(0.0008)	(0.0008)	(0.0009)	(0.0008)	(0.0009)	(0.0008)	(0.0009)	(0.0010)	(0.0010)	(0.0015)
School FE	✓	✓	✓	✓	✓	✓	✓			
GPA decile FE	✓	✓	✓					✓		
Market size		✓								
Market × year FE			✓							
Decile × year FE				✓	✓	✓	✓		✓	
Baseline threshold × year					✓	✓				
Baseline SIMCE × year						✓				
Contemporaneous threshold × year							✓			
School × year FE								✓	✓	
School × decile × year FE										✓
Observations	577,022	577,022	576,239	577,022	539,794	495,754	577,022	576,999	576,999	568,830

Note: This table reports the coefficients from Equation (4) under alternative fixed effects and controls, marked in the lower panel. The outcome is an indicator for switching schools during twelfth grade and the key regressors are the interactions of the standardized median potential gain with year dummies, so coefficients are per standard deviation of the median potential gain within each cohort. Column 1 is the main specification, with fixed effects for the school at the start of twelfth grade and within-school GPA deciles. Column 2 adds the number of schools in the student's market and column 3 market × year fixed effects. Column 4 replaces the decile fixed effects with decile × year; columns 5 and 6 add the school's baseline threshold and then its baseline SIMCE score, each interacted with year, and column 7 instead its contemporaneous threshold × year. Columns 8 and 9 absorb school × year with decile and with decile × year fixed effects, and column 10 school × decile × year. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade, in the cohorts finishing twelfth grade between 2010 and 2018. Coefficients are relative to 2013, the omitted year. Standard errors are clustered at the school-market level, are in parentheses. Stars indicate significance at the 1% (***), 5% (**), and 10% (*) levels.

TABLE A8. Shapley Decomposition of the Variance of the Potential Gain

	2010	2011	2012	2013	2014	2015	2016	2017	2018	Total
Panel A: *Positive potential gain (discrete measure)*										
GPA decile within own school	11.0	11.7	11.4	11.6	12.5	12.0	11.4	11.5	11.8	11.7
Own school threshold	15.2	15.6	16.0	17.0	17.3	18.1	18.5	18.4	18.5	17.0
Market structure	13.4	12.4	12.6	11.2	9.9	10.4	10.3	9.3	8.7	10.6
GPA decile × own school threshold	23.4	22.8	22.7	22.5	23.0	22.7	22.2	22.6	22.6	22.9
GPA decile × market structure	8.1	8.0	7.8	7.7	8.0	7.4	7.3	7.3	7.2	7.8
Own school threshold × market structure	9.9	10.6	10.7	11.2	11.0	11.3	12.1	12.4	12.5	11.3
GPA decile × own school threshold × market structure	19.0	19.0	18.9	18.8	18.4	18.2	18.2	18.6	18.7	18.8
R-squared	0.496	0.488	0.496	0.492	0.513	0.499	0.492	0.499	0.491	0.490
Panel B: *Potential gain (continuous measure)*										
GPA decile within own school	6.3	6.0	6.2	6.0	6.1	6.8	6.7	6.8	7.4	6.4
Own school threshold	18.8	18.6	19.3	19.7	19.9	20.8	20.2	20.1	20.6	19.6
Market structure	7.4	7.7	7.7	8.0	7.9	7.2	7.3	7.0	7.1	7.3
GPA decile × own school threshold	26.0	26.1	25.5	24.8	25.9	25.0	24.3	24.1	23.5	25.0
GPA decile × market structure	12.1	11.6	10.5	10.5	10.0	9.6	10.3	10.1	10.3	10.9
Own school threshold × market structure	11.2	11.6	12.1	12.4	12.0	12.6	13.2	13.7	13.5	12.5
GPA decile × own school threshold × market structure	18.3	18.4	18.6	18.6	18.2	18.0	18.0	18.3	17.6	18.4
R-squared	0.525	0.540	0.570	0.577	0.608	0.579	0.563	0.566	0.556	0.558

Note: This table reports, for each graduating cohort, the Shapley decomposition of the R-squared of a regression of the potential gain on its inputs: the student's GPA decile within their school, deciles of the mean threshold of the school where they started twelfth grade, and market structure, measured by deciles of the mean and dispersion of the alternative schools' thresholds and of their number within 2 km, together with all interactions. Each row is the share of the R-squared attributable to one input or interaction, and shares sum to 100 in each column. Panel A decomposes the indicator for a positive median potential gain and Panel B the median potential gain. The last column pools all cohorts, net of year effects. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade, at schools with a 2010 threshold and SIMCE score, in the cohorts finishing twelfth grade between 2010 and 2018.

TABLE A9. Event Study for Destination School Outcomes Using the Positive Median Gain

	Lower threshold (1)	Lower mean PSU (2)	Fewer admitted to college (3)	Fewer admitted to tier 1 (4)	Lower performance (5)	Lower socioeconomic group (6)	Closer school (7)
Positive gain × 2010	0.0000	0.0001	-0.0000	0.0001	-0.0001	0.0007	0.0005
	(0.0009)	(0.0009)	(0.0008)	(0.0008)	(0.0005)	(0.0005)	(0.0007)
Positive gain × 2011	0.0023**	0.0017	0.0031***	0.0029***	0.0021***	0.0022***	0.0015*
	(0.0011)	(0.0011)	(0.0010)	(0.0011)	(0.0008)	(0.0007)	(0.0008)
Positive gain × 2012	0.0005	-0.0001	-0.0002	0.0004	-0.0005	0.0008	-0.0000
	(0.0009)	(0.0010)	(0.0009)	(0.0009)	(0.0007)	(0.0006)	(0.0007)
Positive gain × 2013	[base]	[base]	[base]	[base]	[base]	[base]	[base]
Positive gain × 2014	0.0250***	0.0246***	0.0246***	0.0253***	0.0196***	0.0210***	0.0126***
	(0.0018)	(0.0018)	(0.0018)	(0.0018)	(0.0015)	(0.0015)	(0.0013)
Positive gain × 2015	0.0005	-0.0006	0.0002	0.0005	0.0005	0.0006	-0.0008
	(0.0009)	(0.0010)	(0.0009)	(0.0009)	(0.0006)	(0.0006)	(0.0008)
Positive gain × 2016	0.0006	-0.0009	0.0005	0.0005	0.0005	0.0001	0.0001
	(0.0010)	(0.0010)	(0.0009)	(0.0008)	(0.0006)	(0.0006)	(0.0008)
Positive gain × 2017	0.0010	-0.0003	-0.0002	0.0002	0.0006	0.0007	0.0010
	(0.0009)	(0.0010)	(0.0009)	(0.0008)	(0.0006)	(0.0006)	(0.0008)
Positive gain × 2018	-0.0006	-0.0009	-0.0011	-0.0008	0.0003	0.0006	-0.0000
	(0.0010)	(0.0010)	(0.0009)	(0.0009)	(0.0006)	(0.0006)	(0.0007)
Pre-policy mean	0.0080	0.0085	0.0069	0.0067	0.0036	0.0029	0.0050
Observations	576,021	557,147	547,833	547,833	511,716	511,716	573,959

Note: This table reports the coefficients from Equation (4) for outcomes describing the destination school. Each column is a separate regression whose outcome is an indicator for switching schools during twelfth grade to a school with the characteristic in the column header, relative to the school of origin; students who did not switch are zero. Rows report the interactions of an indicator for a positive median potential gain with year dummies, relative to 2013, the omitted year. All regressions include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade, in the cohorts finishing twelfth grade between 2010 and 2018. The pre-policy mean is the mean of the outcome over the 2010–2013 cohorts. Standard errors are clustered at the school-market level, are in parentheses. Stars indicate significance at the 1% (***), 5% (**), and 10% (*) levels.

TABLE A10. Event Study for Destination School Outcomes Using the Standardized Median Gain

	Lower threshold	Lower mean PSU	Fewer admitted to college	Fewer admitted to tier 1	Lower performance	Lower socioeconomic group	Closer school
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Potential gain × 2010	-0.0002	0.0005	-0.0004	0.0001	-0.0003	0.0000	0.0002
	(0.0004)	(0.0005)	(0.0005)	(0.0004)	(0.0003)	(0.0003)	(0.0005)
Potential gain × 2011	0.0013***	0.0008	0.0014**	0.0010*	0.0009**	0.0009***	0.0011**
	(0.0005)	(0.0006)	(0.0005)	(0.0005)	(0.0004)	(0.0003)	(0.0005)
Potential gain × 2012	0.0003	0.0000	0.0002	0.0003	-0.0003	0.0003	0.0006
	(0.0004)	(0.0005)	(0.0005)	(0.0005)	(0.0004)	(0.0003)	(0.0005)
Potential gain × 2013	[base]	[base]	[base]	[base]	[base]	[base]	[base]
Potential gain × 2014	0.0150***	0.0157***	0.0165***	0.0165***	0.0132***	0.0140***	0.0086***
	(0.0011)	(0.0012)	(0.0013)	(0.0012)	(0.0011)	(0.0011)	(0.0008)
Potential gain × 2015	0.0003	0.0001	0.0003	0.0004	0.0002	0.0001	0.0004
	(0.0005)	(0.0006)	(0.0005)	(0.0005)	(0.0003)	(0.0004)	(0.0005)
Potential gain × 2016	-0.0001	-0.0006	-0.0001	0.0000	-0.0001	-0.0003	-0.0002
	(0.0004)	(0.0005)	(0.0005)	(0.0004)	(0.0003)	(0.0003)	(0.0005)
Potential gain × 2017	0.0001	0.0000	-0.0004	-0.0002	-0.0003	-0.0003	0.0008*
	(0.0004)	(0.0005)	(0.0004)	(0.0004)	(0.0003)	(0.0003)	(0.0005)
Potential gain × 2018	-0.0001	0.0000	-0.0002	-0.0002	0.0003	0.0002	0.0007
	(0.0004)	(0.0005)	(0.0005)	(0.0005)	(0.0003)	(0.0003)	(0.0004)
Pre-policy mean	0.0080	0.0085	0.0069	0.0067	0.0036	0.0029	0.0050
Observations	576,021	557,147	547,833	547,833	511,716	511,716	573,959

Note: This table reports the coefficients from Equation (4) for outcomes describing the destination school. Each column is a separate regression whose outcome is an indicator for switching schools during twelfth grade to a school with the characteristic in the column header, relative to the school of origin; students who did not switch are zero. Rows report the interactions of the standardized median potential gain with year dummies, relative to 2013, the omitted year, so coefficients are per standard deviation of the median potential gain within each cohort. All regressions include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade, in the cohorts finishing twelfth grade between 2010 and 2018. The pre-policy mean is the mean of the outcome over the 2010–2013 cohorts. Standard errors are clustered at the school-market level, are in parentheses. Stars indicate significance at the 1% (***), 5% (**), and 10% (*) levels.

TABLE A11. Event Study for College Outcomes (Positive Gain): All Programs

	Switched (1)	Applied (2)	Accepted (3)	Enrolled (4)	Dropped out (5)	Graduated (6)	Graduated on time (7)
Positive gain × 2011	0.0023*	-0.0081*	-0.0067	-0.0104**	-0.0149***	0.0074*	0.0063*
	(0.0014)	(0.0042)	(0.0043)	(0.0042)	(0.0031)	(0.0038)	(0.0033)
Positive gain × 2012	-0.0017	-0.0035	-0.0022	-0.0035	-0.0041	-0.0016	0.0037
	(0.0012)	(0.0039)	(0.0040)	(0.0040)	(0.0033)	(0.0036)	(0.0031)
Positive gain × 2013	[base]	[base]	[base]	[base]	[base]	[base]	[base]
Positive gain × 2014	0.0251***	-0.0010	-0.0013	0.0050	-0.0030	0.0089**	0.0002
	(0.0020)	(0.0040)	(0.0041)	(0.0042)	(0.0032)	(0.0038)	(0.0033)
Positive gain × 2015	-0.0017	0.0089**	-0.0041	0.0002	-0.0060*	0.0112***	0.0089**
	(0.0013)	(0.0041)	(0.0041)	(0.0042)	(0.0034)	(0.0040)	(0.0035)
Positive gain × 2016	-0.0004	-0.0066	-0.0168***	-0.0121***	-0.0018	0.0018	0.0132***
	(0.0013)	(0.0044)	(0.0043)	(0.0045)	(0.0034)	(0.0041)	(0.0036)
Pre-policy mean	0.0156	0.3219	0.2843	0.2412	0.0664	0.1184	0.0712
Observations	378,976	381,125	381,125	381,125	380,627	381,125	381,125

Note: This table reports the coefficients from Equation (5). Each column is a separate regression; the outcomes are indicators for switching schools during twelfth grade (column 1), applying to a university in the centralized system (column 2), being admitted (column 3), enrolling (column 4), dropping out of the first program (column 5), graduating from it (column 6), and graduating from it on time (column 7). Outcomes are measured in the student's first admission process and are zero for students who did not take part in one. The first program is the one the student enrolled in through that process; a student drops out if they leave it and never enroll again in a centralized university, and graduates on time if they finish within one year of its nominal length. Rows report the interactions of an indicator for a positive median potential gain with year dummies, relative to 2013, the omitted year. All regressions include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade, in the cohorts finishing twelfth grade between 2011 and 2016. The pre-policy mean is the mean of the outcome over the 2011–2013 cohorts. Standard errors are clustered at the school-market level, are in parentheses. Stars indicate significance at the 1% (***), 5% (**), and 10% (*) levels.

TABLE A12. Event Study for College Outcomes: Highly Selective Programs including County Fixed Effects

	Applied (1)	Accepted (2)	Enrolled (3)	Dropped out (4)	Graduated (5)	Graduated on time (6)
Positive gain × 2011	-0.0037	0.0019	0.0017	-0.0115***	0.0120***	0.0087***
	(0.0043)	(0.0039)	(0.0037)	(0.0026)	(0.0032)	(0.0027)
Positive gain × 2012	-0.0000	0.0039	0.0039	-0.0020	0.0002	0.0034
	(0.0039)	(0.0038)	(0.0037)	(0.0025)	(0.0030)	(0.0025)
Positive gain × 2013	[base]	[base]	[base]	[base]	[base]	[base]
Positive gain × 2014	0.0105***	0.0069*	0.0111***	-0.0007	0.0113***	0.0029
	(0.0040)	(0.0038)	(0.0036)	(0.0025)	(0.0029)	(0.0025)
Positive gain × 2015	0.0161***	-0.0014	0.0037	-0.0055**	0.0117***	0.0094***
	(0.0041)	(0.0040)	(0.0038)	(0.0027)	(0.0036)	(0.0028)
Positive gain × 2016	0.0020	-0.0074*	-0.0028	-0.0002	0.0044	0.0122***
	(0.0046)	(0.0043)	(0.0041)	(0.0027)	(0.0034)	(0.0031)
Pre-policy mean	0.2548	0.1322	0.1401	0.0343	0.0718	0.0429
Observations	380,612	380,921	378,271	379,561	379,772	380,083

Note: This table reports the coefficients from Equation (5) for highly selective programs, adding fixed effects for the county of the student's primary school. Each column is a separate regression; the outcomes are indicators for applying to a highly selective program at any preference (column 1), being admitted to one (column 2), enrolling in one (column 3), dropping out of a highly selective first program (column 4), graduating from one (column 5), and graduating from one on time (column 6). Outcomes are measured in the student's first admission process and are zero for students who did not take part in one. The first program is the one the student enrolled in through that process; a student drops out if they leave it and never enroll again in a centralized university, and graduates on time if they finish within one year of its nominal length. Highly selective programs are those in the top quartile of the mean application score of students admitted in the 2014 admission process. Rows report the interactions of an indicator for a positive median potential gain with year dummies, relative to 2013, the omitted year. All regressions also include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade, in the cohorts finishing twelfth grade between 2011 and 2016. The pre-policy mean is the mean of the outcome over the 2011–2013 cohorts. Standard errors are clustered at the school-market level, are in parentheses. Stars indicate significance at the 1% (***), 5% (**), and 10% (*) levels.

TABLE A13. Regression Discontinuity Validity at the Cutoffs of Programs with Displacement: Alternative Counterfactuals

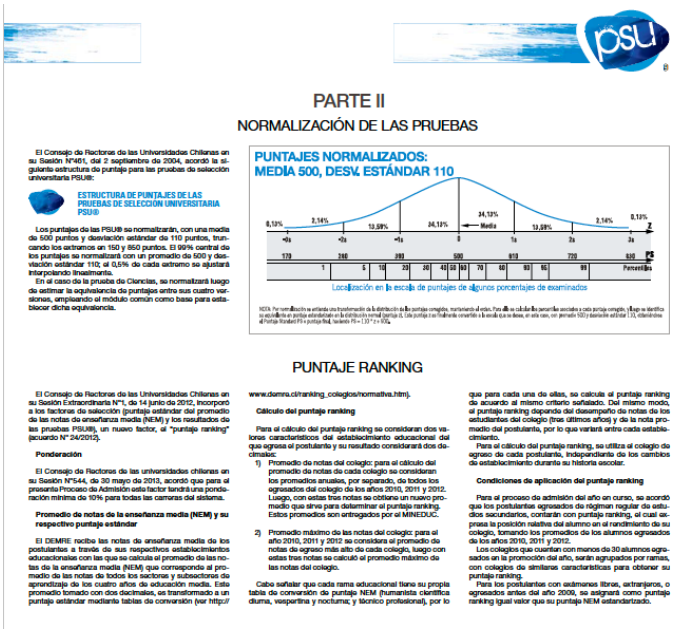
	All programs	Tier 1–2	Highly selective	High employability
Panel A: *Switches of students with a positive gain*				
Positive potential gain	0.022	0.035	0.042*	0.008
	(0.024)	(0.034)	(0.026)	(0.035)
Female	0.002	-0.004	-0.016	0.006
	(0.026)	(0.035)	(0.030)	(0.038)
Mother completed high school	0.010	0.011	0.001	0.006
	(0.016)	(0.023)	(0.019)	(0.021)
SIMCE score	-0.019	-0.015	-0.025	-0.064
	(0.033)	(0.043)	(0.029)	(0.048)
Public school	0.025*	0.031	0.035*	0.037*
	(0.017)	(0.026)	(0.021)	(0.025)
School performance, top quartile	-0.008	-0.009	0.004	0.006
	(0.022)	(0.030)	(0.024)	(0.035)
Gran Santiago	-0.017	-0.030	-0.022	0.002
	(0.024)	(0.031)	(0.026)	(0.031)
Density test (p-value)	0.633	0.826	0.614	0.291
Panel B: *Switches of students with a positive gain in Gran Santiago*				
Positive potential gain	0.032	0.037	0.040*	0.014
	(0.026)	(0.034)	(0.027)	(0.037)
Female	0.009	-0.003	-0.015	0.003
	(0.027)	(0.034)	(0.031)	(0.039)
Mother completed high school	0.005	0.011	-0.006	0.018
	(0.017)	(0.023)	(0.019)	(0.025)
SIMCE score	-0.014	-0.014	-0.019	-0.045
	(0.036)	(0.043)	(0.033)	(0.050)
Public school	0.020	0.029	0.038*	0.027
	(0.018)	(0.026)	(0.024)	(0.027)
School performance, top quartile	-0.006	-0.014	0.004	0.011
	(0.025)	(0.031)	(0.026)	(0.032)
Gran Santiago	-0.022	-0.035	-0.017	-0.001
	(0.024)	(0.031)	(0.025)	(0.034)
Density test (p-value)	0.892	0.906	0.658	0.385

Note: This table reports regression discontinuity estimates of the change in applicant characteristics at the admission cutoffs of programs where at least one student lost a seat to switching in the 2015 admission process. In Panel A only switchers with a positive potential gain are treated as strategic, and in Panel B only those in Gran Santiago with a positive potential gain. Columns use the cutoffs of all such programs and of those at tier 1–2 universities, highly selective programs and high-employability programs. The sample is applications by students who did not switch schools in twelfth grade. Each row is a separate regression of the characteristic on the distance between the application score and the cutoff, within 50 points of it, with a triangular kernel and the mean squared error optimal bandwidth of Calonico et al. (2014). All characteristics are defined only for the class of 2014. The last row of each panel reports the p-value of the density test of Cattaneo et al. (2020). Estimates are conventional; standard errors are robust bias-corrected and clustered by student. Stars indicate significance at the 1% (***), 5% (**), and 10% (*) levels based on robust bias-corrected p-values, so they need not match the ratio of the conventional estimate to its standard error.

TABLE A14. Effect of a Seat Offer at the Cutoffs of Programs with Displacement: All Programs and High-Earnings Programs

	Programs with displaced students when the switchers are:		
	All	With a gain	With a gain, in Gran Santiago
	(1)	(2)	(3)
Panel A: *All affected programs*			
Enrolled in the program	0.491***	0.504***	0.521***
	(0.021)	(0.022)	(0.022)
Enrolled	0.046***	0.046***	0.045***
	(0.013)	(0.014)	(0.014)
Dropped out, first program	0.056***	0.061***	0.062***
	(0.016)	(0.017)	(0.019)
Graduated, first program	-0.020	-0.028*	-0.020
	(0.018)	(0.019)	(0.019)
Graduated on time, first program	0.001	-0.004	0.001
	(0.013)	(0.014)	(0.014)
Graduated, any program	-0.017	-0.022	-0.024
	(0.016)	(0.016)	(0.018)
Panel B: *High-earnings programs*			
Accepted, high-earnings	0.395***	0.393***	0.393***
	(0.014)	(0.015)	(0.014)
Enrolled, high-earnings	0.108***	0.106***	0.110***
	(0.019)	(0.019)	(0.020)
Dropped out, high-earnings first program	0.030**	0.034**	0.027*
	(0.015)	(0.015)	(0.017)
Graduated, high-earnings first program	0.044	0.036	0.051*
	(0.022)	(0.022)	(0.022)
Graduated on time, high-earnings first program	0.015	0.010	0.013
	(0.019)	(0.020)	(0.021)

Note: This table reports regression discontinuity estimates of the effect of a seat offer at the admission cutoffs of programs where at least one student lost a seat to switching. Panel A uses the cutoffs of all such programs and Panel B those of high-earnings programs. Columns use the programs with displaced students when the switchers are all students who switched schools in twelfth grade, those with a positive potential gain, and those with a positive potential gain in Gran Santiago. The sample is applications in the 2014 and 2015 admission processes, at or above the preference where the student was admitted, by students who did not switch schools in twelfth grade. Outcomes are measured in the admission process of the application; the first program is the one the student enrolled in through that process. High-earnings programs are those in the top quartile of graduates' average monthly income four years after graduation, among 2013–2015 graduates. Each row is a separate regression of the outcome on the distance between the application score and the cutoff, within 50 points of it, with a triangular kernel, the mean squared error optimal bandwidth of Calonico et al. (2014), and weights equal to the number of displaced students in the program. Estimates are conventional; standard errors are robust bias-corrected and clustered by student. Stars indicate significance at the 1% (***), 5% (**), and 10% (*) levels based on robust bias-corrected p-values, so they need not match the ratio of the conventional estimate to its standard error.



PARTE II

NORMALIZACIÓN DE LAS PRUEBAS

El Consejo de Rectores de las Universidades Chilenas en su Sesión N°401, del 2 septiembre de 2004, acordó la siguiente estructura de puntaje para las pruebas de selección universitaria PSU®:

ESTRUCTURA DE PUNTAJES DE LAS PRUEBAS DE SELECCIÓN UNIVERSITARIA PSU®

Los puntajes de las PSU® se normalizarán, con una media de 500 puntos y desviación estándar de 110 puntos, truncando los extremos en 150 y 850 puntos. El 99% central de los puntajes se normalizará con un promedio de 500 y desviación estándar 110; el 0,5% de cada extremo se ajustará interpolando linealmente.

En el caso de la prueba de Ciencias, se normalizará luego de estimar la equivalencia de puntajes entre sus cuatro versiones, ampliando el módulo común como base para establecer dicha equivalencia.

PUNTAJES NORMALIZADOS: MEDIA 500, DESV. ESTÁNDAR 110

0,13% 2,14% 13,59% 34,13% 34,13% Media 13,59% 2,14% 0,13%

-3s -2s -1s 0 1s 2s 3s

170 280 390 500 610 720 830 PS

1 5 10 20 40 60 80 90 95 99 Percentiles

Localización en la escala de puntajes de algunos porcentajes de examinados

PUNTAJE RANKING

El Consejo de Rectores de las Universidades Chilenas en su Sesión Extraordinaria N°1, de 14 junio de 2012, incorporó a los factores de selección (puntaje estándar del promedio de las notas de enseñanza media (NEM) y los resultados de las pruebas PSU®), un nuevo factor, el "puntaje ranking" (acuerdo N° 24/2012).

Ponderación

El Consejo de Rectores de las universidades chilenas en su Sesión N°544, de 30 mayo de 2013, acordó que para el presente Proceso de Admisión este factor tendrá una ponderación mínima de 10% para todas las carreras del sistema.

Promedio de notas de la enseñanza media (NEM) y su respectivo puntaje estándar

El DEMRE recibe las notas de enseñanza media de los postulantes a través de sus respectivos establecimientos educacionales con las que se calcula el promedio de las notas de la enseñanza media (NEM) que corresponde al promedio de las notas de todos los sectores y subsectores de aprendizaje de los cuatro años de educación media. Este promedio tomado con dos decimales, es transformado a un puntaje estándar mediante tablas de conversión (ver http://www.demre.cl/ranking_colegios/normativa.htm).

Cálculo del puntaje ranking

Para el cálculo del puntaje ranking se consideran dos valores característicos del establecimiento educacional del que egresa el postulante y su resultado considerará dos dimensiones:

1) Promedio de notas del colegio: para el cálculo del promedio de notas de cada colegio se consideran los promedios anuales, por separado, de todos los egresados del colegio de los años 2010, 2011 y 2012. Luego, con estas tres notas se obtiene un nuevo promedio que sirve para determinar el puntaje ranking. Estos promedios son entregados por el MINEDUC.

2) Promedio máximo de las notas del colegio: para el año 2010, 2011 y 2012 se considera el promedio de notas de egreso más alto de cada colegio, luego con estas tres notas se calculó el promedio máximo de las notas del colegio.

Cabe señalar que cada rama educacional tiene su propia tabla de conversión de puntaje NEM (humanista científica diurna, vespertina y nocturna; y técnico profesional), por lo que para cada una de ellas, se calcula el puntaje ranking de acuerdo al mismo criterio señalado. Del mismo modo, el puntaje ranking dependa del desempeño de notas de los estudiantes del colegio (tres últimos años) y de la nota promedio del postulante, por lo que variará entre cada establecimiento.

Para el cálculo del puntaje ranking, se utiliza el colegio de egreso de cada postulante, independiente de los cambios de establecimiento durante su historia escolar.

Condiciones de aplicación del puntaje ranking

Para el proceso de admisión del año en curso, se acordó que los postulantes egresados de régimen regular de estudios secundarios, contarán con puntaje ranking, el cual expresa la posición relativa del alumno en el rendimiento de su colegio, tomando los promedios de los alumnos egresados de los años 2010, 2011 y 2012.

Los colegios que cuentan con menos de 30 alumnos egresados en la promoción del año, serán agrupados por ramas, con colegios de similares características para obtener su puntaje ranking.

Para los postulantes con exámenes libres, extranjeros, o egresados años del año 2009, se asignará como puntaje ranking igual valor que su puntaje NEM estandarizado.

A. 2013 Admission Process Guidelines

PARTE IV

PUNTAJE RANKING

El CRUCH en su Sesión Extraordinaria N°1, de 14 junio de 2012, incorporó a los factores de selección (puntaje NEM y los resultados de las pruebas PSU®), un nuevo factor, denominado Puntaje Ranking de Notas (acuerdo N° 24/2012).

El Puntaje Ranking expresa la posición relativa del alumno con referencia al rendimiento de su colegio, tomando los promedios de notas de los alumnos egresados del correspondiente establecimiento los años 2011, 2012 y 2013.

CÁLCULO DEL PUNTAJE RANKING

Para el cálculo del Puntaje Ranking se consideran dos valores característicos del establecimiento educacional del que egresa el postulante (independiente de su año de egreso). Estos dos valores característicos son:

1) Promedio de notas del colegio (N_C): para el cálculo del promedio de notas de cada establecimiento educacional se consideran los promedios anuales, por separado, de todos los egresados del establecimiento educacional de los años 2011, 2012 y 2013 (estos promedios son proporcionados por el MINEDUC). Luego, con estas tres notas se obtiene un nuevo promedio que sirve para determinar el Puntaje Ranking.

2) Promedio máximo de las notas del colegio (Max_C): para el cálculo del promedio máximo histórico de notas se considera el promedio de notas de egreso más alto de cada establecimiento educacional por cada año (2011, 2012 y 2013). Luego con estas tres notas se calcula el promedio máximo histórico de las notas del establecimiento educacional.

Ambos valores se calcularán con dos decimales sin aproximación.

El Ranking de Notas se calcula como una bonificación al puntaje NEM cuando el estudiante obtiene un promedio de notas superior al promedio histórico de su establecimiento educacional. Así:

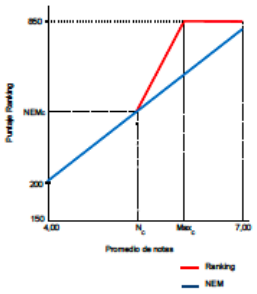
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B. 2014 Admission Process Guidelines

- Los estudiantes con Notas de Enseñanza Media igual o inferior al promedio histórico de su establecimiento, obtienen un puntaje Ranking de Notas igual a su puntaje NEM.
- Los estudiantes con Notas de Enseñanza Media por sobre el promedio histórico de su establecimiento, obtienen un puntaje más alto. Este puntaje se calcula linealmente, con un máximo de 850 puntos cuando el promedio de notas iguala al promedio máximo histórico del establecimiento.
- Los estudiantes con Notas de Enseñanza Media por sobre el promedio máximo histórico de su establecimiento obtienen todos 850 puntos.

En la siguiente figura se grafica el procedimiento previamente descrito:

Puntaje Ranking CRUCH



C. 2014 Admission Process Guidelines: Graphical Explanation

FIGURE A1. Presentation of the Relative Ranking Score in the Official Admission Guidelines, 2013 and 2014 Processes

Note: This image reproduces the sections describing the relative ranking score in the official guidelines published by the centralized admission system for the 2013 and 2014 admission processes. Panel A shows the 2013 guidelines, where the ranking score is described in prose within Part II, "Normalización de las Pruebas", alongside the normalization of test scores. Panels B and C show the 2014 guidelines, where the ranking score appears as its own section, Part IV, "Puntaje Ranking". The 2014 version states separately how the score is computed for students below their school's historic average, between that average and the school's historic maximum, and above the maximum, and includes a figure plotting the ranking score against a student's grade average. The two documents can be dated from the school averages they reference: the 2013 guidelines use cohorts graduating in 2010, 2011 and 2012, and the 2014 guidelines use 2011, 2012 and 2013. The 2014 guidelines were released in 2014.

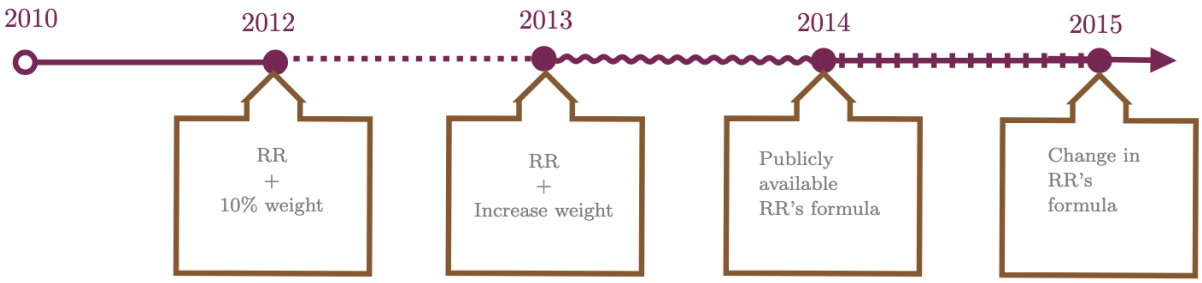


FIGURE A2. Relative Ranking Policy's Timeline

Note: This figure shows the changes to the RR policy over time. In 2012 the RR was introduced with a weight of 10%, taken from the GPA weight. In 2013 universities raised the RR weight, taking it from the GPA or the PSU weights depending on the university. In 2014 the centralized admission system made the formula public, and weights barely changed. In 2015 the RR formula was readjusted.

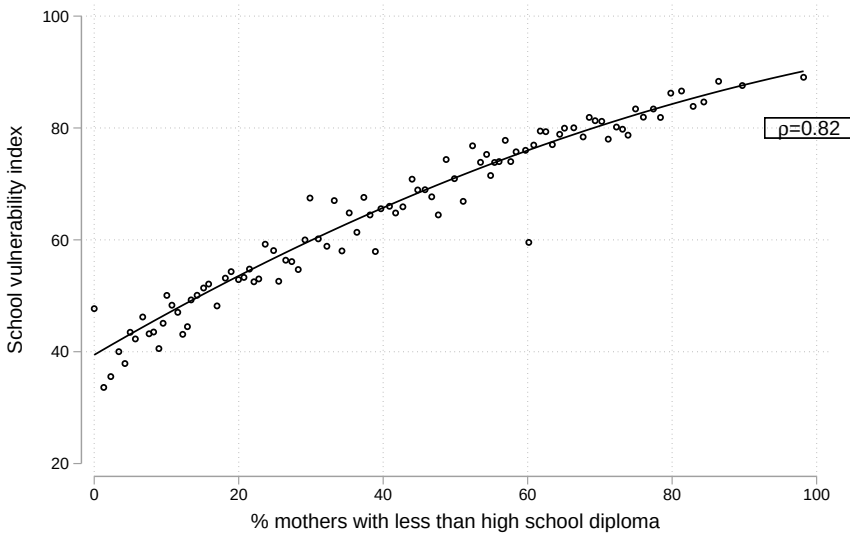
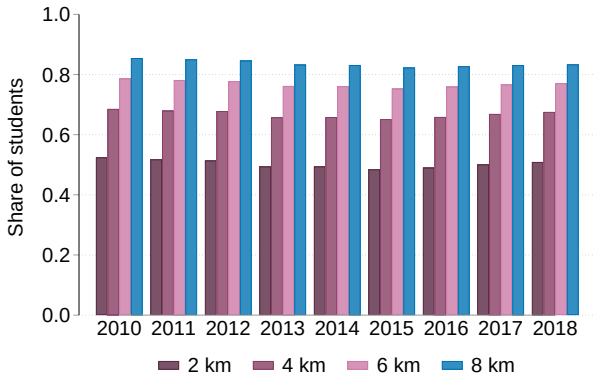
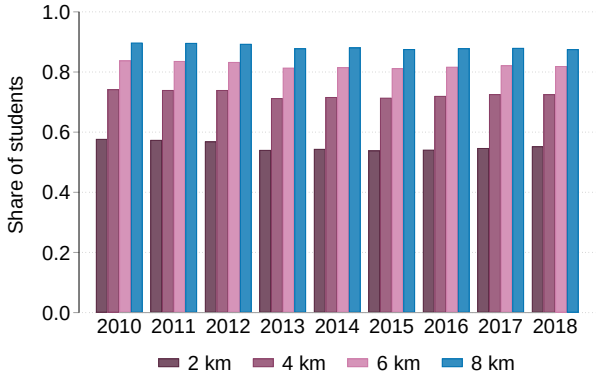


FIGURE A3. School Vulnerability Index and Students' SES

Note: This figure presents the association between the share of a school's students whose mother did not complete high school and the school vulnerability index, IVE, for the class of 2010. Mother's education comes from the PSU registration form and the SIMCE parents' survey. Each dot is the mean IVE within one of 95 equal-sized bins of that share; the line is a quadratic fit. ρ is the correlation across 2,125 schools in Chile.



A. Main sample



B. Same county in primary and secondary school

FIGURE A4. Distance Between Primary and Twelfth-Grade School

Note: This figure shows, for each cohort in twelfth grade between 2010 and 2018, the share of students whose school at the start of twelfth grade lies within 2, 4, 6 and 8 km of their primary school, the first one in which they are observed. Distances are geodesic, and shares are among students whose two schools are at most 50 km apart. Panel A uses students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade; Panel B restricts them to students who lived in the same county at both times.

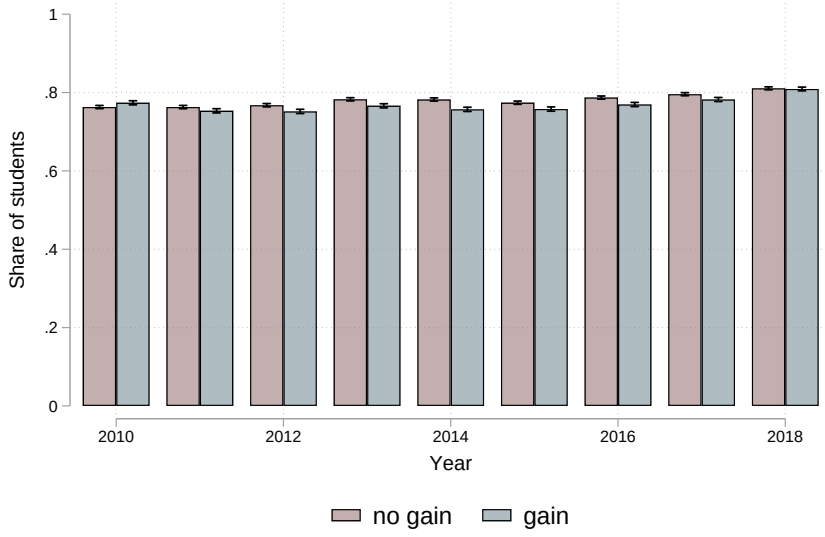


FIGURE A5. Share of Students Living in the Same County During Primary and Secondary School

Note: This figure shows, for each cohort in twelfth grade between 2010 and 2018, the share of students who lived in the same county in primary school, the grade in which they are first observed, and at the start of twelfth grade, separately for students with and without a positive median potential gain. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade. Bars are yearly shares and vertical lines 95% confidence intervals.

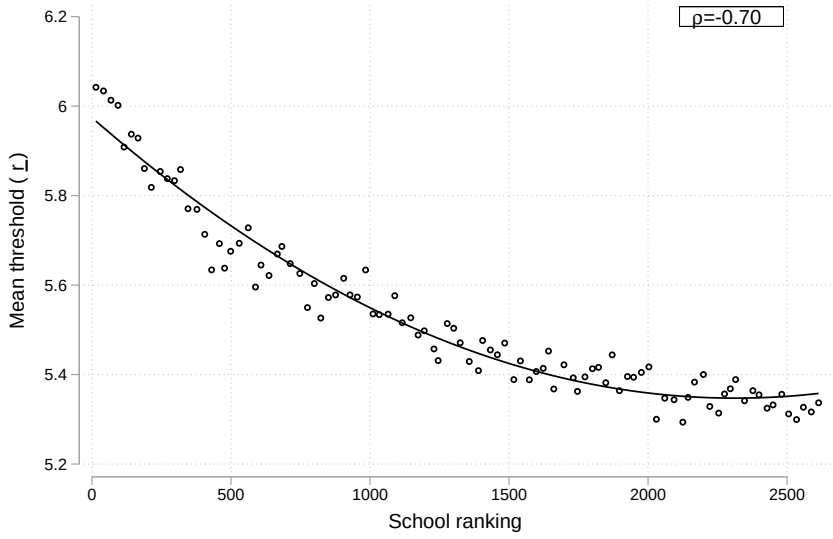


FIGURE A6. School Mean Threshold & School Ranking

Note: This figure presents the associations between school ranking (x-axis) measured using the average on standardized school tests in grade 10 in 2010 and school school mean threshold measured as the average GPA from the three previous cohorts (y-axis) between 2010-2013.

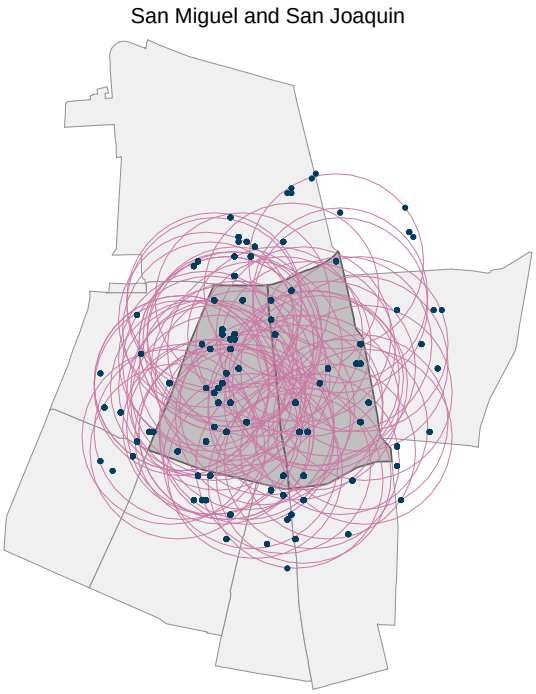


FIGURE A7. School Markets for Two Adjacent Counties

Note: This figure shows the 2 km school markets around the primary schools located in San Miguel, left, and San Joaquín, right, attended by students in the 2013 twelfth-grade cohort. Each circle is the market of one primary school, and dots mark the high schools within those markets. The two counties are in dark gray and the surrounding counties in light gray.

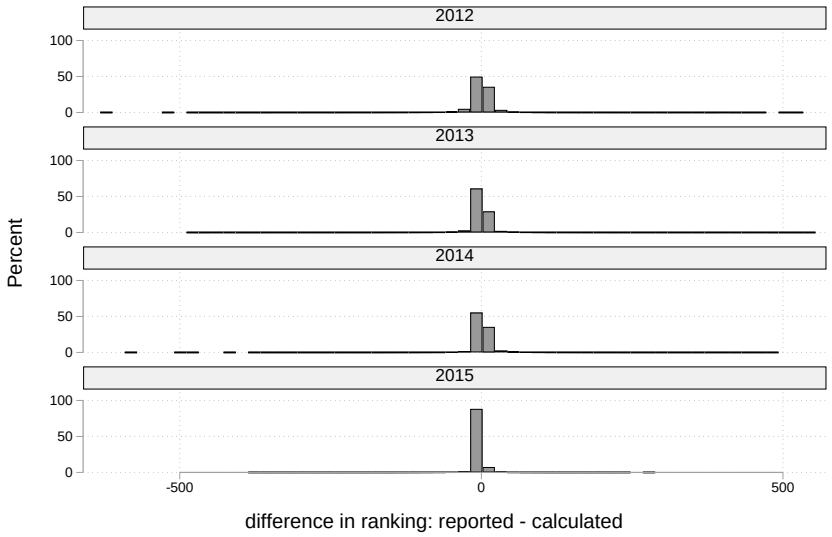


FIGURE A8. Difference between Reported Ranking Score and Calculated Ranking Score

Note: This figure shows, for each graduating cohort from 2012 to 2015, the distribution of the difference between the ranking score reported by DEMRE and the one computed with the formula in Section 2.3, in points of the 208–850 admission score scale. The sample is students in Chile who graduated from high school and sat the PSU for the first time the following year. The computed score uses the school-track thresholds built from the three previous graduating cohorts and the official GPA conversion table for the student's track.

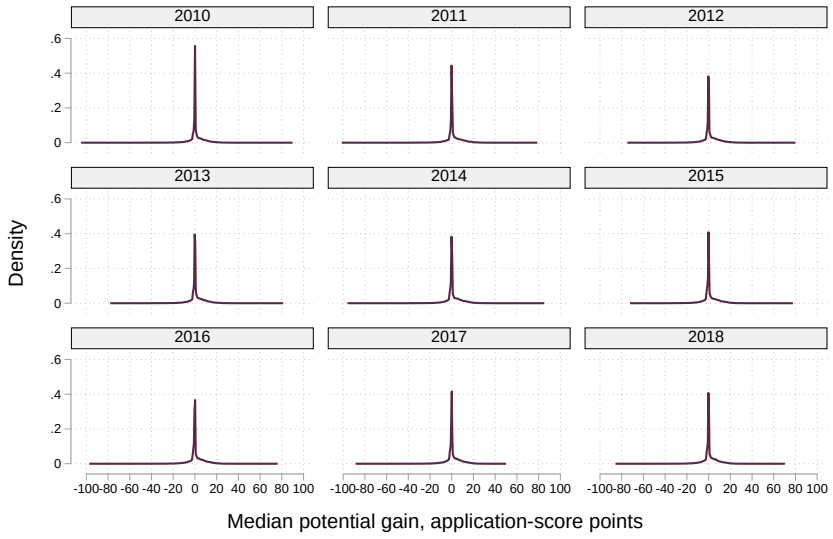


FIGURE A9. Distribution of the Median Potential Gain by Year, Gran Santiago

Note: This figure shows, for each cohort in twelfth grade between 2010 and 2018, the kernel density of the median potential gain in application-score points, the gain in the RR score times the 25% RR weight. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade.

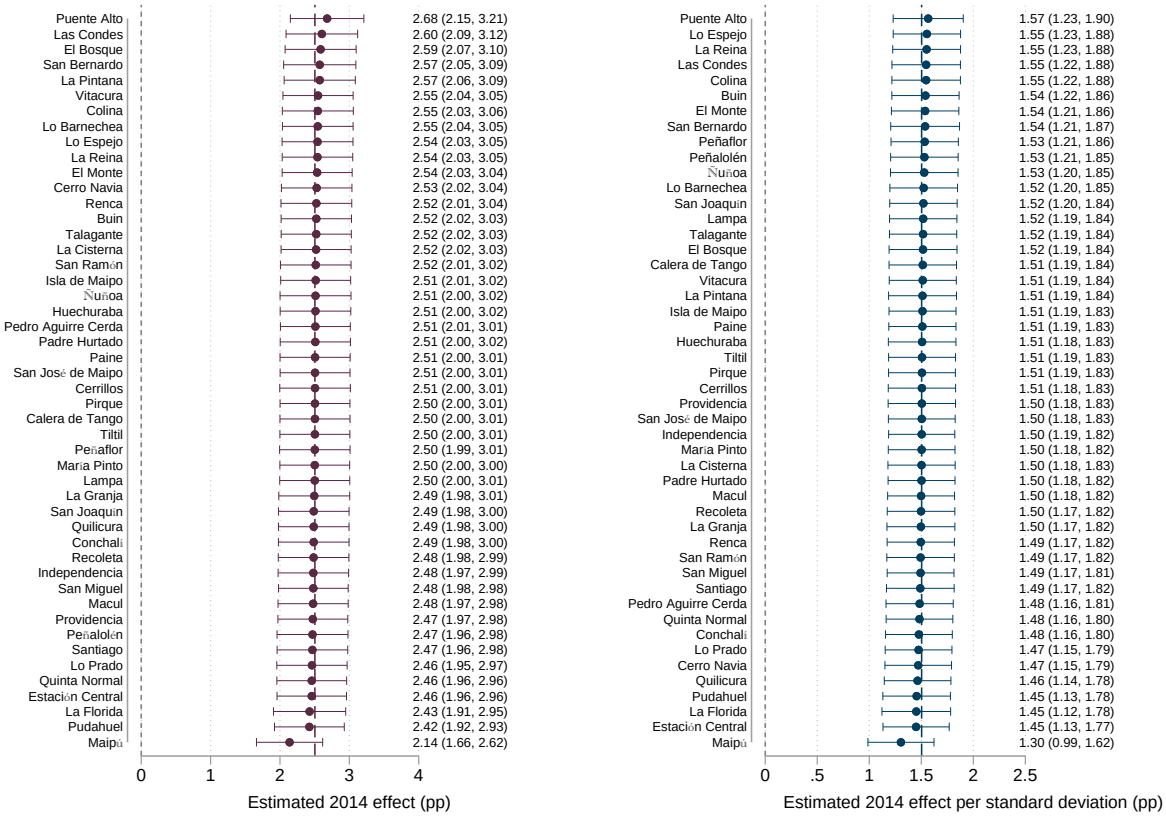


A. Positive median gain

B. Standardized median gain

FIGURE A10. Impact of the RR Formula on Switching at the Beginning of Twelfth Grade

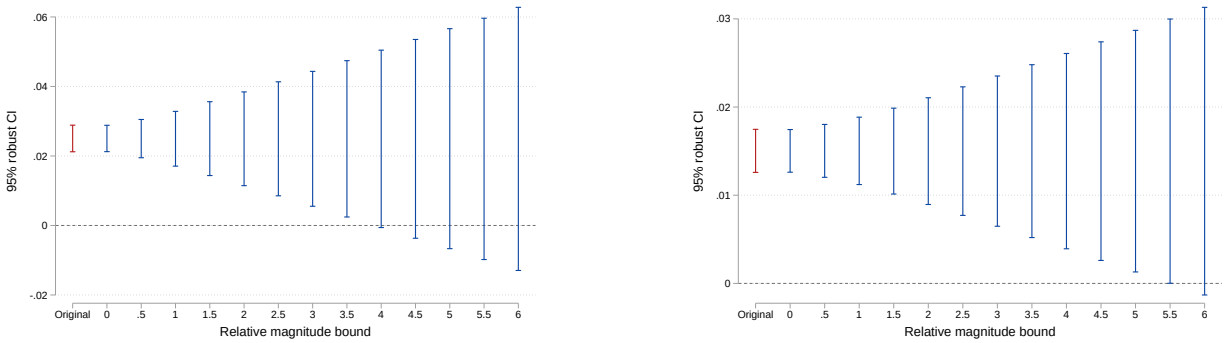
Note: This figure shows the coefficients from Equation (4) for Gran Santiago, where the outcome is an indicator for switching schools at the beginning of twelfth grade. In Panel A the key regressors are the interactions of an indicator for a positive median potential gain with year dummies; in Panel B, the interactions of the standardized median potential gain with year dummies. Both panels include fixed effects for the school attended at the end of eleventh grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade. Coefficients are relative to 2013, the omitted year; in Panel B they are per standard deviation of the median potential gain within each cohort. Standard errors are clustered at the school-market level, and the bars are 99% confidence intervals.



A. Positive Gain B. Standardized Gain

FIGURE A11. Leave-One-County-Out Estimates

Note: This figure shows the 2014 coefficient from Equation (4), in percentage points, re-estimated 48 times, each time excluding the students whose primary school is in one county of Gran Santiago; rows are named after the excluded county. The outcome is an indicator for switching schools during twelfth grade. In Panel A the key regressor is a positive median potential gain, and in Panel B the standardized median potential gain, interacted with year dummies. All regressions include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade. Dots are estimates and horizontal lines 99% confidence intervals, printed at the right; the dashed line is the full-sample estimate and the grey line zero. Standard errors are clustered at the school-market level.



A. Positive gain

B. Standardized gain

FIGURE A12. Sensitivity of the 2014 Estimate to Differential Trends

Note: This figure shows the sensitivity of the 2014 coefficient from Equation (4) to violations of parallel trends, following Rambachan and Roth (2023). Each bar is the 95% robust confidence interval for that coefficient when the post-disclosure difference in trends may be up to $\bar{M}$ times the largest year-on-year difference before it; the leftmost bar is the original interval. Panel A uses a positive median potential gain and Panel B the standardized median potential gain. Both include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade. Standard errors are clustered at the school-market level. The 2014 estimate stays different from zero up to $\bar{M} = 3.9$ in Panel A and 5.5 in Panel B.

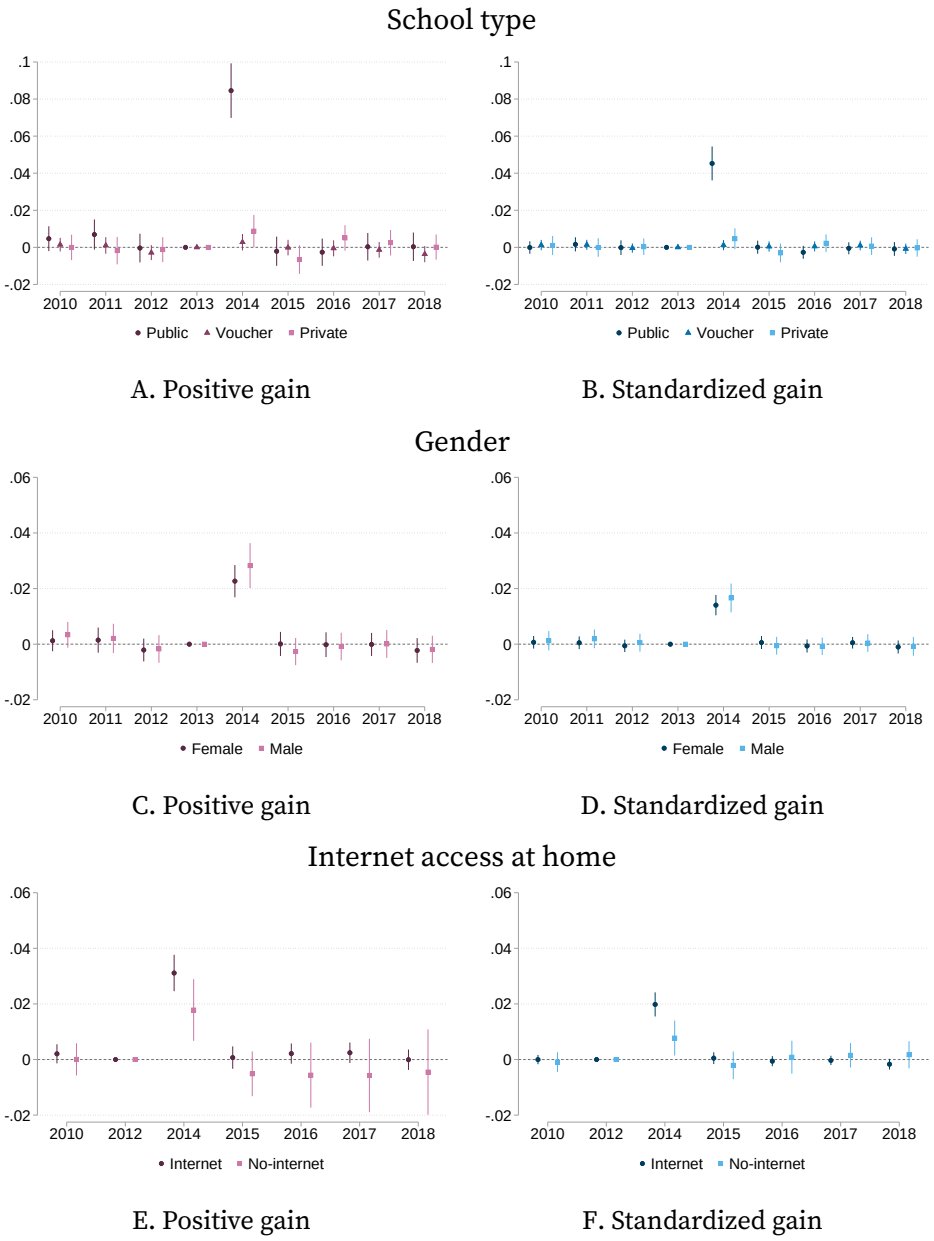
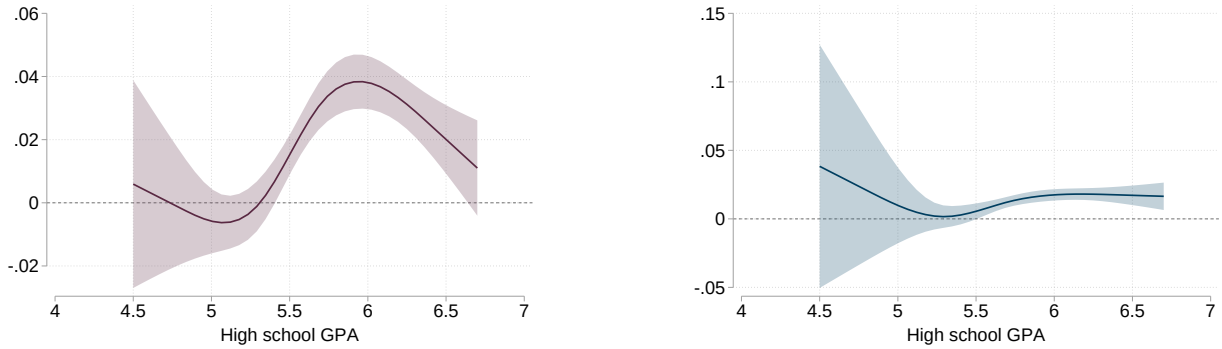


FIGURE A13. Heterogeneous Impact of the Disclosure on School Switching

Note: This figure shows the coefficients from Equation (4) estimated separately by group, where the outcome is an indicator for switching schools during twelfth grade. In the left panels the key regressors are the interactions of an indicator for a positive median potential gain with year dummies; in the right panels, the interactions of the standardized median potential gain with year dummies, so coefficients are per standard deviation of the median potential gain within each cohort. The top panels split students by the funding of their school at the start of twelfth grade: public, circles; voucher, triangles; private, squares. The middle panels split them by sex: female, circles; male, squares. The bottom panels split them by internet access at home: with access, circles; without, squares. Internet access comes from the tenth-grade SIMCE parents' survey, which was not administered to the cohorts finishing twelfth grade in 2011 and 2013, so those panels exclude both cohorts and omit 2012; the other panels use the cohorts finishing twelfth grade between 2010 and 2018 and omit 2013. All regressions include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade. Standard errors are clustered at the school-market level, and the bars are 99% confidence intervals.



A. Positive gain

B. Standardized gain

FIGURE A14. Impact of the Disclosure on School Switching by GPA

Note: This figure shows how the 2014 effect of the disclosure on switching schools during twelfth grade varies with high school GPA. It comes from one estimation of Equation (4) in which the interactions of the treatment with year dummies are further interacted with a restricted cubic spline in GPA, with knots at the 5th, 35th, 65th and 95th percentiles. Lines are the implied 2014 effect at each GPA and shaded areas 99% confidence intervals. Panel A uses an indicator for a positive median potential gain and Panel B the standardized median potential gain. The regressions include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade, with GPA between its 1st and 99th percentiles. Standard errors are clustered at the school-market level.

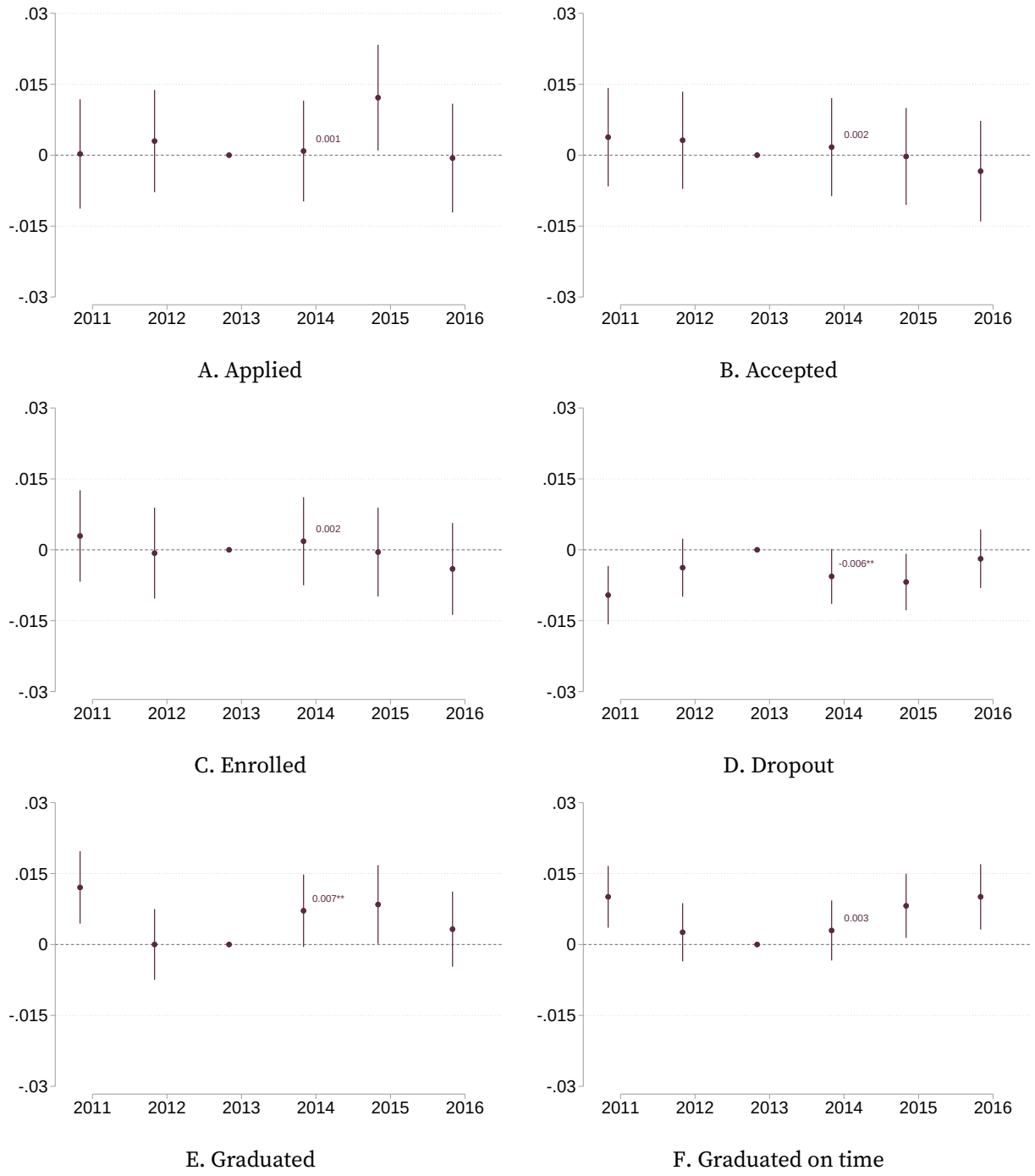


FIGURE A15. Impact of Disclosure on High-Earnings Programs Outcomes

Note: This figure shows the coefficients from Equation (5) for high-earnings programs, using each student's first admission process. The outcomes are indicators for listing such a program among the first three preferences in Panel A, being admitted to one in Panel B, and enrolling in one in Panel C; and, for that first program, dropping out of college in Panel D, graduating from it in Panel E, and graduating from it within one year of its nominal length in Panel F. Students who never applied or enrolled are zero. The key regressors are the interactions of an indicator for a positive median potential gain with year dummies. High-earnings programs are those in the top quartile of graduates' average monthly income four years after graduation, among those who graduated between 2013 and 2015. All panels include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade, in the cohorts finishing twelfth grade between 2011 and 2016. Coefficients are relative to 2013, the omitted year. Standard errors are clustered at the school-market level, and the bars are 99% confidence intervals. The label reports the 2014 coefficient; stars indicate significance at the 1% (***), 5% (**), and 10% (*) levels.

Appendix B. Model Equilibrium, Derivations, and Proofs

Equilibrium and Policy Effects

Students' GPAs in schools H and L, before any switching, follow distributions $F_H(g)$ and $F_L(g)$, respectively. Let g_e^* denote the GPA that yields a score exactly equal to AS^* in school e. Two conditions characterize the equilibrium. First, the centralized system uses a single ranking with no quotas, so the cutoff is unique across schools:

(Unique Score) $$AS_L(g_L^*) \;=\; AS_H(g_H^*) \;=\; AS^*.$$

Second, the total mass of students admitted to the selective college must equal the available capacity:

(Capacity) $$\mu_L\big[1 - F_L(g_L^*)\big] \;+\; \mu_H\big[1 - F_H(g_H^*)\big] \;=\; K.$$

I now characterize the equilibrium under three scenarios.

PROPOSITION A1 (Equilibrium before the policy). *Under Equation* (1)*, scores do not depend on school, so* $g_L^* = g_H^* = g_0^*$*. No student has an incentive to switch schools.*

PROPOSITION A2 (Equilibrium after the policy without switching). *When the policy is implemented but students cannot switch, schools H and L generically have different GPA cutoffs for admission to the selective college:* $g_H^* \neq g_L^*$*. The share of students admitted to the selective college increases in the school with the lower cutoff and decreases in the other. The equilibrium cutoff rises:* $AS_1^* > AS_0^*$.

To build intuition, assume $\underline{r}_L < \underline{r}_H$ and $\bar{r}_L \leq \bar{r}_H$, as in Figure 3. Then $AS_L(g_i) \geq AS_H(g_i)$ for all g_i, so by (Unique Score) we have $g_L^* < g_H^*$: a lower GPA suffices at school L. The capacity constraint then implies that school L gains seats and school H loses them. Under $F_H = F_L$ this follows immediately; under general distributions, the result holds whenever the score function in school L weakly dominates that in school H, which the threshold ordering guarantees. The formal proof is below.

PROPOSITION A3 (Equilibrium after the policy with switching). *When the policy is implemented and students are allowed to switch, the equilibrium cutoff rises further:* $AS_0^* \leq AS_1^* \leq AS_2^*$*. The effect of the policy on the composition of students in the selective college depends on how costly it is for students to switch.*

When switching is allowed, students satisfying (3) relocate to the school where the score function is more generous, pushing AS^* upward. When $\tilde{c}_i = 0$ for all students, every student with a positive gain switches and the policy's intended redistribution is fully undone. Students displaced from the selective college are pushed into the less selective college. When $\tilde{c}_i$ is prohibitively high, the outcome coincides with Proposition A2. The formal proof is below. Note that with multiple selective colleges, full attenuation generically fails even when switching costs are zero, because switchers who upgrade across tiers couple the capacity constraints. The single-college case therefore provides an upper bound on the degree of policy undoing.

Proof of Proposition A1

Before the policy, $AS(g_i) = \gamma g_i + \theta$ does not depend on school. By (Unique Score), $\gamma g_L^* + \theta = \gamma g_H^* + \theta$, so $g_L^* = g_H^* \equiv g_0^*$. Since $AS_e(g_i) = AS_s(g_i)$ for all g_i, switching can never move a student from below the cutoff to above it, and no student switches. The capacity constraint (Capacity) becomes

$$\mu_L\big[1 - F_L(g_0^*)\big] + \mu_H\big[1 - F_H(g_0^*)\big] = K,$$

which pins down g_0^*. School e's share of students in the selective college is $\mu_e[1 - F_e(g_0^*)]/K$, which depends on the school-specific distribution F_e. □

Proof of Proposition A2

After the policy, under $\underline{r}_L < \underline{r}_H$ and $\bar{r}_L \le \bar{r}_H$, we have $AS_L(g) \ge AS_H(g)$ for all g, with strict inequality for $g \in (\underline{r}_L, \underline{r}_H)$. The unique-score condition (Unique Score) requires $AS_L(g_L^*) = AS_H(g_H^*) = AS^*$. Since $AS_L(g) \ge AS_H(g)$, a lower GPA suffices in school L: $g_L^* \le g_H^*$, with strict inequality whenever AS^* falls in the range where AS_L strictly dominates AS_H.

With $g_L^* < g_H^*$, school L's mass of students in the selective college is $\mu_L[1 - F_L(g_L^*)]$. Before the policy, it was $\mu_L[1 - F_L(g_0^*)]$. Since $g_L^* < g_0^*$ (shown below), $F_L(g_L^*) < F_L(g_0^*)$, so school L's share of the selective college increases. The capacity constraint is unchanged at K, so school H's share must decrease. Students displaced from the selective college move to the less selective college.

Before the policy, the cutoff solves $\mu_L[1 - F_L(g_0^*)] + \mu_H[1 - F_H(g_0^*)] = K$ with a common GPA threshold. After the policy, $g_L^* < g_0^* < g_H^*$ and the score function AS_L weakly dominates AS (the pre-policy function) in the bonus region. Therefore $AS_1^* = AS_L(g_L^*) > AS(g_0^*) = \gamma g_0^* + \theta = AS_0^*$. □

Proof of Proposition A3

When switching is permitted, some students from school H relocate to school L and obtain $AS_L(g_i) > AS_H(g_i)$. These switchers increase the mass of students with scores above any given threshold, which (holding K fixed) requires AS^* to rise weakly. If no student switches, the cutoff remains at its Proposition A2 value. Hence $AS_0^* \le AS_1^* \le AS_2^*$.

Set $\tilde{c}_i = 0$ for all i. All students from school H with $g_i \in [g_L^*, g_H^*)$ switch to school L, because $AS_L(g_i) \ge AS^* > AS_H(g_i)$. After switching, every student with $g_i \ge g_L^*$—regardless of original school—has a score at or above AS^*: either they were already in the selective college from school H (if $g_i \ge g_H^*$) or they switched to school L (if $g_L^* \le g_i < g_H^*$).

The capacity constraint after switching becomes:

$$\mu_L\big[1 - F_L(g_L^*)\big] + \mu_H\big[1 - F_H(g_L^*)\big] = K.$$

Before the policy, the constraint was $\mu_L[1 - F_L(g_0^*)] + \mu_H[1 - F_H(g_0^*)] = K$. Both equations have the same form with a common GPA cutoff, so $g_L^* = g_0^*$. The composition of students in the selective college—$\mu_e[1 - F_e(g_0^*)]$ from school e—is therefore identical to the pre-policy allocation, regardless

of whether $F_H = F_L$. Students displaced from the selective college by switchers are not excluded from higher education; they move to the less selective college.

In general, if a fraction q of students with a positive gain have switching costs exceeding U_i^S, the policy's compositional effect is proportional to q: the composition lies between the pre-policy allocation ($q = 0$) and the no-switching outcome of Proposition A2 ($q = 1$). □

Appendix C. Analysis Using Different Treatment Measures and School Market Characteristics

This appendix examines whether the results depend on two choices: how the incentive a student faces is measured, and the market over which that incentive is computed. The first part reports the event study under alternative definitions of the potential gain, binary and continuous. The second describes how many schools students have within reach and estimates the response separately for students with few and many alternatives.

C.1. Alternative Definitions of the Potential Gain

Figure A16 presents the main event-study estimates for the probability of switching schools during twelfth grade, from Equation (4), under alternative definitions of the treatment variable. The baseline specification (used throughout the paper) uses the median potential gain across schools in the student's local market.

As described in Section 4, the baseline discrete treatment variable is a binary indicator equal to one if student i has a positive median potential gain across schools in their local market. Each of the binary alternatives is a threshold on the fraction of the schools in the student's market that would raise their ranking, and they differ only in how large that fraction must be. The baseline median potential gain requires at least half. I consider four binary alternatives: (i) *maximum potential gain*, $Max > 0$, equal to one if the highest potential gain in the student's market is positive, which requires at least one school; (ii) *75th percentile*, $p75 > 0$, equal to one if the 75th percentile of the potential gains in their market is positive, which requires at least a quarter of the schools; (iii) *mean potential gain*, $Mean > 0$, equal to one if the average potential gain is positive; and (iv) *25th percentile*, $p25 > 0$, equal to one if the 25th percentile of the potential gains is positive, which requires at least three quarters of the schools in the market. Figure A16, Panel A, presents the event study for the discrete measures.

The results under any of the measures follow a similar shape: flat before 2014, a sharp increase in switching in 2014, when the policy's formula was disclosed in the guidelines, and a return to zero thereafter. What changes is the magnitude, which rises as the definition requires more of the market to be worth moving to, while the treated group shrinks. The 2014 estimate is 1.5 pp under the maximum, which treats 89.7% of students; 1.8 pp under the 75th percentile, with 65.5% of students treated; 1.5 pp under the mean (70.3%), 2.5 pp under the baseline median (38.4%); and 4.1 pp under the 25th percentile (17.7%). Joint tests of each measure against the baseline report



A. Positive gain

B. Standardized gain

FIGURE A16. Impact under Alternative Treatment Definitions

Note: This figure shows the coefficients from Equation (4) under alternative treatment definitions, where the outcome is an indicator for switching schools during twelfth grade. In Panel A the treatment is an indicator that a statistic of the potential gains across the schools in the student's market is positive: the median, the baseline, circles; the maximum, triangles; the 75th percentile, squares; the mean, diamonds; and the 25th percentile, hollow triangles. In Panel B it is the median potential gain, circles, or the share of schools in the market with a positive gain, triangles, each standardized within cohort. All regressions include fixed effects for the school at the start of twelfth grade and for within-school GPA deciles. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade. Coefficients are relative to 2013, the omitted year. Standard errors are clustered at the school-market level, and the bars are 99% confidence intervals.

differences of −0.96 pp for the maximum, −0.67 pp for the 75th percentile, −0.98 pp for the mean, and +1.59 pp for the 25th percentile, all significant at a 1% level. Since the definitions are nested, a looser one moves students from the comparison group into the treated group, so the estimate falling as the definition loosens means those students respond less than the ones already included. The response is nonetheless positive and statistically significant under every definition, including the one that asks only for a single school in the market to raise the student's ranking.

Panel B of Figure A16 presents the event study estimates under two continuous measures. The baseline measure also enters continuously, as the median potential gain in the student's market, and I report one alternative to it: the share of schools in the market that give the student a positive gain. The two summarize different features of the same distribution of gains: the median measures how much the typical school in the market would raise the student's ranking, while the share of schools with a positive gain measures how much of the market would raise it at all. They nonetheless deliver very similar point estimates: a one standard deviation increase in the median gain raises the probability of switching in 2014 by 1.47 pp, and a one standard deviation increase in the share of schools with a positive gain by 1.59 pp; the difference of 0.12 pp is not statistically different from zero ($p = 0.134$).

C.2. Responses under Different Sizes of School Markets

The measures above describe the gains a market offers. The number of schools a student has within reach and the number worth moving to are different objects, and only the latter should affect students' responses. In Gran Santiago the number of schools in the market and the potential gain are in fact essentially uncorrelated (correlation of −0.07). Whether a nearby school raises a

student's ranking depends on their own GPA, on that school's threshold and on the threshold of the school they attend, so a market can be large without offering much.

Figure A17 reports the distribution of market size, including lines for the 25th, 50th and 75th percentile. The dashed line shows the mean of the distribution, that is 18.1 schools, slightly higher than the median of 16 schools. Figure A18 tests the importance of the number of alternatives by estimating the event study separately above and below the median, dividing students into groups facing 9.3 and 26.2 schools on average. Having a positive median gain raises the probability of switching in 2014 by 2.2 pp in the smaller-market group and 2.9 pp in the larger-market group, and equality is not rejected at the 5% level; under the continuous measure the estimates are 1.4 pp and 1.6 pp per standard deviation ($p = 0.27$). The size of a market does not predict the response because it does not predict how much of that market is worth moving to.

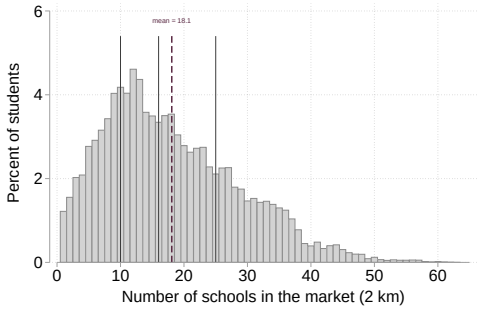
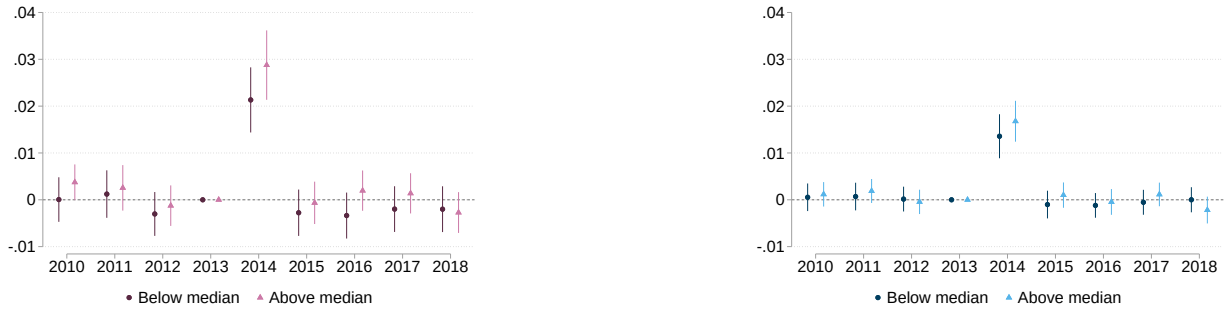


FIGURE A17. Distribution of the Number of Alternative Schools

Note: This figure shows the distribution of the number of high schools within 2 km of the student's primary school, pooling the cohorts finishing twelfth grade between 2010 and 2018. The sample is students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there at the start of twelfth grade. Bars are percentages of students; solid lines mark the 25th, 50th and 75th percentiles and the dashed line the mean.



A. Positive gain

B. Standardized gain

FIGURE A18. Impact School Switching by the Number of Alternatives Available

Note: This figure shows the coefficients from Equation (4) for switching schools during twelfth grade, estimated separately for students below the within-cohort median number of high schools within 2 km of their primary school, circles, and at or above it, triangles. Panel A uses a positive median potential gain and Panel B the standardized median potential gain, per standard deviation. All regressions include school at the start of twelfth grade and within-school GPA decile fixed effects. Sample: students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there. Coefficients are relative to 2013. Standard errors are clustered at the school-market level; bars are 99% confidence intervals.

Appendix D. Students' Relevant Educational Choice Set

A student's potential gain depends on the set of schools they could switch to. My main analysis uses a 2-kilometer buffer around each student's primary school. As a sensitivity analysis, Figure A19 presents the median potential gain distribution for buffers of 2 to 8 km. The distributions are similar across radii.

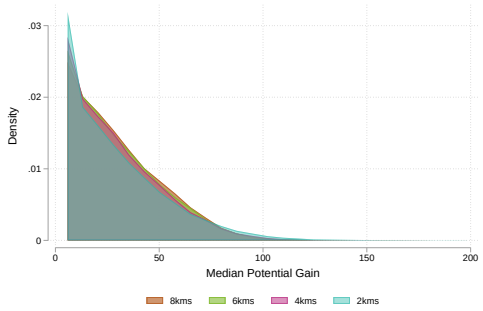
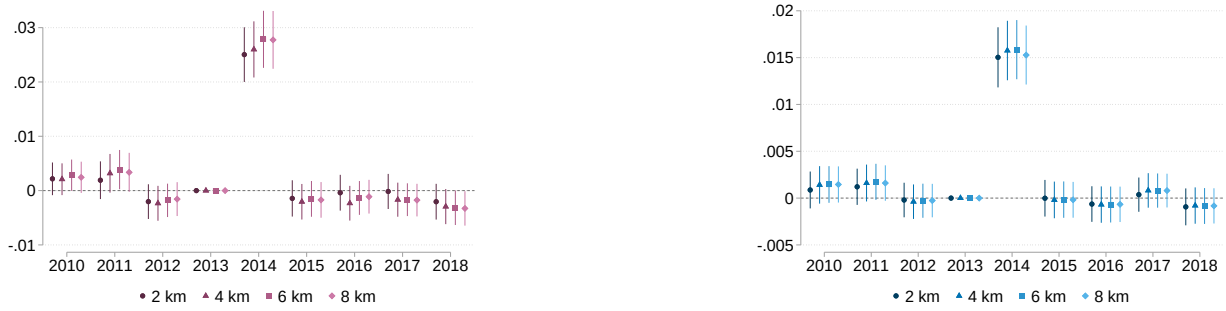


FIGURE A19. Positive Median Potential Gain Distribution.

Note: This figure shows the density of the median potential gain, in RR-score points, among students with a positive gain, for markets defined as the high schools within 2, 4, 6 and 8 km of the student's primary school. Sample: students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there, cohorts finishing twelfth grade 2010–2014.

Figure A20 re-estimates Equation (4) for Gran Santiago with the potential gain computed on each buffer. The estimates barely change with the radius. A positive gain raises the probability of switching during twelfth grade in 2014 by 2.5 pp with the 2-km buffer, and by 2.6, 2.8, and 2.8 pp with the 4-, 6-, and 8-km buffers. Under the continuous measure, the estimates range from 1.5 to 1.6 pp per standard deviation.



A. Positive gain

B. Standardized gain

FIGURE A20. Effect of the Disclosure on Switching Schools by School-Market Radius

Note: This figure shows the coefficients from Equation (4) for switching schools during twelfth grade, with the median potential gain computed over the high schools within 2 km, circles; 4 km, triangles; 6 km, squares; and 8 km, diamonds, of the student's primary school. Panel A uses a positive median potential gain and Panel B the standardized median potential gain. All regressions include school at the start of twelfth grade and within-school GPA decile fixed effects. Sample: students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there. Coefficients are relative to 2013. Standard errors are clustered at the school-market level; bars are 99% confidence intervals.

Appendix E. Grade Inflation and the Treatment Measures

This appendix reproduces the main results of Section 6.1 using treatment measures adjusted for grade inflation. The first part describes how the measures are reconstructed and reports a balance table comparing treated and control students before and after the adjustment. The second part reports the estimates of Equation (4) under each. Deflating the grades leaves them largely unchanged.

E.1. Constructing the grade-inflation-adjusted measures

To construct the adjusted measures, I use the results presented in Fajnzylber et al. (2019). They document that the ranking bonus raised recorded grades unevenly across schools: schools whose students stood to gain more from lenient grading, or with a larger share of students in the bonus zone, inflated more. Within schools, however, they find no evidence that inflation was targeted at the students with larger gains, pointing to schools raising grades uniformly across their own students but differentially across schools. Their estimates imply 0.164 grade points (on a scale of 1.0 to 7.0) for the cohort graduating in 2014, whose 10th, 11th, and 12th grades were affected, and they extrapolate 0.219 points once all four high school grades fall under the policy — roughly 0.055 points per exposed grade, and about twice that in the schools facing the strongest incentives.[49]

The treatment measures used in this paper are built from grades (see Section 4.3), so if grade inflation differs across schools it could change which students are recorded as having a potential gain or as having any gain at all. For inflation to confound the estimates it would have to be that (i) students are misclassified through changes in their own GPA and in the thresholds, so they count as treated only because inflation occurred, and (ii) switchers target schools that inflated less in the past, and so post lower thresholds, while expecting those schools to grade them no less generously than their origin school would have during the second semester, given the timing of switching (see Figure 8). Condition (i) alone would bias the estimates only if the resulting misclassification created non-classical measurement error; condition (ii) describes one behavior that would make that possible.

Following Fajnzylber et al. (2019), the first step in recovering the deflated inputs of the treatment measure, the student's GPA and high schools' thresholds, is to classify schools by the strength of their incentive to inflate. That strength has two parts: how much a marginal grade point is worth to a school's students, and how many of its students are positioned to use it. Let $\underline{r}_s$ be the mean GPA of school s over its three most recent graduating cohorts and $\bar{r}_s$ its highest GPA across those cohorts, the two points in the GPA interval between which the ranking bonus rises. Then, a grade point inside that interval is worth

$$b_s = \frac{850 - z(\underline{r}_s)}{\bar{r}_s - \underline{r}_s} - z' \tag{A1}$$

[49] Grade inflation enters additively in their model, and it does so here.

points of ranking score, where $z(\cdot)$ is the national conversion from the GPA scale to the standardized score scale and z' its slope.[50] Letting π_s be the share of students in school s whose grades fall between the two thresholds, the school's incentive is $I_s = \pi_s \cdot b_s$. I compute I_s on the 2011 graduating cohort, the last one to finish high school before the policy was announced.

In a second step, I take the national distribution of I and sort schools into three groups: 737 schools are in the bottom quartile, 1,472 between the 25th percentile and the 75th percentiles, and 736 in the top quartile. Four schools have no bonus zone, so I drop them. Students attending a school that cannot be matched to a 2011 value are assigned to the middle group; they account for 5.6 percent of student-years nationally over the period, and 5.5 percent of the estimation sample (Gran Santiago). Each group $g \in \{l, m, h\}$ is then assigned a rate of inflation per exposed grade δ_g, following the gradient in Fajnzylber et al. (2019).[51]

In the third step I deflate each input in turn: the students' GPA and the thresholds of their starting and ending school for those who switched. The three carry different amounts of inflation given how they are constructed. A student's GPA is the average of their four high school grades, and each year's grade is inflated only by what had accumulated by the year it was taken, so the average carries less than the final year alone. A school's thresholds are computed from its three previous graduating cohorts, so they carry the average of what those cohorts' own GPAs carried, which is less again.

TABLE A15. Exposure Multipliers for the Treatment Inputs, by Graduating Cohort

	Inflation carried, in units of δ_g	
Graduating cohort	Student's NEM	School's thresholds
2010	0.00	0.000
2011	0.00	0.000
2012	0.25	0.000
2013	0.75	0.083
2014	1.50	0.333
2015	2.50	0.833
2016	2.50	1.583
2017	2.50	2.167
2018	2.50	2.500

Note: This table reports, for each graduating cohort, how much grade inflation the student's GPA and the school's thresholds carry, in units of δ_g, the school's inflation rate per exposed grade. The adjusted GPA subtracts the first column times δ_g, and the adjusted thresholds the second, which averages the first over the three previous cohorts. The multipliers are the same for all three inflation groups.

To fix ideas, take the cohort graduating in 2014 in a school in the middle group of incentives to inflate grades, therefore $\delta = 0.058$. Their four years of grades carry zero, one, two, and three units of δ_g respectively, so their GPA carries 1.50 δ_g or 0.087 points. The thresholds they face are built from the 2011, 2012, and 2013 graduates, whose own four-year GPAs carried 0, 0.25 and 0.75 units

[50] Following DEMRE official information, the slope is 206 in the academic track, 202 for adult students in the academic track, and 204 for students in the vocational track.

[51] The gradient is 0.033 grade points per exposed grade in the low group, 0.058 in the middle group and 0.070 in the high group.

of δ, so the thresholds carry 0.333 δ, or 0.019 points. Table A15 reports both multipliers for every cohort.[52]

Finally, with the deflated grades and thresholds, I recompute each student's potential gain at every school in their market exactly as in Section 4.3, and rebuild the discrete and the continuous measures. The two versions of the continuous measure correlate at 0.99. Of the 588,733 student-years in the estimation sample, 2.4% fall in a different treatment group under the adjusted measure: 0.8% acquire a positive potential gain, and 1.6% lose one, so the treated share falls from 38.1 to 37.3%. Reclassification is zero for the 2010 and 2011 cohorts by construction, reaches 4.0% in 2014 and peaks at 5.0% in 2015.

Table A16 reports means and standard deviations for students with and without a positive potential gain, computed first on the original data and then on the deflated data, together with the difference between the two. Grade point average is mechanically lower under the deflated data, by 0.075 points among students with no positive gain and 0.056 among those with one. Apart from it, the two groups look the same under both measures.

TABLE A16. Balance of Characteristics under the Original and the Deflated Discrete Treatment Measure

	Original measure				Deflated measure				Difference	
	No positive gain		Positive gain		No positive gain		Positive gain		No positive	Positive
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	gain	gain
Panel A: *Outcomes*										
Switched schools in twelfth grade, beginning or during the year	0.05	0.22	0.03	0.16	0.05	0.22	0.03	0.16	-0.000	-0.000
Switched schools during twelfth grade	0.02	0.14	0.01	0.10	0.02	0.14	0.01	0.10	-0.000	-0.000
Panel B: *GPA and test score*										
GPA (non-standardized)	5.30	0.44	5.86	0.44	5.23	0.43	5.80	0.44	-0.075	-0.056
Tenth-grade SIMCE score, standardized	-0.06	0.94	0.67	0.95	-0.06	0.94	0.68	0.95	-0.000	0.014
Panel C: *Other characteristics*										
Public school	0.29	0.46	0.26	0.44	0.29	0.45	0.26	0.44	-0.001	0.000
Voucher school	0.60	0.49	0.51	0.50	0.60	0.49	0.51	0.50	0.000	-0.002
Private school	0.11	0.31	0.22	0.42	0.11	0.31	0.22	0.42	0.001	0.001
Academic track	0.42	0.49	0.69	0.46	0.42	0.49	0.70	0.46	0.002	0.002
Vocational track	0.41	0.49	0.26	0.44	0.41	0.49	0.26	0.44	-0.001	-0.001
School performance, quartile 1 (lowest)	0.21	0.41	0.10	0.30	0.21	0.40	0.10	0.30	-0.001	-0.001
School performance, quartile 2	0.27	0.45	0.18	0.38	0.27	0.44	0.18	0.38	-0.001	0.000
School performance, quartile 3	0.27	0.45	0.26	0.44	0.27	0.45	0.26	0.44	0.000	-0.001
School performance, quartile 4 (highest)	0.25	0.43	0.46	0.50	0.25	0.43	0.46	0.50	0.002	0.001
Mother's education: less than high school	0.29	0.46	0.20	0.40	0.29	0.46	0.20	0.40	-0.001	-0.000
Mother's education: high school	0.47	0.50	0.43	0.50	0.47	0.50	0.43	0.50	0.000	-0.001
Mother's education: more than high school	0.24	0.43	0.37	0.48	0.24	0.43	0.37	0.48	0.001	0.001
Male	0.52	0.50	0.43	0.49	0.52	0.50	0.43	0.49	-0.000	-0.001
Female	0.48	0.50	0.57	0.49	0.48	0.50	0.57	0.49	0.000	0.001
Low aspirations	0.39	0.49	0.18	0.39	0.38	0.49	0.18	0.38	-0.001	-0.002
High aspirations	0.61	0.49	0.82	0.39	0.62	0.49	0.82	0.38	0.001	0.002
No internet access	0.24	0.43	0.17	0.37	0.24	0.42	0.17	0.37	-0.001	-0.000
Internet access	0.76	0.43	0.83	0.37	0.76	0.42	0.83	0.37	0.001	0.000
Observations	359,339		220,975		363,725		216,589			

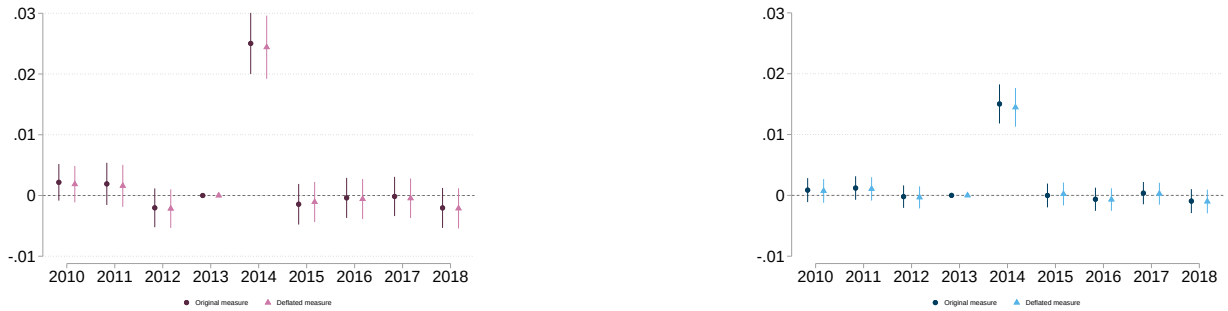
Note: This table reports means and standard deviations for students with and without a positive median potential gain, under the original treatment measure, columns 1 to 4, and under one rebuilt from grades deflated following Fajnzylber et al. (2019), columns 5 to 8. Columns 9 and 10 report the deflated minus the original mean for each group. Rows differ only because the deflation reclassifies students, except GPA, which the deflated block reports deflated. Sample: students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there, cohorts finishing twelfth grade 2010–2018.

E.2. Main Switching Results under the Adjusted Measures

Figure A21 plots the estimates of Equation (4) under the original and the adjusted measures. Panel A presents the estimation coefficients using the discrete treatment measure, while Panel B presents

[52] Deflated grades are censored below at 4.0, the minimum passing grade. The floor binds for at most two students in ten thousand in any cohort.

the results under the continuous measure. In both panels circles report the estimates from the main text and triangles the estimates obtained after deflating grades and thresholds.



A. Positive gain

B. Standardized gain

FIGURE A21. Effects Using the Original and Grade-Inflation-Adjusted Measures of Potential Gain

Note: This figure shows the coefficients from Equation (4) for switching schools during twelfth grade, with the treatment built from the original grades, circles, and from grades and school thresholds deflated for the grade inflation implied by Fajnzylber et al. (2019), triangles. Panel A uses a positive median potential gain and Panel B the standardized median potential gain. All regressions include school at the start of twelfth grade and within-school GPA decile fixed effects. Sample: students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there. Coefficients are relative to 2013. Standard errors are clustered at the school-market level; bars are 99% confidence intervals.

The two estimated coefficients track each other in every cohort. The 2014 estimate under the deflated discrete measure is 2.44 percentage points, against 2.50 with the original measure, a fall of 2.5%. Under the deflated continuous measure the estimate is 1.45 against 1.50, a fall of 3.7%. Deflation barely moves the point estimates and leaves the standard errors unchanged to the third decimal.

Back of the envelope calculations give a sense of how much inflation would be needed to account for the effect in full. What displaces a student across the gain boundary is not the level of inflation but the gap between what their own GPA carries and what the thresholds they face carry, which for a middle-group school in the 2014 cohort is $(1.50 - 0.33).058 \approx 0.07$ grade points. That gap reclassifies 2.4 percent of the sample. Taking reclassification as locally proportional to the gap, accounting for the estimate in full would require reclassifying essentially the entire treated group, 38% of students, or roughly sixteen times the inflation Fajnzylber et al. (2019) document. That is a change of about 1.1 grade points, two standard deviations of the GPA distribution, and would imply an average recorded GPA inflated from 5.5 to 6.6 on a scale that ends at 7.

Appendix F. The 2011 Student Protests and Their Effect on School Switching

This appendix describes the 2011 protests, which raised twelfth-grade switching, and separates that response from the strategic switching of 2014.

From May to December 2011, secondary and higher education students staged protests and school occupations lasting from hours to months (UNICEF Chile 2014; Penaglia and Mejías 2019). The protests opposed the funding and governance of the education system (Somma 2012). By June 2011 466 schools were occupied (Penaglia and Mejías 2019), with the occupations concentrated in public schools, although voucher and private schools were occupied as well (UNICEF Chile 2014). Although the mobilizations were concentrated in Santiago, of the 124 days of protest recorded in 2011, 52 were simultaneous in Santiago and other regions, 47 in Santiago alone and 25 in other regions (Penaglia and Mejías 2019).

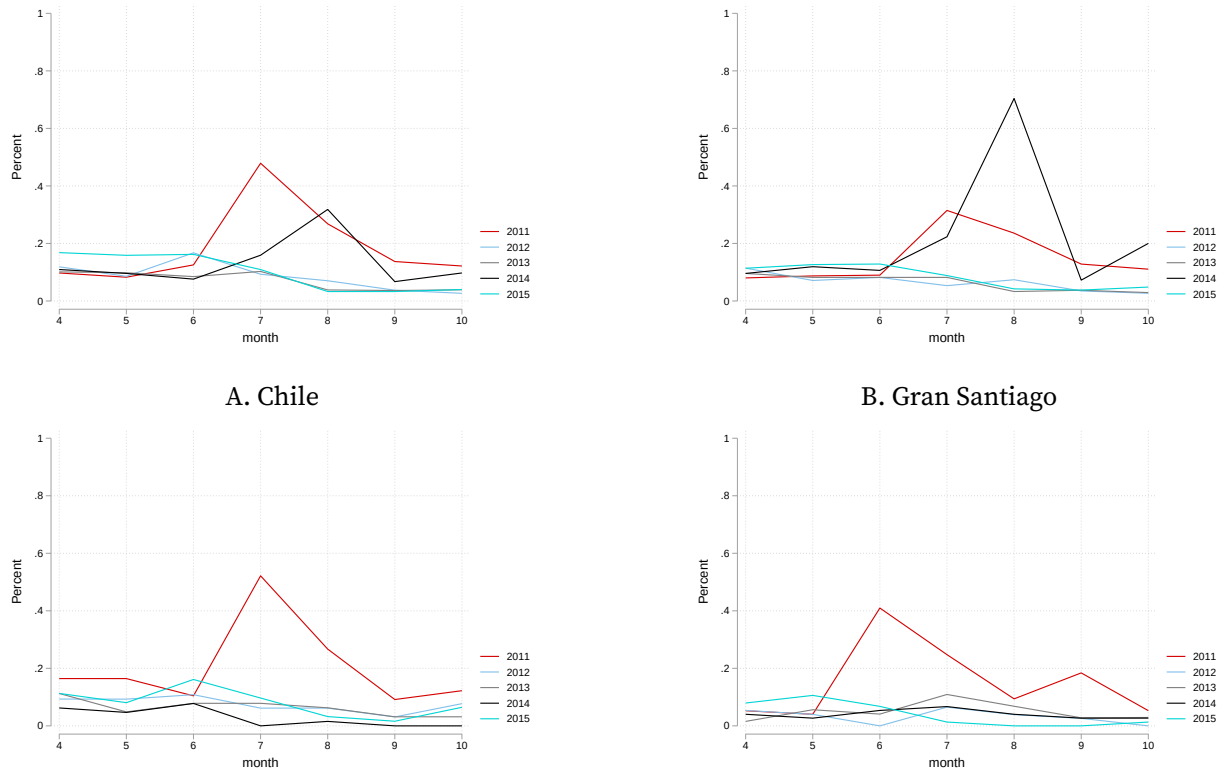
There are two main aspects that separate the 2011 and 2014 moves in twelfth grade and their underlying motives:

Timing. The 2011 moves concentrate right after the occupations began in May and June.[53] Several patterns tie the 2011 moves to the protests. First, nationwide, the number of changes during the year spikes in 2011, while the number at the beginning of the year peaks in 2012 (5 shows the same split for Gran Santiago), which suggests students changed schools for the academic year right after the protests. Second, the monthly changes confirm that students switched schools earlier in the year in 2011 than in 2014: the monthly probability of changing school peaks in July 2011 and in August 2014. Finally, the event study shows that students with a gain switched more in July 2011 (Figure A23), the month after the occupations began, and in August in 2014 (Figure 8). In 2011 students with a gain also switched more than those without, but by less than in 2014, consistent with a shock that hit all students in occupied schools (Figure 8).

Geography. The protests in 2011 were concentrated in Gran Santiago but extended to the whole country by June 2011. Gran Concepción and Gran Valparaíso were also heavily affected, but the most affected counties were spread across the country. Figure A22 shows that these two metropolitan areas saw substantial switching in 2011 but not in 2014. H explains why neither area shows a spike in 2014. Using newspaper coverage, I list the thirty counties most affected in 2011 and split the country outside Gran Santiago into those counties and the rest.[54] The July 2011 increase appears only in the affected counties: it is there that students with a positive potential gain are more likely to change schools in July 2011 than in March 2013, while in the rest of the country the same difference is not distinguishable from zero. The continuous measure gives the same pattern (Figure A23).

[53] Emol, 13 June 2011, reports close to one hundred schools occupied across the country.

[54] The thirty counties are Concepción, Valparaíso, Punta Arenas, Antofagasta, Calama, Temuco, Valdivia, Arica, La Serena, Iquique, Rancagua, Talca, Puerto Montt, Viña del Mar, Coquimbo, San Antonio, Talcahuano, Curicó, Osorno, Chillán, Copiapó, Coronel, Los Ángeles, Quillota, Tocopilla, San Felipe, Linares, Ancud, Coyhaique and Ovalle. A student is in the affected group when the counties of the primary school, the twelfth-grade school and the school attended that month are all on this list. The comparison group is the remaining counties outside Gran Santiago, together with the students of the thirty affected counties whose primary or twelfth-grade school lies outside them.



A. Chile

B. Gran Santiago

C. Gran Valparaiso

D. Gran Concepción

FIGURE A22. Percent of Students Switching Schools in Twelfth Grade by Month

Note: This figure shows the percent of twelfth graders who switch schools in each month from April to October, one line per year from 2011 to 2015. A student switches in a month when their school the following month differs; the denominator is all twelfth graders observed that month. Panel A covers Chile. Panels B to D cover students who lived in the metropolitan area in primary school and at the start of twelfth grade and whose school that month is in the area.

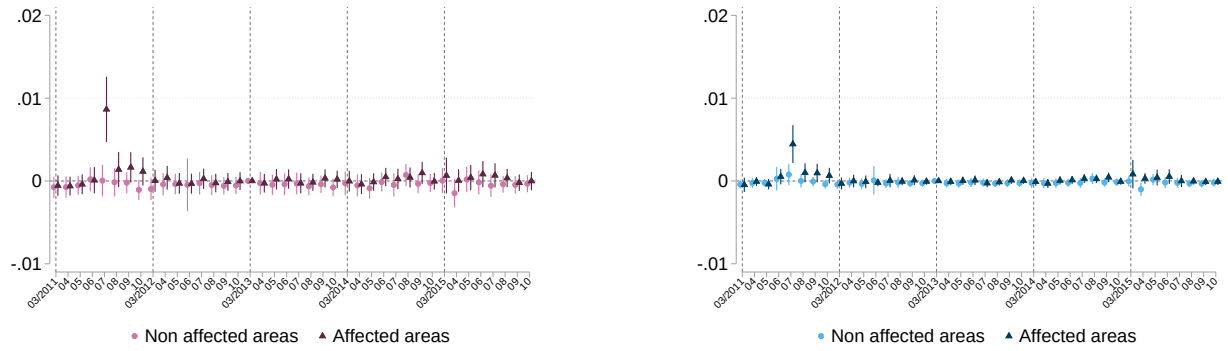
Appendix G. Construction of the Variables

This appendix describes how the switching outcome and the college outcomes used in Section 6 are built, and the characteristics used to classify the programs students apply to, enroll in, and graduate from.

G.1. Switching Schools during Twelfth Grade

From MINEDUC, I use the yearly student performance records that contain one record for each student in each school they attend during the academic year, with the school identifier, grade, class, GPA, attendance, and end-of-year status. A student who switches schools during the year therefore has one record in each school.

I build the switching variables from these records as follows. Switching during the academic year equals one if the student has records in two different schools in the same year. Changing class within the same school does not count as a switch. Switching at the beginning of the year equals



A. Positive gain

B. Standardized gain

FIGURE A23. Monthly School Switching during Twelfth Grade outside Gran Santiago

Note: This figure shows the coefficients from a monthly version of Equation (4), March to October of 2011 to 2015, where the outcome equals one in the last month a student attends a school before switching to another. Panel A uses a positive median potential gain and Panel B the standardized median potential gain. Triangles are students in the 30 counties most affected by the 2011 protests, listed in footnote 54, and circles all other students; each group is estimated separately. Sample: students who lived outside Gran Santiago in primary school and at the start of twelfth grade and attended a school outside it. All regressions include school at the start of twelfth grade and within-school GPA decile fixed effects. Coefficients are relative to March 2013. Standard errors are clustered at the school-market level; bars are 99% confidence intervals.

one if the school where the student starts year t differs from the school where they ended year $t-1$. I drop students with more than two records in a year (who switched more than once), since their initial school is ambiguous, and students with no recorded grades. The sample keeps students in twelfth grade between 2010 and 2018. Figure 5 eports both types of switch by year in Gran Santiago, for all schools (Panel A) and for high-performing schools (Panel B).

For monthly switches, I use a similar process with the monthly attendance records collected by MINEDUC, available from March 2011. For each student, the monthly switch equals one in month m if the school they attend in month m differs from the school they attend in month $m+1$.

G.2. From the Admission Process to the Degree

I use the records of the Departamento de Evaluación, Medición y Registro Educacional (DEMRE) and of the Servicio de Información de Educación Superior of the Ministry of Education (SIES) to construct the college outcomes.

From DEMRE, I obtain, for every admission process, the PSU registration, the rank-ordered list of program-university pairs, the preference to which the student was admitted, the program in which they enrolled if they used the regular process,[55] and the program catalogue with each program's admission weights. From SIES, I obtain the yearly enrollment panel of every student in higher education, the yearly file of degrees awarded, and the average earnings and employability of graduates by generic field within university.[56] I keep students who graduated from high school

[55]The centralized system has a few additional routes through which a student can be admitted to a university. All require the PSU, but some, such as PACE, admit students through a different process.

[56]The DEMRE catalogue lists 1,419 programs in the 2014 admission process, while SIES records 1,851 programs in Table 1 once campuses and non-centralized programs, mostly evening programs, are counted. Earnings are published for 800

between 2011 and 2016 and follow them until they graduate from college or until 2025, the last year of SIES degree records.

Since students can apply more than once through the centralized system, either with the same PSU score or by retaking the test, I first create the outcomes at the student-process level and then at the student level. A process is a PSU sitting or, for a student who applies and enrolls using a PSU score from an earlier process, the year they enroll.[57] Each student-process records whether the student applied, the programs on their list, whether and where they were admitted, the program they enrolled in, and any program they moved into afterwards without a break of more than one year. For students observed in more than one process, a degree is matched to the process whose programs it came from.

For a student graduating from high school in cohort $t \in \{2011, \ldots, 2016\}$, I use admission processes $t+1$ to T, where $T = 2020$ is the last process available in the data. Applied equals one if the student lists at least one program in the centralized system in their first admission process, admitted if they are admitted to a program on their list, and enrolled if they enroll in a program in the centralized system. An enrollment counts only if the program was in the DEMRE offer in the year the student entered it, which excludes evening, distance, and special programs for degree holders at universities that use the system. For universities that joined the centralized system during the period of analysis, enrollments before the institution started using the system are not counted. A common-entry program (plan común, bachillerato, or ingreso común) and the degree it feeds are treated as one program.

For dropout and switching, I use the yearly enrollment panel to follow the first program the student enrolled in. Leaving the first program equals one if the student leaves it without a degree within two years of entry. A student who leaves switches programs if they enroll in a different program in the centralized system, in the same admission process or in a later one, and drops out if they never enroll in a university in the centralized system again. For graduation, I compare the program from which the student graduated with the program they first enrolled in. Graduated equals one if the student obtains a degree from the first program they enrolled in, and graduated on time equals one if, in addition, the semesters from entry to the degree exceed the program's nominal length, including the titling period, by at most two semesters. Students who never enroll take the value zero in dropout, switching, and graduation.

I measure each outcome for all programs and for programs that are highly selective, high-employability, or high-earnings, as defined in Section G.3. For applications, admission, and enrollment, the outcome equals one if the program the student applies to, is admitted to, or enrolls in is of that type. For dropout, switching, and graduation, it equals one if the first program the student enrolled in is of that type.

university-generic-field cells, so programs at the same university in the same generic field share one figure.

[57] I drop 458 students whose first enrollment is not preceded by a PSU sitting in the data.

G.3. Program Characteristic Definitions

I fix program characteristics before the disclosure of the RR formula. For university selectivity, I follow Kapor et al. (2024) and classify universities into four tiers using the 2012 admission process. For program selectivity, I use the 2014 admission process, which contains 1,333 of the 1,423 programs offered in the 2015 admission process, the first to which students exposed to the disclosure could apply.

Selectivity is the mean admission scores of the students admitted to the program, and a program is highly selective if it is in the top quartile of that measure.

For program earnings, I use the mean monthly income of the program's graduates in the fourth year after the degree, calculated by SIES pooling the 2013, 2014, and 2015 university graduating cohorts.[58] For employability, I use the share of the program's graduates employed in the first year after the degree, also published by SIES. For programs in the 2014 DEMRE catalog without a published figure (SIES suppresses small cells, and some figures are not reported), earnings and employability are predicted from a linear model with graduation-year, generic-field, and university fixed effects. Of the 1,419 programs in the 2014 admission process, 744 have a reported earnings figure, 547 a predicted one, 45 the mean of the degrees they feed, and 83 none. A program is high-earnings if it is in the top quartile of earnings and high-employability if it is in the top quartile of employability.

For STEM and traditional programs, I use the field classifications SIES assigns to each program. A program is STEM if SIES classifies it in the natural sciences, mathematics and statistics, engineering, or information technology fields of the CINE 2013 classification, and traditional if its generic field is business administration, medicine, law, or civil engineering. Of the 1,419 programs offered in the 2014 admission process, 14.2% are at a tier 1 or 2 university, 24.8% are highly selective, 26% are high-employability, 26.5% are high-earnings, 17.4% are traditional, and 32.1% are STEM.

Appendix H. Metropolitan Areas in Chile and the Conditions for Switching Schools

This appendix describes the school and college markets of Chile's metropolitan areas. Before the disclosure, the areas already differ in the two things that matter: the school market, which makes a move possible, and the college market, which makes it worth making. Among the three main metropolitan areas, which together cover more than 50% of students in twelfth grade, only Gran Santiago meets both, and its scale is part of why: of the next largest area, it has more schools within reach and more selective programs to compete for.

Table A17 reports the population of each area and the size of its twelfth-grade cohort in 2013, the last cohort to graduate before the disclosure of the RR policy formula.[59] Gran Santiago is by far

[58] A student exposed to the disclosure finishes high school in 2014 at the earliest, enters college in 2015, after the last cohort the figure is built from.

[59] Metropolitan areas follow the government's definition under Law 21.074. Gran Santiago comprises Santiago, Cerrillos, Cerro Navia, Conchalí, El Bosque, Estación Central, Huechuraba, Independencia, La Cisterna, La Florida, La Granja, La Pintana, La Reina, Las Condes, Lo Barnechea, Lo Espejo, Lo Prado, Macul, Maipú, Ñuñoa, Pedro Aguirre Cerda,

the biggest metropolitan area in the country, accounting for 39.5% of the nationwide population and 43.4% of twelfth graders. Gran Concepción and Gran Valparaíso, the next two largest areas in the country, hold roughly a million people each, representing 5.7% and 5.4% of the country respectively, while the fourth largest area, Gran Coquimbo, holds less than half a million people. The rest of the appendix presents characteristics for four groups: Gran Santiago, Gran Concepción, Gran Valparaíso, and the rest of the country.

TABLE A17. Population and Twelfth-Grade Students by Metropolitan Area

Metropolitan area	Population	Share (%)	Students in 12th grade	Share (%)
Gran Santiago	6,958,284	39.5	64,712	43.4
Gran Concepcion	1,008,512	5.7	9,190	6.2
Gran Valparaiso	952,990	5.4	8,640	5.8
Gran Coquimbo	432,786	2.5	3,715	2.5
Gran Rancagua	340,270	1.9	2,963	2.0
Gran Iquique	291,003	1.7	2,328	1.6
Gran Puerto Montt	284,465	1.6	2,108	1.4
Remaining areas	7,343,592	41.7	55,479	37.2
Total	17,611,902	100.0	149,135	100.0

Note: This table reports the 2013 population and number of twelfth graders in each metropolitan area defined by Law No. 21.074, with all other counties grouped as the remaining areas. Population comes from INE's county projections, base 2017 census. Students are the 2013 twelfth-grade cohort, assigned to an area when they lived in it both in primary school and at the start of twelfth grade. Shares are relative to the column totals.

Section 3 implies that switching pays only if (i) a school within reach would raise the student's score and (ii) the student applies to programs selective enough for that change in the admission score to matter (3).

Table A18 reports school market characteristics at three radii around each student's market centroid, defined as the geolocation of their primary school, by area group. Within 2 km a student in Gran Santiago has on average 18 alternative schools while a student in Gran Valparaíso has 17, and students in Gran Concepción and the rest of Chile have 10 alternative schools. The areas separate more as the market widens; at 6 km the counts are 111, 59, 30, and 20, respectively. The alternatives that would actually raise a student's ranking score follow the same pattern, rising from 7.4, 6.7, 4.1 and 3.8 at 2 km to 45.3, 22.4, 11.3 and 7.7 at 6 km for Gran Santiago, Gran Valparaíso, Gran Concepción, and the rest of Chile, respectively. Gran Santiago has the lowest mean threshold at every radius, so a student with good grades at a demanding school finds nearby schools that would rank them higher: 41% of the alternatives within 2 km give a positive gain, and 39% of students in the 2013 cohort have a positive median gain, the highest of all areas. Gran Valparaíso and Gran Concepción differ: Gran Valparaíso has many more schools within reach, but they offer a median

Peñalolén, Providencia, Pudahuel, Quilicura, Quinta Normal, Recoleta, Renca, San Joaquín, San Miguel, San Ramón, Vitacura, Puente Alto, Pirque, San José de Maipo, Colina, Lampa, Tiltil, San Bernardo, Buin, Calera de Tango, Paine, María Pinto, Talagante, El Monte, Isla de Maipo, Padre Hurtado and Peñaflor. Gran Valparaíso comprises Valparaíso, Viña del Mar, Concón, Quilpué and Villa Alemana. Gran Concepción comprises Concepción, Coronel, Chiguayante, Hualqui, Lota, Penco, San Pedro de la Paz, Santa Juana, Talcahuano, Tomé and Hualpén.

gain of only 0.89 RR points, against 5.43 in Gran Concepción.

TABLE A18. The School Alternatives and Potential Gains by Metropolitan Area

	Gran Santiago			Gran Concepción			Gran Valparaíso			Rest of Chile		
	2 km	4 km	6 km	2 km	4 km	6 km	2 km	4 km	6 km	2 km	4 km	6 km
Panel A: Size of the market												
Alternatives in market	18.01	56.77	111.49	10.31	19.42	29.99	16.94	39.97	58.99	9.90	17.18	20.26
Schools with positive gain	7.37	23.27	45.32	4.10	7.44	11.28	6.67	15.38	22.35	3.83	6.59	7.72
Panel B: Thresholds and gain												
Mean threshold in market	5.44	5.45	5.45	5.56	5.56	5.57	5.55	5.54	5.55	5.52	5.53	5.53
SD of threshold in market	0.23	0.24	0.25	0.24	0.25	0.26	0.24	0.26	0.26	0.21	0.22	0.22
Median gain	4.87	4.01	3.80	5.43	4.82	2.87	0.89	0.48	0.22	4.99	3.96	3.70
SD of median gain	27.31	26.42	26.44	29.01	27.93	26.64	30.37	29.07	28.86	29.39	28.28	28.25
Share with positive median gain	0.39	0.36	0.35	0.38	0.37	0.34	0.31	0.29	0.29	0.36	0.34	0.34
Number of markets		1,677			265			338			2,361	
Number of students		64,712			9,190			8,640			66,593	

Note: This table describes students' school markets of 2, 4 and 6 km around their primary school, by metropolitan area; the rest of Chile includes all other counties. Panel A reports the number of high schools in the market and the number that give the student a positive potential gain. Panel B reports the mean of the alternative schools' mean thresholds and its standard deviation across those schools, and the student's median potential gain in RR points, its standard deviation across students, and the share with a positive median gain; all are averaged over students. Sample: the 2013 twelfth-grade cohort, students who lived in the same area in primary school and at the start of twelfth grade.

For a student to switch schools, alternatives with positive gain are worth having only if a better ranking changes the student's perception of where they can be admitted. Although applications are national, students apply locally, so the selective programs in their own area are what matter. Table A19 describes the college market of each area. Panel A presents the characteristics of the programs offered in each area. Gran Santiago leaves only 7% of its seats unfilled, against 18% in Gran Concepción, 12% in Gran Valparaíso, and 22% in the rest of Chile. This matters because in a program that does not fill, every applicant is admitted and a better ranking changes nothing. Gran Santiago also offers far more that is worth competing for: 14% of the programs in Gran Santiago are among the hundred most selective in the country against 6% in Gran Concepción, 5% in Gran Valparaíso and 2% in the rest of Chile. The gap persists further down the ranking: among the top 1,000 of the 1,419 programs, the share is 80% in Gran Santiago, 65% in Gran Concepción and 73% in Gran Valparaíso.

Panel B shows how students respond to what their area offers. Among the first five preferences,[60] at least 98% of applicants list at least one program in their own area when living in one of the three main metropolitan areas. Applicants in the three areas are about equally likely to list a tier 1–3 university; tier 3 is where Gran Concepción and Gran Valparaíso have most of their universities.[61] The difference is at the top tiers: 71% of applicants in Gran Santiago list a program at a tier 1 or 2 university, against 22% in Gran Valparaíso and 7% in Gran Concepción.

[60] More than 95% of admitted students are placed in one of their first five preferences (Table 3).

[61] One tier 2 university, the Universidad Adolfo Ibáñez, has a campus in Gran Valparaíso.

TABLE A19. College Programs and Applications by Metropolitan Area

	Gran Santiago	Gran Concepción	Gran Valparaíso	Rest of Chile
***Panel A**: What is available*				
Number of programs (N=1,419)	475	175	210	559
Number of seats (N=110,380)	47,501	14,473	16,076	32,330
Share of seats left unfilled	0.07	0.18	0.12	0.22
Share of programs in the national top 100	0.14	0.06	0.05	0.02
Share of programs in the national top 500	0.53	0.33	0.34	0.22
Share of programs in the national top 1000	0.80	0.65	0.73	0.63
At least one campus from a tier 1 university (N=2)	2	0	0	1
At least one campus from a tier 2 university (N=3)	3	0	1	0
At least one campus from a tier 3 university (N=14)	8	3	3	7
At least one campus from a tier 4 university (N=14)	3	3	2	12
***Panel B**: What students do*				
Share of students who apply in the same year they graduate	0.32	0.40	0.38	0.30
Among the first three preferences				
At least one program in own area	0.98	0.98	0.97	0.64
At least one program at a tier 1 university	0.47	0.05	0.09	0.15
At least one program at a tier 1–2 university	0.67	0.06	0.20	0.22
At least one program at a tier 1–3 university	0.90	0.85	0.88	0.81
Among the first five preferences				
At least one program in own area	0.99	0.98	0.98	0.69
At least one program at a tier 1 university	0.49	0.06	0.10	0.17
At least one program at a tier 1–2 university	0.71	0.07	0.22	0.25
At least one program at a tier 1–3 university	0.92	0.88	0.89	0.83
Number of students	64,712	9,190	8,640	66,593
Number of applicants	20,700	3,720	3,287	20,305

Note: This table describes college programs and applications by metropolitan area; the rest of Chile includes all other counties. Panel A covers the programs in the 2014 admission process, assigned to areas by campus; national totals are in parentheses. National rankings order programs by the mean application score of admitted students, and shares are over the programs in each area. Unfilled seats are posted seats minus admitted students. Universities are counted in every area where they have a campus, so the tier rows sum to more than the national total. Panel B covers the 2013 twelfth-grade cohort, students who lived in the same area in primary school and at the start of twelfth grade. Applying means a valid application in the 2014 process; the other rows are among applicants and report the share listing at least one program of each kind among their first three or five preferences. Tiers follow Kapor et al. (2024).

Appendix I. Policy Intensity and the Timing of the Response

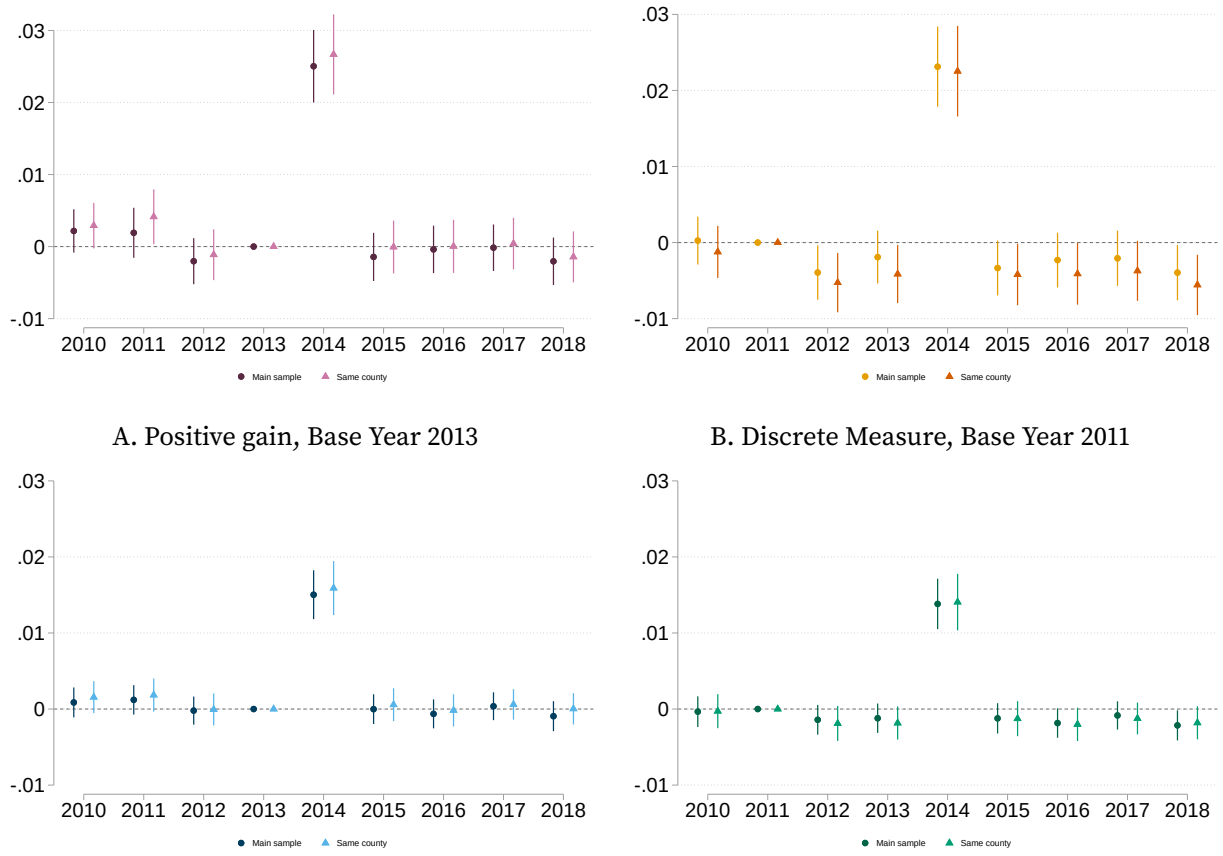
The 2014 spike in switching during twelfth grade admits two competing interpretations: policy salience and policy bite. Figure A24 and Table A2 present evidence in favor of the salience interpretation.

Throughout the paper, policy salience means a shock that makes students more aware of how the RR is computed. The RR had been publicly discussed since 2012, when it was first implemented, and had appeared in the official guidelines since 2013, but its construction was poorly explained. I argue that the relevant shock is the change in how it was presented in the guidelines. Figure A1 reproduces the relevant sections of the guidelines for the 2014 and 2015 admission processes; Section 2.3 describes the change.*Policy bite*, by contrast, refers to an increase in the weight the RR carries in the admission score.

To separate the two, I compute the average weights of the RR and the GPA in the centralized system. Table A2 shows the averages for all participating universities and separately by selectivity tier. The RR weight rises twice: for the 2012 cohort, when the component is introduced at 10 percent, and for the 2013 cohort, when it reaches 22.2 percent on average. In both cases the increase comes mostly from a reduction in the weight on GPA. Between the 2013 cohort, which is the base year in the main analysis, and the 2014 cohort, the RR weight changes by 0.76 pp across all programs, by

0.10 pp in tier 1 and by 0.27 pp in tier 2. The change is statistically indistinguishable from zero in the two selective tiers, and while the overall difference is marginally significant ($p = 0.052$), it is less than one point on a base of 22. The incentive facing the 2014 cohort was therefore essentially the one facing the cohort before it.

Because the policy was introduced for the 2012 cohort (admission process 2013), and the weights rose for the 2013 cohort, I also re-estimate the main specification using 2011 as the base year. Figure A24 shows no increase in switching in either 2012 or 2013. If anything, the coefficients for those years are negative and far smaller in magnitude than the 2014 estimate. Panels A and B use the positive-gain indicator and Panels C and D the standardized gain.



A. Positive gain, Base Year 2013

B. Discrete Measure, Base Year 2011

C. Standardized gain, Base Year 2013

D. Continuous Measure, Base Year 2011

FIGURE A24. Switching Schools during Twelfth Grade, Alternative Base Years

Note: This figure shows the coefficients from Equation (4) for switching schools during twelfth grade, relative to 2013 in Panels A and C and to 2011 in Panels B and D, so that the 2013 cohort, the first to face the higher ranking weight, is estimated. Panels A and B use a positive median potential gain and Panels C and D the standardized median potential gain. Circles are students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there; triangles restrict them to students who lived in the same county at both times. All regressions include school at the start of twelfth grade and within-school GPA decile fixed effects. Standard errors are clustered at the school-market level; bars are 99% confidence intervals.

Appendix J. The Switching Response of Emblematic and Other High-Performing Public Schools

This appendix examines whether the switching response documented in Gran Santiago is concentrated among the public high schools commonly known as *liceos emblemáticos* (emblematic schools). I first define the category, list its schools, and count the high-performing public schools outside it. I then compare the university application patterns of students in the two groups before the policy, and finally estimate the response separately for each. I do not find evidence that the effects are driven only by emblematic schools.

Emblematic schools in Chile. They are long-established public high schools associated with academic excellence, tradition and prestige. The Ministry of Education recognizes no single definition of the category and has produced at least two lists that do not entirely coincide. The first, in a 2020 Centro de Estudios report, names 19 schools, all in the metropolitan region of Santiago. The second, the 2025 Plan de Recuperación de Liceos Emblemáticos, names 28 schools across the country. They overlap on 9, so I use their union, 38 schools, which A20 lists with their source. Fifteen of the 38 schools lie outside Gran Santiago, so the category is not specific to the main metropolitan area.

Within Gran Santiago, the emblematic schools are a subset of high-performing public schools: among the 45 high-performing public schools, 18 carry an emblematic designation and 27 do not. Comparing the two categories holds performance and financing fixed, so a difference between the two sets reflects the designation and other characteristics correlated with it.

One such characteristic is whether their students apply to university at all, and where they aim when they do. Among the cohorts graduating from high school between 2010 and 2011, 62.1% of students at the emblematic schools submitted a valid application, against 25.2% at the other top-quartile public schools; among those who applied, 82.1% listed a program at a tier 1 or 2 university among their first five preferences, against 64.8% (see Table A21). A student already aiming at the most selective universities has more to gain from a higher ranking score, and so may be more likely to switch.

Figure A25 presents the results for each type of high-performing public school separately. Students with a positive median gain at the 27 non-emblematic schools increase their probability of switching in 2014 by 7.41 pp, and students at the 18 emblematic schools by 13.85 pp. Estimated jointly, the difference is 6.44 pp, significant at the 1% level. Under the continuous measure the estimates are 4.15 pp and 6.85 pp with a difference of 2.70 pp at the 5% significance level. Students at emblematic schools respond more than any other group, but they do not drive the results: students with a gain at other high-performing public schools also respond to the disclosure.

The separation appears only once the formula is public. Taking 2012 as the omitted year, the effect of a positive gain at emblematic schools is 0.71 pp in 2013, not distinguishable from zero, against 14.44 pp in 2014; under the continuous measure it is 0.14 pp and 6.75 pp. In the year in which a school with advance knowledge of the formula could have acted on it, these schools moved no more than any other.

TABLE A20. Schools Identified as *Liceos Emblemáticos*

School	Comuna	Region	2020 list	2025–26 plan
Metropolitan Region(Santiago)				
Instituto Nacional	Santiago	Metropolitana	•	•
Javiera Carrera	Santiago	Metropolitana	•	•
Isaura Dinator de Guzmán	Santiago	Metropolitana	•	•
Liceo de Aplicación Rector Jorge E. Schneider	Santiago	Metropolitana	•	•
Manuel Barros Borgoño	Santiago	Metropolitana	•	•
Internado Nacional Barros Arana	Santiago	Metropolitana	•	•
Miguel Luis Amunátegui	Santiago	Metropolitana	•	
Confederación Suiza	Santiago	Metropolitana	•	
Darío Salas	Santiago	Metropolitana	•	
Miguel de Cervantes y Saavedra	Santiago	Metropolitana	•	
Teresa Prats	Santiago	Metropolitana		•
Valentín Letelier	Recoleta	Metropolitana	•	
Liceo N° 7 Luisa Saavedra de González	Providencia	Metropolitana	•	•
Carmela Carvajal de Prat	Providencia	Metropolitana	•	•
José Victorino Lastarria	Providencia	Metropolitana	•	•
Polivalente Arturo Alessandri P.	Providencia	Metropolitana	•	
Tajamar	Providencia	Metropolitana	•	
Carmela Silva Donoso	Ñuñoa	Metropolitana	•	
República de Siria	Ñuñoa	Metropolitana		•
Augusto D'Halmar	Ñuñoa	Metropolitana		•
José Toribio Medina	Ñuñoa	Metropolitana		•
Andrés Bello	San Miguel	Metropolitana	•	
Instituto Superior de Comercio de Chile	San Miguel	Metropolitana	•	
Outside the Metropolitan Region				
Octavio Palma Pérez	Arica	Arica y Parinacota		•
Bernardo O'Higgins	Iquique	Tarapacá		•
Mario Bahamondes Silva	Antofagasta	Antofagasta		•
José Antonio Carvajal	Copiapó	Atacama		•
Gregorio Cordovez	La Serena	Coquimbo		•
Eduardo de la Barra	Valparaíso	Valparaíso		•
Óscar Castro	Rancagua	O'Higgins		•
Abate Molina	Talca	Maule		•
Narciso Tondreau	Chillán	Ñuble		•
Enrique Molina Garmendia	Concepción	Biobío		•
Pablo Neruda	Temuco	La Araucanía		•
Armando Robles Rivera	Valdivia	Los Ríos		•
Manuel Montt	Puerto Montt	Los Lagos		•
Josefina Aguirre Montenegro	Coyhaique	Aysén		•
Luis Alberto Barrera	Punta Arenas	Magallanes		•

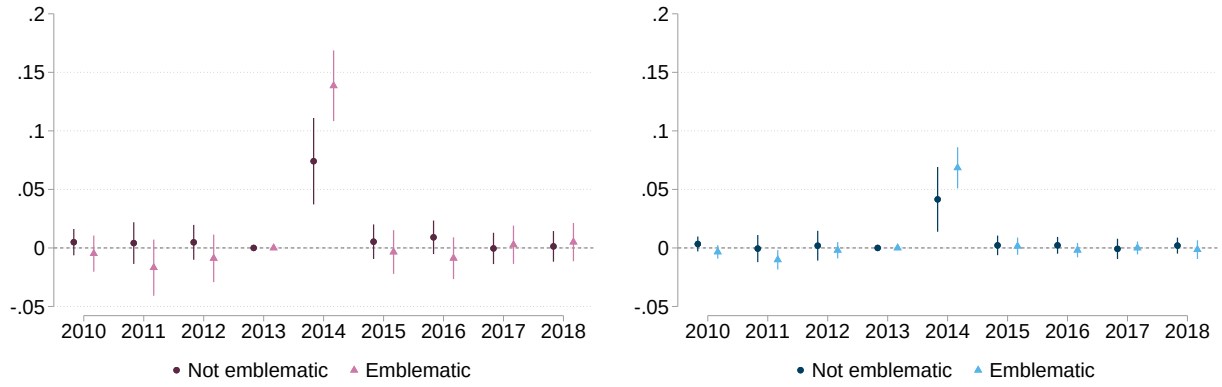
Note: This table lists the schools identified as *liceos emblemáticos*. The category has no single official definition, and MINEDUC has published two lists that partly overlap: *Apuntes* 1-2020, Table A1, which reproduces a 2016 list covering only the Metropolitan Region, and the Plan de Recuperación de Liceos Emblemáticos, ministerial resolution of 3 March 2025, with 19 schools entering in 2025 and 9 in 2026. The last two columns mark the source; nine schools appear on both, and the analysis uses the union of 38. Schools are grouped by whether they are located in Gran Santiago.

Because the estimates compare students in the same school and the same GPA decile who differ in whether their own market offers a gain, a school that informed all of its students is captured within these fixed effects. The design cannot, however, identify how the information reached students, possibly through teachers or peers. That would bias the estimates only if schools treated their students differently, and the one documented school response in this setting, grade inflation, was uniform within schools (Appendix E), which reduces the concern that the response reflects schools behaving strategically rather than students reacting to the policy.

TABLE A21. University Applications Before the Policy, Emblematic and Other High-Performing Public Schools

	Share applying to university (%)	Share listing a tier 1 or 2 university in the first five preferences (%)	Students	Applicants	Schools
Emblematic	62.1	82.1	7,721	4,797	18
Not emblematic	25.2	64.8	5,666	1,430	26
Difference	36.9	17.3			
	(8.1)	(5.7)			
p-value	<0.001	0.004			

Note: This table describes the applications of the 2010 and 2011 twelfth-grade cohorts, the last before the RR policy was announced, at public schools in the top performance quartile by funding type and region. Sample: students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there; 44 schools, 18 emblematic and 26 not. A student applies if they submit an application in the following admission process; the second column is among applicants and reports the share listing a program at a tier 1 or 2 university, following Kapor et al. (2024), among their first five preferences. A school is emblematic if it appears on either list in Table A20. Differences come from regressing each indicator on the emblematic designation, with standard errors clustered by school in parentheses.



A. Positive gain

B. Standardized gain

FIGURE A25. Switching Schools Emblematic and Other High-Performing Public Schools

Note: This figure shows the coefficients from Equation (4) for switching schools during twelfth grade, estimated separately for public schools in the top performance quartile by funding type and region without an emblematic designation, circles, and with one, triangles. Panel A uses a positive median potential gain and Panel B the standardized median potential gain. Sample: students who lived in Gran Santiago in primary school and at the start of twelfth grade and attended a school there; 45 schools, 27 not emblematic and 18 emblematic. A school is emblematic if it appears on either list in Table A20. All regressions include school at the start of twelfth grade and within-school GPA decile fixed effects. Coefficients are relative to 2013. Standard errors are clustered at the school-market level; bars are 99% confidence intervals.